SCONTENTS

STORES LaTeX CONTENTS

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CTAN: https://www.ctan.org/pkg/scontents
https://github.com/pablgonz/scontents

Abstract

This package allows to store LaTeX code, including "*verbatim*", in ⟨*sequences*⟩ using the l3seq module of expl3. The ⟨*stored content*⟩ can be used as many times as desired in the document, additionally you can write to ⟨*external files*⟩ or show it in ⟨*verbatim style*⟩.

Contents

1 Description of the package 1
2 Motivation and Acknowledgments . . . 1
3 License and Requirements 2
4 The scontents package 2
4.1 Installation 2
4.2 Loading and options 2
4.3 The TAB character 2
4.4 Configuration of the options 3
4.5 Options Overview 4
5 User interface 4
5.1 The environment scontents 4
5.2 The command \newenvsc 5
5.3 The command \Scontents 5
5.4 The command \getstored 6
5.5 The command \foreachsc 6
5.6 The command \typestored 7
5.7 The command \meaningsc 7
5.8 The command \mergesc 7
5.9 The environment verbatimsc 8
6 Other commands provided 8
6.1 The command \countsc 8
6.2 The command \cleanseqsc 8
6.3 Programming interface 9
7 The scontents package in action 9
8 Examples 10
8.1 From answers package 10
8.2 From filecontentsdef package . . 10
8.3 From TeX-SX 11
8.4 Customization of verbatimsc 13
8.5 The command \mergesc in action . . 15
8.6 Stored contents before storing them . 16
8.7 The tagged PDF example 16
9 References 17
10 Change history 18
11 Index of Documentation 20
12 Implementation 21
13 Index of Implementation 63

1 Description of the package

The SCONTENTS package allows to ⟨*store contents*⟩ in ⟨*sequences*⟩ or ⟨*external files*⟩. In some ways it is similar to the filecontentsdef package, with the difference in which the ⟨*content*⟩ is stored. The idea behind this package is to get an approach to ConTEXt *"buffers"* by making use of ⟨*sequences*⟩.

2 Motivation and Acknowledgments

In LaTeX there is no direct way to record content for later use, although you can do this using \macro, recording ⟨*verbatim content*⟩ is a problem, usually you can avoid this by creating external files or boxes.

The general idea of this package is to try to imitate this implementation "*buffers*" that has ConTEXt which allows you to save content in memory, including *verbatim*, to be used later. The filecontentsdef[1] package solves this problem and since expl3[2] has an excellent way to manage data, ideas were combined giving rise to this package.

This package would not be possible without the great work of Jean François Burnol who was kind enough to take my requirements into account and add the filecontentsdefmacro environment to filecontentsdef package. Also a special thanks to Phelype Oleinik who has collaborated and adapted a large part of the code and all LaTeX team for their great work and to the different members of the TeX-SX community who have provided great answers and ideas. Here a note of the main ones:

1. Stack datastructure using LaTeX
2. LaTeX equivalent of ConTeXt buffers
3. Storing an array of strings in a command
4. Collecting contents of environment and store them for later retrieval

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†E-mail: «pablgonz@educarchile.cl».

5. Collect contents of an environment (that contains verbatim content)

Starting with version 2.3 the SCONTENTS package is fully compatible with *tagged* PDF[3] and will have LaTeX release 2026-06-01 (TeX Live 2026) and update expl3 release as a minimum requirement.

3 License and Requirements

Permission is granted to copy, distribute and/or modify this software under the terms of the LaTeX Project Public License (lppl), version 1.3 or later (https://www.latex-project.org/lppl.txt). The software has the status "maintained".

The package SCONTENTS requires an updated version of LaTeX to work (minimum required to LaTeX release 2026-06-01 and expl3 update). This package can be used with plain, context, xelatex, lualatex, pdflatex and the classical workflow latex»dvips»ps2pdf.

4 The scontents package

4.1 Installation

The package SCONTENTS is present in TeX Live and MiKTeX, use the package manager to install. For manual installation, download scontents.zip and unzip it, run luatex scontents.ins and move all files to appropriate locations, then run mktexlsr. To produce the documentation with source code run luatex scontents.ins and lualatex scontents.dtx three times.

```
scontents.tex       »  TDS:tex/generic/scontents/
scontents-code.tex  »  TDS:tex/generic/scontents/
scontents.sty       »  TDS:tex/latex/scontents/
t-scontents.mkiv    »  TDS:tex/context/third/scontents/
scontents.pdf       »  TDS:doc/latex/scontents/
README.md           »  TDS:doc/latex/scontents/
scontents.dtx       »  TDS:source/latex/scontents/
scontents.ins       »  TDS:source/latex/scontents/
```

4.2 Loading and options

The package is loaded in the usual way:

For LaTeX users

```
\usepackage[⟨key = val⟩]{scontents}
```

For plain TeX users

```
\input scontents.tex
```

For ConTeXt users

```
\usemodule{scontents}
```

The package options are not available for plain TeX and ConTeXt and must be passed using \setupsc (§4.4). ConTeXt users should use -luatex, the implementation does not support LuaMetaTeX.

4.3 The TAB character

Some users use the horizontal TAB '⇥' character from keyboard to indent the source code of the document and depending on the text editor used, some will use real TAB (*hard tabs*), others with *soft tabs* (␣␣ or ␣␣␣␣) or both.

At first glance it may seem the same, but the way in which TAB character (*hard tabs*) are processed according to the context in which they are found within a file, both in ⟨*reading*⟩[1] and ⟨*writing*⟩[2] are different and may have adverse consequences.

In a standard LaTeX document, the character TAB '⇥' are treated as explicit spaces (in most contexts) and is the behavior when ⟨*stored content*⟩, but when ⟨*writing files*⟩ these are preserved.

With a TeX Live distribution, the TAB character is *printable* for latex, pdflatex and lualatex, but if you use xelatex you must add the -8bit option on the command line, otherwise you will get TeX character TAB '^^I' in the ⟨*output file*⟩.

[1] Check the answer given by Ulrich Diez in Keyboard TAB character in argument v (xparse).
[2] Check the answer given by Enrico Gregorio in How to output a tabulation into a file.

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As a general recommendation *"Do not use TAB character unless strictly necessary"*, for example within a *"verbatim environment"* that supports this character such as `Verbatim` of the packages `fancyvrb`[4], `fvextra`[5] or `lstlisting` of the package `listings`[6] or when you want to generate a `MakeFile` file.

4.4 Configuration of the options

Most of the options can be passed directly to the package or using the command `\setupsc`. All boolean keys can be passed using the equal sign '=' or just the name of the key, all unknown ⟨*keys*⟩ will return an error. In this section are described some of the options, a summary of all options is shown in §4.5.

`\setupsc` `\setupsc{`⟨*key = val*⟩`}`

The command `\setupsc` takes the mandatory argument {⟨*key = val*⟩} and sets the ⟨*keys*⟩ for the rest of the document, or until the end of the current group, it can be used both in the preamble and in the body of the document as many times as desired. However, options set in the declaration of an environment (with `\newenvsc`) have precedence over options set with `\setupsc`.

Options in the optional arguments of environments and commands have the highest precedence, overriding both options in `\newenvsc` and in `\setupsc`.

`debug-var` = {⟨*true* | *false*⟩} default: *true*

Determines whether the ⟨*argument*⟩ passed to commands or the ⟨*body*⟩ passed to provided environments is written to the `.log` file. This ⟨*key*⟩ is only available as a package option or using `\setupsc`.

`verb-font` = {⟨*font family*⟩} default: `\ttfamily`

Sets the ⟨*font family*⟩ used to display the ⟨*stored content*⟩ for `\typestored`, `\mergesc` and `\meaningsc` commands. This ⟨*key*⟩ is only available as a package option or using `\setupsc`.

`store-all` = {⟨*seq name*⟩} default: *not used*

It is a ⟨*meta-key*⟩ that sets the `store-env` key of the `scontents` environment and the `store-cmd` key of the `\Scontents` command. This ⟨*key*⟩ is only available as a package option or using `\setupsc`.

`remember` = {⟨*seq names*⟩} default: *not used*

Sets the *sequences*, a comma separated list of {⟨*seq names*⟩}, whose ⟨*stored contents*⟩ can be used before they are stored, as `\label` and `\ref` do with the references. At the end of the document the ⟨*stored contents*⟩ of these *sequences* are written to the file `\jobname.sct` and in the next compilation `\getstored`, `\foreachsc`, `\typestored`, `\meaningsc`, `\mergesc` and `\countsc` take the ⟨*stored contents*⟩ from it, so they are available before they are stored. As with the references, two compilations are needed: the first one prints nothing where a ⟨*stored content*⟩ is not available yet and a "warning" at the end of the document asks to compile again. If a ⟨*stored content*⟩ is still not available, for example with an ⟨*index*⟩ greater than the number of ⟨*stored contents*⟩, a "warning" says that there were undefined ⟨*stored contents*⟩. This ⟨*key*⟩ is only available in LaTeX, in the preamble, as a package option or using `\setupsc`. For example:

```
\setupsc{remember=objectives} % in the preamble
\section{Objectives}
\getstored[1]{objectives}
\section{Methods}
\Scontents[store-cmd=objectives]{Measure the effect of the method.}
```

A complete example is in §8.6. The file `\jobname.sct` is an auxiliary file, as the `.aux` file, and it can be removed at any time. To remove it with `latexmk -c` add this line to the `latexmkrc` file:

```
push @generated_exts, 'sct';
```

`overwrite` = {⟨*true* | *false*⟩} default: *false*

Sets whether the ⟨*files*⟩ generated by `write-out` and `write-env` from the `scontents` environment will be rewritten. This ⟨*key*⟩ is available as a package option, for `\setupsc`, for `\Scontents` and `\Scontents*`, for the base environment `scontents` and all environments created using `\newenvsc`.

`print-all` = {⟨*true* | *false*⟩} default: *false*

It is a ⟨*meta-key*⟩ that sets the `print-env` key of the `scontents` environment and the `print-cmd` key of the `\Scontents` command. This ⟨*key*⟩ is only available as a package option or using `\setupsc`.

`force-eol` = {⟨*true* | *false*⟩} default: *false*

Sets if the *last end of line* for the ⟨*stored content*⟩ is hidden or not. This ⟨*key*⟩ is necessary only if the *last line* is the closing of some environment defined by `fancyvrb`[4] or `fvextra`[5] packages as `\end{Verbatim}` or another environment that does not support a comments '%' after closing `\end{«env»}%`. This ⟨*key*⟩ is available for the base environment `scontents`, the `\Scontents` and `\Scontents*` commands and all environments created using `\newenvsc`.

`width-tab` = {⟨*integer*⟩} default: *1*

Sets the equivalence in ⟨*spaces*⟩ for the character `TAB` '↹' used when displaying ⟨*stored content*⟩ in *verbatim style*. The value must be a ⟨*positive integer*⟩. This ⟨*key*⟩ is available for `\typestored`, `\mergesc` and `\meaningsc` commands.

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4.5 Options Overview

key	package	\setupsc	scontents	\Scontents	\Scontents*	\typestored	\meaningsc	\mergesc
debug-var	✓	✓	✗	✗	✗	✗	✗	✗
verb-font	✓	✓	✗	✗	✗	✗	✗	✗
store-all	✓	✓	✗	✗	✗	✗	✗	✗
remember	✓	✓	✗	✗	✗	✗	✗	✗
overwrite	✓	✓	✓	✓	✓	✓	✓	✓
print-all	✓	✓	✗	✗	✗	✗	✗	✗
force-eol	✓	✓	✓	✓	✓	✗	✗	✗
width-tab	✓	✓	✗	✗	✗	✓	✓	✓
store-env	✓	✓	✓	✗	✗	✗	✗	✗
store-cmd	✓	✓	✗	✓	✓	✗	✗	✗
print-env	✓	✓	✓	✗	✗	✗	✗	✗
print-cmd	✓	✓	✗	✓	✓	✓	✓	✓
write-env	✗	✗	✓	✗	✗	✗	✗	✗
write-cmd	✗	✗	✗	✓	✓	✗	✗	✗
write-out	✗	✗	✓	✓	✓	✓	✓	✓
typestored	✗	✗	✗	✗	✗	✗	✗	✓
meaningsc	✗	✗	✗	✗	✗	✗	✗	✓
getstored	✗	✗	✗	✗	✗	✗	✗	✓

5 User interface

The user interface consists in scontents environment, \Scontents and \Scontents* commands to ⟨*stored content*⟩, \getstored command to get the ⟨*stored content*⟩, \typestored and \mergesc commands to print *verbatim style* the ⟨*stored content*⟩ along with other utilities described in this documentation.

5.1 The environment scontents

scontents

```
\begin{scontents}[⟨key=val⟩]
    ⟨body env⟩
\end{scontents}
```

The scontents environment takes the optional argument [⟨*key = val*⟩] and the {⟨*body env*⟩}, which it *"stores"* in a ⟨*sequence*⟩ or *"writes"* to an ⟨*external file*⟩ if desired, including *verbatim material.* After the package has been loaded, the environment can be used both in the preamble and in the body of the document.

For the correct operation \begin{scontents} and \end{scontents} must be in *different lines*, all ⟨*keys*⟩ must be passed separated by commas and *without spaces* '␣' of the start of the environment.

Comments '%' or *any character* after \begin{scontents} or [⟨*key = val*⟩] on the *same line* are NOT supported, the package will return an "error" message if this happens. In a similar way comments '%' or *any character* after \end{scontents} on the *same line* the package will return a "warning" message.

The environment can be *"nested"* if it is properly balanced and does not appear *"literally"* in commented lines or in some *verbatim* environment or command. As an example:

```
\begin{scontents}[store-env=outer]
This text is in the outer environment (before nested).
\begin{scontents}[store-env=inner]
This text is found in the inner environment (inside of nested).
\end{scontents}
This text is in the outer environment (after nested).
\end{scontents}
```

Of course, the ⟨*stored content*⟩ in the *sequence* {⟨*inner*⟩} is *only available* after ⟨*stored content*⟩ in the *sequence* {⟨*outer*⟩} one has been retrieved, either by using the key print-env or \getstored command.

It is advisable to store the ⟨*content*⟩ in *sequences* with different names, so as not to get lost in the order (position) in which it is stored.

Notes for plain TEX and ConTEXt users

In plain TEX there are no environments as in LATEX. Instead of using the environment scontents, one should use a *"pseudo environment"* delimited by \scontents and \endscontents.

\scontents
\endscontents

```
\scontents[⟨key=val⟩]
    ⟨body env⟩
\endscontents
```

The *pseudo environment* takes the same optional argument [⟨*key = val*⟩] as the scontents environment, on the same line as \scontents, and the {⟨*body env*⟩} up to \endscontents.

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ConTEXt users should use \startscontents and \stopscontents.

\startscontents
\stopscontents

\startscontents[⟨key=val⟩]
⟨body env⟩
\stopscontents

The *pseudo environment* takes the same optional argument [⟨key = val⟩] as the scontents environment, on the same line as \startscontents, and the {⟨body env⟩} up to \stopscontents.

Options for environment

The environment options can be configured globally using option in package or the \setupsc command and locally using [⟨key = val⟩] in the environment. The key force-eol is available for this environment.

store-env = {⟨seq name⟩} default: *contents*

Sets the *name* of the *sequence* in which the {⟨body env⟩} will be stored. If the *sequence* does not exist, it will be created globally.

print-env = {⟨true | false⟩} default: *false*

Sets if the ⟨stored content⟩ is displayed or not at the time of running the environment. The ⟨stored content⟩ is extracted from the *sequence* {⟨seq name⟩} set by the key store-env at the time it is executed.

write-env = {⟨file.ext⟩} default: *not used*

Sets the *name* of the ⟨external file⟩ in which the {⟨body env⟩} of the environment will be written. The ⟨file.ext⟩ will be created in the working directory and the {⟨body env⟩} will be *stored* in the *sequence* set by the key store-env at the time it is executed. If ⟨file.ext⟩ does not exist, it will be created or overwritten if the overwrite key is used.

write-out = {⟨file.ext⟩} default: *not used*

Sets the *name* of the ⟨external file⟩ in which the {⟨body env⟩} of the environment will be written. The ⟨file.ext⟩ will be created in the working directory and the {⟨body env⟩} will NOT be *stored* in the *sequence* set by the key store-env at the time it is executed. If ⟨file.ext⟩ does not exist, it will be created or overwritten if the overwrite key is used.

In the keys write-env and write-out the character TAB will be written in ⟨file.ext⟩, relative or absolute paths are not supported. XELATEX users using character TAB must add -8bit at the command line, otherwise you will get TEX character TAB '^^I' in ⟨file.ext⟩.

5.2 The command \newenvsc

\newenvsc

\newenvsc{⟨env name⟩}[⟨initial keys⟩]

The command \newenvsc takes the mandatory argument {⟨env name⟩} and the optional argument [⟨initial keys⟩], and creates the ⟨new environment⟩ {⟨env name⟩} with the same characteristics of the scontents environment. The values entered in [⟨initial keys⟩] will be considered as the default values for this new environment and the valid ⟨keys⟩ are store-env and print-env. If {⟨env name⟩} already exists, an "error" is returned and its definition is not altered.

In plain TEX and ConTEXt the ⟨new environment⟩ is a *pseudo environment* as the one of scontents (§5.1): for an environment named myenv it is \myenv ... \endmyenv in plain TEX and \startmyenv ... \stopmyenv in ConTEXt. For example:

```
\newenvsc{myenvstore}[store-env=myseq,print-env=false]
```

created the environment myenvstore that ⟨stored content⟩ in the *sequence* {⟨myseq⟩} and will NOT display the ⟨stored content⟩ when the environment it is executed.

5.3 The command \Scontents

\Scontents

\Scontents[⟨key = val⟩]{⟨argument⟩}
\Scontents*[⟨key = val⟩]{⟨argument⟩}
\Scontents*[⟨key = val⟩]⟨del⟩⟨argument⟩⟨del⟩

The command \Scontents takes an optional *starred argument* '*', the optional argument [⟨key = val⟩] and the mandatory {⟨argument⟩}, between *braces* or, in the starred version, between two ⟨del⟩, and stores the {⟨argument⟩} in the *sequence* set by the key store-cmd.

The command \Scontents reads the {⟨argument⟩} in standard mode and *"stores"* it in the *sequence* set by the key store-cmd at the time it is executed. It is not possible to pass *verbatim things*, but it is possible to use the implementation of \Verb delimited by *braces* '{ }' provided by the fvextra[5] package for *verbatim one line* and \lstinline from listings[6] package, but it is preferable to use the *starred argument* '*'.

The \Scontents* command reads the {⟨argument⟩} under *verbatim catcode* regimen and *"stores"* it in the *sequence* set by the key store-cmd at the time it is executed. If its *"first"* delimiter is a *brace* '{',

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it will be assumed that the {⟨*argument*⟩} is nested within *braces*. Otherwise it will be assumed that {⟨*argument*⟩} is delimited by that first delimiter ⟨*del*⟩ like command `\verb`. Blank lines are preserved, escaped braces '`\{`' and '`\}`' must also be balanced if the {⟨*argument*⟩} is used with *braces* and character `TAB` typed from the keyboard are converted into spaces.

The *starred argument* '`*`' and [⟨*key* = *val*⟩] must NOT be separated by *spaces* '␣' between them and the command. Both versions can be used anywhere in the document and cannot be used as an {⟨*argument*⟩} for other command.

Options for command

The command options can be configured globally using option in package or the `\setupsc` command and locally using [⟨*key* = *val*⟩].

`store-cmd` = {⟨*seq name*⟩} default: *contents*

Sets the *name* of the *sequence* in which the {⟨*argument*⟩} is stored. If the *sequence* does not exist, it will be created globally.

`print-cmd` = {⟨*true* | *false*⟩} default: *false*

Sets if the ⟨*stored content*⟩ is displayed or not at the time of running the command. The ⟨*stored content*⟩ is extracted from the *sequence* {⟨*seq name*⟩} set by the key `store-cmd` at the time it is executed.

Options for writing files

The following ⟨*keys*⟩ and the `force-eol` and `overwrite` keys are available for both versions of the command. The `\Scontents*` command writes the {⟨*argument*⟩} to ⟨*file.ext*⟩ exactly as it was typed. The `\Scontents` command reads the {⟨*argument*⟩} in standard mode, so it is written as a list of tokens, in the same way as `\detokenize` does: a space is added after each control word, the character '`#`' is doubled, the line breaks become spaces and the blank lines become `\par`. To write the {⟨*argument*⟩} literally, use the *starred argument* '`*`'.

`write-cmd` = {⟨*file.ext*⟩} default: *not used*

Sets the *name* of the ⟨*external file*⟩ in which the {⟨*argument*⟩} will be written. The ⟨*file.ext*⟩ will be created in the working directory and the {⟨*argument*⟩} will be *stored* in the *sequence* set by the key `store-cmd` at the time it is executed. If ⟨*file.ext*⟩ does not exist, it will be created or overwritten if the `overwrite` key is used.

`write-out` = {⟨*file.ext*⟩} default: *not used*

Sets the *name* of the ⟨*external file*⟩ in which the {⟨*argument*⟩} will be written. The ⟨*file.ext*⟩ will be created in the working directory and the {⟨*argument*⟩} will NOT be *stored* in any *sequence* set by the key `store-cmd` at the time it is executed. If ⟨*file.ext*⟩ does not exist, it will be created or overwritten if the `overwrite` key is used.

In the keys `write-cmd` and `write-out` the character `TAB` will be written in ⟨*file.ext*⟩, relative or absolute paths are not supported. XeLaTeX users using character `TAB` must add `-8bit` at the command line, otherwise you will get TeX character `TAB` '`^^I`' in ⟨*file.ext*⟩.

5.4 The command `\getstored`

`\getstored` `\getstored`[⟨*index*⟩]{⟨*seq name*⟩}

The command `\getstored` takes the optional argument [⟨*index*⟩] and the mandatory argument {⟨*seq name*⟩}, and retrieves the ⟨*stored content*⟩ whose *position* in the *sequence* {⟨*seq name*⟩} is the *integer value* set in ⟨*index*⟩. A *negative* ⟨*index*⟩ counts from the last ⟨*stored content*⟩ and, if the *optional argument* is not passed, the *"last"* ⟨*stored content*⟩ is retrieved.

The command is robust and can be used as an {⟨*argument*⟩} for another command.

5.5 The command `\foreachsc`

`\foreachsc` `\foreachsc`[⟨*key* = *val*⟩]{⟨*seq name*⟩}

The command `\foreachsc` takes the optional argument [⟨*key* = *val*⟩] and the mandatory argument {⟨*seq name*⟩}, and executes the command `\getstored` on each ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}, or `\typestored` or `\meaningsc` with the `typestored` or `meaningsc` key. If the *optional argument* is not passed, `\getstored` is executed on all the ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}.

Options for command

`getstored` ⟨*without value*⟩ default: *active*

Executes `\getstored` on each ⟨*stored content*⟩, as without ⟨*keys*⟩.

`typestored` ⟨*without value*⟩ default: *not active*

Executes `\typestored` on each ⟨*stored content*⟩, which is shown in *verbatim style* (§5.6).

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`meaningsc` ⟨*without value*⟩ default: *not active*

Executes `\meaningsc` on each ⟨*stored content*⟩ (§5.7).

`sep` = {⟨*code*⟩} default: *empty*

Establishes the *separation* between each ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}. For example, you can use `sep={\\[10pt]}` for vertical separation of ⟨*stored contents*⟩.

`step` = {⟨*integer*⟩} default: *1*

Sets the *increment* (⟨*step*⟩) applied to the value set by key `start` for each ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}. The value must be a ⟨*positive integer*⟩.

`start` = {⟨*integer*⟩} default: *1*

Set the *position* of the ⟨*stored content*⟩ within the *sequence* {⟨*seq name*⟩} from which to *start* executing. The value must be a ⟨*positive integer*⟩.

`stop` = {⟨*integer*⟩} default: *total*

Set the *position* of the ⟨*stored content*⟩ within the *sequence* {⟨*seq name*⟩} at which execution ends. The value must be a ⟨*positive integer*⟩.

`before` = {⟨*code*⟩} default: *empty*

Sets the {⟨*code*⟩} that will be executed *before* each ⟨*stored content*⟩ within the *sequence* {⟨*seq name*⟩}. The {⟨*code*⟩} must be passed *between braces* '{ }'.

`after` = {⟨*code*⟩} default: *empty*

Sets the {⟨*code*⟩} that will be executed *after* each ⟨*stored content*⟩ within the *sequence* {⟨*seq name*⟩}. The {⟨*code*⟩} must be passed *between braces* '{ }'.

`wrapper` = {⟨*code* `{#1}` *more code*⟩} default: *empty*

Wraps the ⟨*stored content*⟩ within the *sequence* {⟨*seq name*⟩} referenced by `{#1}`. The {⟨*code*⟩} must be passed *between braces* '{ }'. For example `\foreachsc[wrapper={\makebox[1em][l]{#1}}]{contents}`.

5.6 The command `\typestored`

`\typestored` `\typestored[`⟨*index, start-stop, 1-end, keys*⟩`]{`⟨*seq name*⟩`}`

The command `\typestored` takes the optional argument with the ⟨*index*⟩, the ⟨*start-stop*⟩, the ⟨*1-end*⟩ and the ⟨*keys*⟩, and the mandatory argument {⟨*seq name*⟩}, and prints the ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩} in the internal `verbatimsc` environment (§5.9). If the *optional argument* is not passed, the *first* ⟨*stored content*⟩ is printed.

The *integer value* set in ⟨*index*⟩ is the *position* of the ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}, a *negative* ⟨*index*⟩ counts from the last ⟨*stored content*⟩ as in `\getstored`. The *integer values* set in ⟨*start-stop*⟩ define a *range* of positions, where `end` is the last ⟨*stored content*⟩ and a ⟨*start*⟩ greater than ⟨*stop*⟩ prints the *range* in reverse order, and ⟨*1-end*⟩ prints *"all"* the ⟨*stored content*⟩. Several of them can be given separated by commas. The rest of the accepted ⟨*keys*⟩ are `print-cmd` with default value `true`, `write-out`, `width-tab` and `overwrite`.

In the key `write-out` the character `TAB` will be written in ⟨*file.ext*⟩, relative or absolute paths are not supported. XeLaTeX users using character `TAB` must add `-8bit` at the command line, otherwise you will get TeX character `TAB` '`^^I`' in ⟨*file.ext*⟩.

5.7 The command `\meaningsc`

`\meaningsc` `\meaningsc[`⟨*index, start-stop, 1-end, keys*⟩`]{`⟨*seq name*⟩`}`

The command `\meaningsc` takes the same optional argument as `\typestored` (§5.6) and the mandatory argument {⟨*seq name*⟩}, and prints the *meaning* of the ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}, as `\meaning` does, on a single line. The ⟨*index*⟩, ⟨*start-stop*⟩, ⟨*1-end*⟩ and ⟨*keys*⟩ have the same behavior as in `\typestored` and, if the *optional argument* is not passed, the *first* ⟨*stored content*⟩ is printed.

5.8 The command `\mergesc`

`\mergesc` `\mergesc[`⟨*typestored* | *meaningsc* | *getstored*
keys⟩`]{{`⟨*seq A*⟩`}[`⟨*index*⟩`], {`⟨*seq B*⟩`}[`⟨*start - stop*⟩`], {`⟨*seq C*⟩`}[`⟨*1-end*⟩`]}`

The command `\mergesc` takes the optional argument with the ⟨*keys*⟩, one of them `typestored`, `meaningsc` or `getstored`, and the mandatory argument with a *comma separated list* of *sequences*, and assembles the ⟨*stored content*⟩ in them, as {⟨*seq A*⟩}`[1]`, {⟨*seq B*⟩}`[2-5]` and {⟨*seq C*⟩}`[1-end]` in the syntax, and then executes `\typestored` (§5.6) if the `typestored` key is active, `\meaningsc` (§5.7) if the `meaningsc` key is active, or rescans it as `\getstored` (§5.4) does if the `getstored` key is active.

The {⟨*argument*⟩} taken by this command is a *comma separated list* of the form {⟨*seq name*⟩} followed by either [⟨*index*⟩], [⟨*start-stop*⟩] or [⟨*1-end*⟩], a {⟨*seq name*⟩} alone assembles all its ⟨*stored content*⟩. The use of the keys `typestored`, `meaningsc` or `getstored` are *"mandatory"* and disjoint from each other,

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the rest of the accepted ⟨*keys*⟩ are `print-cmd`, `write-out`, `width-tab` and `overwrite`, which are not accepted with the `getstored` key.

The use of the `write-out` key with this command follows the same rules already described, the main advantage is that allows joining ⟨*stored content*⟩ *without rewriting* the file over and over again, by design TEX does not have an *append mode* for writing files, this effectively allows you to write chunks of code and then merge them into a single file.

5.9 The environment `verbatimsc`

`verbatimsc`

The environment used by `\typestored` and `\mergesc` to display the ⟨*stored content*⟩ in *verbatim style*. The environment is compatible with *tagged* PDF and can be customized in the following ways after loading the SCONTENTS package:

Using the packages `fvextra`[5] or `fancyvrb`[4]:

```
\ExplSyntaxOn
\cs_undefine:N \verbatimsc
\cs_undefine:N \endverbatimsc
\ExplSyntaxOff
\usepackage{fancyvrb}
\DefineVerbatimEnvironment{verbatimsc}{Verbatim}{numbers=left}
```

Using the package `minted`[7]:

```
\ExplSyntaxOn
\cs_undefine:N \verbatimsc
\cs_undefine:N \endverbatimsc
\ExplSyntaxOff
\usepackage{minted}
\newminted{tex}{linenos}
\newenvironment{verbatimsc}{\VerbatimEnvironment\begin{texcode}}{\end{texcode}}
```

Using the package `listings`[6]:

```
\ExplSyntaxOn
\cs_undefine:N \verbatimsc
\cs_undefine:N \endverbatimsc
\ExplSyntaxOff
\usepackage{listings}
\lstnewenvironment{verbatimsc}
  {
   \lstset{
           basicstyle=\small\ttfamily,
           columns=fullflexible,
           language=[LaTeX]TeX,
           numbers=left,
           numberstyle=\tiny\color{gray},
           keywordstyle=\color{red}
          }
  }{}
```

At the moment, the `fvextra`[5] and `fancyvrb`[4] packages partially support *tagged* PDF.

6 Other commands provided

6.1 The command `\countsc`

`\countsc` `\countsc{`⟨*seq name*⟩`}`

The command `\countsc` takes the mandatory argument {⟨*seq name*⟩} and returns the number of ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}. The command is expandable, so it can be used in an *integer expression*, and if the *sequence* does not exist an "error" is returned along with `0`.

6.2 The command `\cleanseqsc`

`\cleanseqsc` `\cleanseqsc{`⟨*seq name*⟩`}`

The command `\cleanseqsc` takes the mandatory argument {⟨*seq name*⟩} and removes all ⟨*stored content*⟩ in the *sequence* {⟨*seq name*⟩}. If the *sequence* does not exist, it is created empty.

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6.3 Programming interface

For other packages and for the code written with expl3, the ⟨*stored contents*⟩ can be read without printing them with the following functions. They work in LaTeX, plain TeX and ConTeXt and also read the *sequences* set by the remember key.

The ⟨*stored contents*⟩ are given as they were stored, without the mark that SCONTENTS adds to hide the last end of line: the content of an environment is a string whose lines are separated by the end of line character ^^J, the content of \Scontents* is also a string and the content of \Scontents keeps its tokens.

\scontents_if_exist_p:n ⋆ \scontents_if_exist_p:n {⟨*seq name*⟩}
\scontents_if_exist:nTF ⋆ \scontents_if_exist:nTF {⟨*seq name*⟩} {⟨*true code*⟩} {⟨*false code*⟩}

Tests if the *sequence* {⟨*seq name*⟩} exists.

\scontents_count:n ⋆ \scontents_count:n {⟨*seq name*⟩}

Leaves the number of ⟨*stored contents*⟩ in the *sequence* {⟨*seq name*⟩} in the input stream, or 0 if it does not exist, without an "error". It can be used in an *integer expression*.

\scontents_get:nnN \scontents_get:nnN {⟨*seq name*⟩} {⟨*index*⟩} ⟨*tl var*⟩

Sets the ⟨*tl var*⟩, locally, to the ⟨*stored content*⟩ at the position {⟨*index*⟩} of the *sequence* {⟨*seq name*⟩}. The {⟨*index*⟩} is an *integer expression*, a negative one counts from the end. If the *sequence* does not exist or the {⟨*index*⟩} is out of range, the ⟨*tl var*⟩ is empty and an "error" is returned.

\scontents_map_inline:nn \scontents_map_inline:nn {⟨*seq name*⟩} {⟨*inline function*⟩}

Applies the {⟨*inline function*⟩} to each ⟨*stored content*⟩ of the *sequence* {⟨*seq name*⟩}, in order, the ⟨*stored content*⟩ is #1. The {⟨*inline function*⟩} can use the other functions and \scontents_map_inline:nn again. If the *sequence* does not exist an "error" is returned. For example, this shows in the .log file each ⟨*stored content*⟩ of the *sequence* data converted to uppercase:

```
\ExplSyntaxOn
\scontents_map_inline:nn { data }
  { \iow_log:e { \text_uppercase:n {#1} } }
\ExplSyntaxOff
```

7 The SCONTENTS package in action

Remember the abstract on the first page? This is it:

Abstract

This package allows to store LaTeX code, including "*verbatim*", in ⟨*sequences*⟩ using the l3seq module of expl3. The ⟨*stored content*⟩ can be used as many times as desired in the document, additionally you can write to ⟨*external files*⟩ or show it in ⟨*verbatim style*⟩.

And the description of the package?

The SCONTENTS package allows to ⟨*store contents*⟩ in ⟨*sequences*⟩ or ⟨*external files*⟩. In some ways it is similar to the filecontentsdef package, with the difference in which the ⟨*content*⟩ is stored. The idea behind this package is to get an approach to ConTeXt *"buffers"* by making use of ⟨*sequences*⟩.

I've only written:

```
\begin{abstract}
This package allows to store \hologo{LaTeX} code, including \enquote{\emph{verbatim}},
in \mymeta{sequences} using the \mypkg{l3seq} module of \mypkg{expl3}. The \mymeta{stored
content} can be used as many times as desired in the document, additionally you can write
to \mymeta{external files} or show it in \mymeta{verbatim style}.
\end{abstract}
```

and

```
The \mypkg*[no-index]{scontents} package allows to \mymeta{store contents} in \mymeta{sequences}
or \mymeta{external files}. In some ways it is similar to the \mypkg{filecontentsdef}
package, with the difference in which the \mymeta{content} is stored. The idea behind
this package is to get an approach to \hologo{ConTeXt} \emph{\enquote{buffers}} by
making use of \mymeta{sequences}.
```

Of course, I didn't copy and paste. The real code they were written with is:

```
1 \begin{scontents}[store-env=abstract,print-env=true]
2 \begin{abstract}
3 This package allows to store \hologo{LaTeX} code, including \enquote{\emph{verbatim}},
4 in \mymeta{sequences} using the \mypkg{l3seq} module of \mypkg{expl3}. The \mymeta{stored
```

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```
content} can be used as many times as desired in the document, additionally you can write
to \mymeta{external files} or show it in \mymeta{verbatim style}.
\end{abstract}
\end{scontents}
```

and

```
\begin{scontents}[store-env=description, print-env=true]
The \mypkg*[no-index]{scontents} package allows to \mymeta{store contents} in \mymeta{sequences}
or \mymeta{external files}. In some ways it is similar to the \mypkg{filecontentsdef}
package, with the difference in which the \mymeta{content} is stored. The idea behind
this package is to get an approach to \hologo{ConTeXt} \emph{\enquote{buffers}} by
making use of \mymeta{sequences}.
\end{scontents}
```

I stored the content in memory and then ran `\getstored` and `\typestored`. This is one of the ways you can use SCONTENTS.

8 Examples

These are some adapted examples that have served as inspiration for the creation of this package. The examples are attached to this documentation and can be extracted from your PDF viewer or from the command line by running:

```
$ pdfdetach -saveall scontents.pdf
```

and then you can use the excellent arara[3] tool to compile them.

8.1 From answers package

Example 1

Adaptation of example 1 of the package answers[8]. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage[store-cmd=solutions]{scontents}
\newtheorem{ex}{Exercise}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}
\section{Problems}
\begin{ex}
First exercise
\Scontents{First solution.}
\end{ex}

\begin{ex}
Second exercise
\Scontents{Second solution.}
\end{ex}

\section{Solutions}
\foreachsc[sep={\\[10pt]}]{solutions}
\end{document}
```

8.2 From filecontentsdef package

Example 2

Adaptation of example from package filecontentsdef[1]. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage[store-env=defexercise,store-cmd=defexercise]{scontents}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}
% not starred
```

[3] The cool TEX automation tool: https://www.ctan.org/pkg/arara

```
\Scontents{
Prove that \[x^n+y^n=z^n\] is not solvable in positive integers if $n$ is at
most $-3$.\par
}
% starred
\Scontents*|Refute the existence of black holes in less than $140$ characters.|
% write environment to \jobname.txt
\begin{scontents}[write-env=\jobname.txt]
\def\NSA{NSA}%
Prove that factorization is easily done via probabilistic algorithms and
advance evidence from knowledge of the names of its employees in the
seventies that the \NSA\ has known that for 40 years.\par
\end{scontents}
% see all stored
\begin{itemize}
\foreachsc[before={\item }]{defexercise}
\end{itemize}
% \getstored are robust :)
\section{\getstored[2]{defexercise}}
\end{document}
```

8.3 From TeX-SX

Example 3

Adapted from LaTeX equivalent of ConTeXt buffers. Source file .

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage[store-cmd=tikz]{scontents}
\usepackage{tikz}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\Scontents{\matrix{ \node (a) {$a$} ; & \node (b) {$b$} ; \\ } ;}
\Scontents{\matrix[ampersand replacement=\&]
{ \node (a) {$a$} ; \& \node (b) {$b$} ; \\ } ;}
\Scontents{\matrix{\node (a) {$a$} ; & \node (b) {$b$} ; \\ } ; }
\begin{document}
\section{tikzpicture}
\begin{tikzpicture}
\getstored[1]{tikz}
\end{tikzpicture}

\begin{tikzpicture}
\getstored[2]{tikz}
\end{tikzpicture}

\begin{tikzpicture}
\getstored{tikz}
\end{tikzpicture}

\begin{scontents}[store-env=buffer]
Hello World!

This is a \verb*|fake poor man's buffer :)|.
\end{scontents}

\section{source tikz}
\typestored[1]{tikz}
\typestored[2]{tikz}
\typestored[3]{tikz}

\section{fake buffer}
\subsection{real content}
\getstored[1]{buffer}
\subsection{verbatim style}
\typestored[1]{buffer}
\subsection{meaning}
\meaningsc[1]{buffer}

\section{tikz again}
```

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```
\foreachsc[before={\begin{tikzpicture}},after={\end{tikzpicture}},sep={\\[10pt]}]{tikz}
\end{document}
```

Example 4

Adapted from Collecting contents of environment and store them for later retrieval. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage{scontents}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}
\begin{scontents}[store-env=main]
Something for main A.
\end{scontents}

\begin{scontents}[store-env=main]
Something for \verb|main B|.
\end{scontents}

\begin{scontents}[store-env=other]
Something for \verb|other|.
\end{scontents}

\textbf{Let's print them}

This is first stored in main: \getstored[1]{main}\par
This is second stored in main: \getstored{main}\par
This is stored in other: \getstored{other}

\textbf{Print all of stored in main}\par
\foreachsc[sep={\\[10pt]}]{main}
\end{document}
```

Example 5

Adapted from Collect contents of an environment (that contains verbatim content). Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage{scontents}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}
\section{Problem stated the first time}
\begin{scontents}[print-env=true,store-env=problem]
This is normal text.
\verb|This is from the verb command.|
\verb*|This is from the verb* command.|
This is normal text.
\begin{verbatim}
This is from the verbatim environment:
&%{}~
\end{verbatim}
\end{scontents}
\section{Problem restated}
\getstored[1]{problem}
\section{Problem restated once more}
\getstored[1]{problem}
\end{document}
```

Example 6

Adapted from Environment hiding its content. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass[10pt]{article}
\usepackage{scontents}
```

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```
\newenvsc{forshort}[store-env=forshort,print-env=false]
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}

Something in the whole course.

\begin{forshort}
    Just a summary...
\end{forshort}

\end{document}
```

8.4 Customization of verbatimsc

Example 7

Customization of verbatimsc using the fvextra[5] and tcolorbox[9] packages. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage{scontents}
\ExplSyntaxOn
\cs_undefine:N \verbatimsc
\cs_undefine:N \endverbatimsc
\ExplSyntaxOff
\usepackage{fvextra}
\usepackage{xcolor}
\definecolor{mygray}{gray}{0.9}
\usepackage{tcolorbox}
\newenvironment{verbatimsc}%
{\VerbatimEnvironment
\begin{tcolorbox}[colback=mygray, boxsep=0pt, arc=0pt, boxrule=0pt]
\begin{Verbatim}[fontsize=\scriptsize, breaklines, breakafter=*, breaksymbolsep=0.5em,
breakaftersymbolpre={\,\tiny\ensuremath{\rfloor}}]}%
{\end{Verbatim}%
\end{tcolorbox}}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}
\section{Test \texttt{\textbackslash begin\{scontents\}} with \texttt{fancyvrb}}
Test \verb+{scontents}+ \par

\begin{scontents}
Using \verb+scontents+ env no \verb+[key=val]+, save in seq \verb+contents+
with index 1.

Prove new \Verb*{ fancyvrb with braces } and environment \verb+Verbatim*+
\begin{verbatim}
 verbatim  environment
\end{verbatim}
\end{scontents}

\section{Test \texttt{\textbackslash Scontents} with \texttt{fancyvrb}}
\Scontents{ We have coded this in \LaTeX: $E=mc^2$.}

\section{Test \texttt{\textbackslash getstored}}
\getstored[1]{contents}\par
\getstored{contents}

\section{Test \texttt{\textbackslash meaningsc}}
\meaningsc[1]{contents}\par
\meaningsc[2]{contents}

\section{Test \texttt{\textbackslash typestored}}
\typestored[1]{contents}
\typestored[2]{contents}
\end{document}
```

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Example 8

Customization of verbatimsc using the listings[6] package. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage{scontents}
\ExplSyntaxOn
\cs_undefine:N \verbatimsc
\cs_undefine:N \endverbatimsc
\ExplSyntaxOff
\usepackage{xcolor}
\usepackage{listings}
\lstnewenvironment{verbatimsc}
  {
   \lstset{
           basicstyle=\small\ttfamily,
           breaklines=true,
           columns=fullflexible,
           language=[LaTeX]TeX,
           numbers=left,
           numbersep=1em,
           numberstyle=\tiny\color{gray},
           keywordstyle=\color{red}
          }
  }{}
\setlength{\parindent}{0pt}
\pagestyle{empty}
\begin{document}
\section{Test \texttt{\textbackslash begin\{scontents\}} with \texttt{listings}}
Test \verb+{scontents}+ \par

\begin{scontents}
Using \verb+scontents+ env no \verb+[key=val]+, save in seq \verb+contents+ with index 1.\par

Prove \lstinline[basicstyle=\ttfamily]| lstinline | and environment \verb+Verbatim*+
\begin{verbatim}
  verbatim  environment
\end{verbatim}
\end{scontents}

\section{Test \texttt{\textbackslash Scontents*} with \texttt{listings}}

\Scontents*+We have coded this in \lstinline[basicstyle=\ttfamily]|\LaTeX: $E=mc^2$|
and more.+

\section{Test \texttt{\textbackslash getstored}}
\getstored{contents}\par
\getstored[1]{contents}

\section{Test \texttt{\textbackslash typestored}}
\typestored[1]{contents}
\typestored[2]{contents}
\end{document}
```

Example 9

Customization of verbatimsc using the minted[7] package. Source file.

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage{scontents}
\ExplSyntaxOn
\cs_undefine:N \verbatimsc
\cs_undefine:N \endverbatimsc
\ExplSyntaxOff
\usepackage{minted}
\newminted{tex}{linenos}
\newenvironment{verbatimsc}{\VerbatimEnvironment\begin{texcode}}{\end{texcode}}
\pagestyle{empty}
```

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```
\setlength{\parindent}{0pt}
\begin{document}
\section{Test \texttt{\textbackslash begin\{scontents\}} with \texttt{minted}}
Test \verb+{scontents}+ \par

\begin{scontents}[overwrite,write-env=\jobname.tsc,force-eol=true]
Using \verb+scontents+ env no \verb+[key=val]+, save in seq \verb+contents+
with index 1.\par

Prove new \Verb*{ new fvextra with braces } and environment \verb+Verbatim*+
\begin{Verbatim}[obeytabs, showtabs, tab=\rightarrowfill, tabcolor=red]
No tab
        One real tab
                Two real Tab plus       one tab
\end{Verbatim}
\end{scontents}

\section{See \Verb{\jobname.tsc}}
Read \Verb{\jobname.tsc} (shows TABs as red arrows):
\VerbatimInput[obeytabs, showtabs, tab=\rightarrowfill, tabcolor=red]{\jobname.tsc}

\section{Test \texttt{\textbackslash Scontents} with \texttt{minted}}

\Scontents{ We have coded \par this in \LaTeX: $E=mc^2$.}

\section{Test \texttt{\textbackslash getstored}}
\getstored[1]{contents}\par
\getstored{contents}

\section{Test \texttt{\textbackslash typestored}}
\typestored[1]{contents}
\typestored[2]{contents}
\end{document}
```

8.5 The command \mergesc in action

The command \mergesc in action, adapted from Denis Bitouzé request at https://github.com/pablgonz/scontents/issues/2. Source file .

```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\documentclass{article}
\usepackage{scontents}
% Fix part of a MCE that should go before babel's loading
\begin{scontents}[store-env=mce]
\documentclass[french]{article}
\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{lmodern}
\usepackage[a4paper]{geometry}
\end{scontents}
% Fix part of a MCE that should go after (>=) babel's loading
\begin{scontents}[store-env=mce]
\usepackage{babel}
\begin{document}
\end{scontents}
% Fix part of a MCE that should go after its body
\begin{scontents}[store-env=mce]
\end{document}
\end{scontents}
\begin{document}
\section{First answer}
% Variable part of a MCE that should added to the fixed preamble, before babel's loading
\begin{scontents}[store-env=mce-1]
\usepackage{amsmath}
\end{scontents}
% Variable part of a MCE being the code snippet
\begin{scontents}[store-env=mce-1]
\begin{align}
  0 & \neq 1 \\
  1 & \neq 0
\end{align}
```

```
\end{scontents}
\begin{description}
\item[Preamble's addition]\leavevmode
  \typestored[1]{mce-1}
\item[Code snippet]\leavevmode
  \typestored[2]{mce-1}
\item[MCE]\leavevmode
  \mergesc[typestored, print-cmd=true]
    {
      {mce}[1], {mce-1}[1], {mce}[2], {mce-1}[2], {mce}[3]
    }
\end{description}
\section{Second answer}
% Variable part of a MCE that should added to the fixed preamble, before babel's loading
\begin{scontents}[store-env=mce-2]
\usepackage{amsmath}
\end{scontents}
% Variable part of a MCE being the code snippet
\begin{scontents}[store-env=mce-2]
\begin{flalign}
  0 & \neq 1 \\
  1 & \neq 0
\end{flalign}
\end{scontents}

\begin{description}
\item[Preamble's addition]\leavevmode
  \typestored[1]{mce-2}
\item[Code snippet]\leavevmode
  \typestored[2]{mce-2}
\item[MCE]\leavevmode
  \mergesc[typestored, print-cmd=true, write-out=mce.txt, overwrite=true]
    {
      {mce}[1], {mce-2}[1], {mce}[2], {mce-2}[2], {mce}[3]
    }
\end{description}
\end{document}
```

8.6 Stored contents before storing them

The `remember` key in action: the ⟨*stored contents*⟩ of the *sequence* `objectives` are used in the section "Objectives" before they are stored in the section "Methods", as requested at https://github.com/pablgonz/scontents/issues/1. It needs two compilations. Source file.

```
% arara: lualatex: { branch: developer }
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log, sct ] }
\documentclass{article}
\usepackage[remember=objectives]{scontents}
\begin{document}
\section{Objectives}
The project has \countsc{objectives} objectives, developed in the
methods: \foreachsc[sep={; }]{objectives}.

The first one is the key of the project: \getstored[1]{objectives}.

\section{Methods}
\subsection{First objective}
\Scontents[store-cmd=objectives, print-cmd]{measure the effect of the method on small samples}

The samples are taken from the previous studies.

\subsection{Second objective}
\begin{scontents}[store-env=objectives, print-env]
compare the results with the ones of the previous studies
\end{scontents}
\end{document}
```

8.7 The tagged PDF example

This example is just to show the compatibility of SCONTENTS with *tagged* PDF using `lualatex`. The attached files are only for testing. Source file and *tagged* PDF.

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```
% arara: lualatex: { branch: developer }
% arara: clean: { extensions: [ aux, log ] }
\DocumentMetadata{tagging=on, lang=en-US, pdfversion=2.0, pdfstandard=ua-2}
\documentclass{article}
\usepackage{scontents,unicode-math,hyperref}
\hypersetup{pdftitle = {Test scontents package},}
\begin{document}
Some

\begin{scontents}[print-env=true]
    First code \verb|\foo|

    And more code \verb|\bar|
\end{scontents}

Text

\begin{scontents}[print-env=true]
    Second code \verb|\foo|

    And more code \verb|\bar|
\end{scontents}

Text

\Scontents*{code \verb|\baz|}

% \typestored
\typestored[1]{contents}

% \mergesc
\mergesc[typestored]{ {contents}[1-end] }

% \getstored
\getstored[2]{contents}
\end{document}
```

This example has been checked using veraPDF together with ngpdf.

9 References

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[16] The LaTeX Project. "The LaTeX3 Interfaces". Available from CTAN, https://www.ctan.org/pkg/l3kernel, 2026.

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10 Change history

In this section you will find some (not all) of the changes in SCONTENTS development, from the first public implementation using the filecontentsdef[1] package to the current version with only expl3[2].

v2.8 (ctan), 2026-11-01
- Fix warning for empty argument in \Scontents and \Scontents*.
- Add error for empty sequence names and undefined sequences.
- Check positive integers in start, step, stop and width-tab keys.
- The messages for writing and overwriting files only go to the .log file.
- Fix wrapper key and allow \par in the optional argument of \foreachsc.
- Fix negative ⟨*index*⟩ in \typestored, \meaningsc and \mergesc.
- Fix the check of existing ⟨*output files*⟩ in ConTEXt for files written in the same run.
- Fix loading in ConTEXt when expl3 was already loaded by another module.
- The messages of the environments show their start and end as written in plain TEX and ConTEXt.
- The "error" index-out-of-range shows the limit of the sequence and the "errors" index-out-of-range and empty-variable have a help text.
- Fix {⟨*seq name*⟩} without ⟨*index*⟩, undefined sequences and ⟨*index*⟩ out of range in \mergesc.
- Fix TAB at the beginning of a line in \typestored and \mergesc.
- Fix empty line at the beginning of \typestored and \mergesc in plain TEX.
- Unknown ⟨*keys*⟩ in the package options, \setupsc and the ⟨*initial keys*⟩ of \newenvsc return an "error" that names the command.
- Added regression tests for LaTeX, plain TEX and ConTEXt, run with l3build check.
- New remember key to use the ⟨*stored contents*⟩ before they are stored, only in LaTeX (GitHub issue #1).
- New getstored key for \mergesc and getstored, typestored and meaningsc keys for \foreachsc (GitHub issue #10).
- New programming interface for expl3: \scontents_if_exist:nTF, \scontents_count:n, \scontents_get:nnN and \scontents_map_inline:nn (GitHub issue #10).

v2.7 (ctan), 2026-06-01
- Internal environment verbatimsc update for *tagged* PDF.
- Update requirement to LaTeX release 2026-06-01.
- Add debug-var key for writing variables to the .log file.
- Update and improvements documentation and examples.

v2.6 (ctan), 2025-11-20
- Internal environment verbatimsc update for *tagged* PDF.
- Update requirement to LaTeX release 2025-11-01.

v2.5 (ctan), 2025-11-01
- Replacing \scantokens (primitive) with \tl_retokenize:n.
- Cleanup warnings and details returned by expltools.
- Update minimum required of expl3 to 2025-07-08.

v2.4 (ctan), 2025-05-15
- Optimization of expansion code from 'x' to 'e'.
- Restructuring code for documentation and implementation.
- Add new keys for \typestored and \meaningsc.
- Check the version of expl3 in plain TEX and ConTEXt.

v2.3 (ctan), 2025-04-23
- Adapting the verbatimsc environment for *tagged* PDF.
- Update minimum required to LaTeX release 2024-11-01.
- Safer code for replacement \obeyedline.

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v2.2 (ctan), 2025-03-26
- Fix internal definition for some functions.
- Replace `\peek_charcode_ignore_spaces:NTF` by `\peek_charcode:NTF`.
- Set correct code for `\obeyedline` implement in LaTeX release 2024-06-01.

v2.1 (ctan), 2024-06-14
- Fix `\cleanseqsc` command.
- Add `\mergesc` command.
- Fix internal definition for seq var.
- Fix internal code for `\typestored`.
- Replace `\cs_argument_spec:N` by `\cs_parameter_spec:N`.
- Detect `l3keys2e` package (obsolete in june 2022 LaTeX release).
- Minor adjustments in the documentation.

v2.0 (ctan), 2022-04-04
- Adapting the `verbatimsc` environment (compatibility `verbatim` package).
- Removed compatibility layer for older LaTeX releases.
- Fix loader in plain TeX and ConTeXt.
- Minor adjustments in the documentation.

v1.9 (ctan), 2020-01-21
- Update and improvements in the internal code.
- Updating the generic code for `I/O` verification.
- Add `write-cmd` and `write-out` keys for `\Scontents*`.
- Fix `sep` key in `\foreachsc`.

v1.8 (ctan), 2019-11-18
- Add `\newenvsc` command.
- Fix nested environment in plain TeX and ConTeXt.
- Modified default value in `\getstored`.
- Add `overwrite` key to reduce `I/O` operations.
- Deleted an unnecessary group in the code.

v1.7 (ctan), 2019-10-29
- The `verbatimsc` environment was rewritten.
- Minor adjustments in documentation.

v1.6 (ctan), 2019-10-26
- The internal behavior of `\getstored` has been modified.
- The internal behavior of `\foreachsc` has been modified.
- Corrected file extension for ConTeXt.
- Remove spurious warning.

v1.5 (ctan), 2019-10-24
- Add support for plain TeX and ConTeXt.
- Split internal code for optimization.
- Add support for vertical spaces in `[`⟨*key* = *val*⟩`]`.
- Add `\foreachsc` command.
- Check if `verbatim` package is loaded.

v1.4 (ctan), 2019-10-03
- Add `store-all` key.
- Messages and keys were separated.
- Restructuring of documentation.
- Now the version of `expl3` is checked instead of `xparse`.
- The internal behavior of `force-eol` has been modified.

v1.3 (ctan), 2019-09-24
- The environment `scontents` can now nest.
- Added `force-eol`, `verb-font` and `width-tab` keys.
- The extra space has been removed when you run `\getstored`.
- Internal code has been rewritten more efficiently.
- Remove *starred argument* '`*`' for `\typestored`.
- Remove `filecontentsdef` dependency.
- Changing `\regex_replace_all:` for `\tl_replace_all:`.

v1.2 (ctan), 2019-08-28
- Restructuring of documentation.
- Added copy of `\tex_scantokens:`.

v1.1 (ctan), 2019-08-12
- Extension of documentation.
- Replace `\tex_endinput:D` by `\file_input_stop:`.

v1.0 (ctan), 2019-07-30
- First public release.

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11 Index of Documentation

The italic numbers denote the pages where the corresponding entry is described.

C

Commands provided by SCONTENTS:

\Scontents 5
\cleanseqsc 8
\countsc 8
\foreachsc 6
\getstored 6
\meaningsc 7
\mergesc 7
\newenvsc 5
\setupsc 3
\typestored 7

E

Environments :

filecontentsdefmacro 1
lstlisting 3
Verbatim 3

Environments provided by SCONTENTS:

scontents 4

K

Keys for \foreachsc provided by SCONTENTS:

after 7
before 7
getstored 6
meaningsc 7
sep 7
start 7
step 7
stop 7
typestored 6
wrapper 7

Keys for \Scontents provided by SCONTENTS:

print-cmd 6
store-cmd 6
write-cmd 6
write-out 6

Keys for \setupsc provided by SCONTENTS:

debug-var 3
force-eol 3
overwrite 3
print-all 3
remember 3
store-all 3
verb-font 3
width-tab 3

Keys for environments provided by SCONTENTS:

print-env 5
store-env 5
write-env 5
write-out 5

P

Packages :

answers 10
expl3 . . . . . 1, 2, 9, 18, 19
fancyvrb 3, 8
filecontentsdef 1, 9, 10, 18, 19
fvextra 3, 5, 8, 13
l3keys2e 19
l3seq 1, 9
listings 3, 5, 8, 14
minted 8, 14
tcolorbox 13
verbatim 19
xparse 19

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12 Implementation

The most recent publicly released version of SCONTENTS is available at CTAN: https://www.ctan.org/pkg/scontents. Historical and developmental versions are available at https://github.com/pablgonz/scontents. While general feedback via email is welcomed, specific bugs or feature requests should be reported through the issue tracker: https://github.com/pablgonz/scontents/issues.

All variables and functions defined in this package are private and are NOT intended to work or be used by another package or module.

12.1 General conventions

The package is split into a *loader* and a *core*. The loader is different for each format and generates scontents.sty for LaTeX, scontents.tex for plain TeX and t-scontents.mkiv for ConTeXt, the core is shared by the three formats and generates scontents-code.tex, which is loaded by each loader.

Since the core runs in plain TeX and ConTeXt, where the names of some primitives may be redefined by the format, the core uses the \tex_...:D names for the primitives that have no expl3 interface, such as \tex_kern:D, \tex_lastskip:D or \tex_spacefactor:D.

Each command and environment is implemented in the same order: its variables, its messages, its ⟨*keys*⟩, its internal functions and finally the public command or environment.

12.2 Declaration of the package

First we set up the module name for DocStrip l3doc class:

```
⟨@@=scontents⟩
```

Now we define some common macros to hold the package date and version:

```
⟨*loader⟩
\def\ScontentsFileDate{2026-11-01}%
\def\ScontentsFileVersion{2.8}%
\def\ScontentsFileDescription{Stores LaTeX contents in memory or files}%
```

The LaTeX loader is quite simple, we just need to make sure of the minimum version for correct operation and then set interfaces up. The choice of LaTeX release 2026-06-01 is the latest available in TeX Live 2026 and is necessary to be able to implement the package's full compatibility with *tagged* PDF.

```
⟨*latex⟩
\NeedsTeXFormat{LaTeX2e}[2026-06-01]
\ProvidesExplPackage
  {scontents} {\ScontentsFileDate} {\ScontentsFileVersion}
  {\ScontentsFileDescription}
⟨/latex⟩
```

12.2.1 Load expl3-generic and xparse-generic in plain TeX and ConTeXt

\c__scontents_version_str
\l__scontents_char_value_int

The plain TeX and ConTeXt loaders are similar. We define the constant string \c__scontents_version_str with the version info and add \ExplSyntaxOn to be able to load the frozen xparse[10]. In ConTeXt the module is loaded after \unprotect, which makes '!' a letter, but expl3 and xparse-generic.tex use '!' in boolean expressions and expl3-generic.tex does not set its category code, so '!' is set to *other* before loading it and is a letter again at the end (§12.27).

```
⟨*!latex⟩
⟨context⟩\writestatus{loading}{User Module scontents v\ScontentsFileVersion}
⟨context⟩\unprotect
⟨context⟩\catcode`\!=12\relax
\input expl3-generic.tex
\ExplSyntaxOn
\str_const:ce { c__scontents_version_str } { \ScontentsFileDate\space
  v\ScontentsFileVersion\space \ScontentsFileDescription }
\iow_log:e { Package: ~ scontents ~ \str_use:c { c__scontents_version_str } }
```

Outside of LaTeX we can't use ltcmd[11, 12], part of the LaTeX kernel, but the old xparse[10] provides xparse-generic.tex, which is loadable in any format. The integer variable \l__scontents_char_value_int saves the category code of '@' while it is loaded.

```
\int_new:N  \l__scontents_char_value_int
\int_set:Nn \l__scontents_char_value_int { \char_value_catcode:n { `\@ } }
\char_set_catcode_letter:N \@
\file_input:n { xparse-generic.tex }
\char_set_catcode:nn { `\@ } { \l__scontents_char_value_int }
⟨/!latex⟩
```

(*End of definition for* \c__scontents_version_str *and* \l__scontents_char_value_int.)

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12.2.2 Checking expl3 version

For plain TEX, LATEX and ConTEXt we must check the minimum requirement, in this case \tl_retokenize:n which was added in release 2025-07-08 of expl3 included in TEX Live 2025.

```
\cs_if_exist:NF \tl_retokenize:n
  {
    \msg_new:nnn { scontents } { expl-too-old }
      {
        Please~install~an~up~to~date~TeX~distribution~or~update~using~
        your~TeX~package~manager~or~from~CTAN. \\
        See~documentation.~Loading~scontents~will~abort!
      }
    \msg_fatal:nn { scontents } { expl-too-old }
    \ExplSyntaxOff
    \file_input_stop:
  }
```

12.2.3 Preventing double loading in plain TEX

In plain TEX, check that the package isn't being loaded twice, LATEX and ConTEXt already defend against that:

```
⟨*plain⟩
\tl_if_exist:NT \l__scontents_cmd_name_tl
  {
    \msg_new:nnn { scontents } { already-loaded }
      {
        The~'scontents'~package~is~already~loaded.~
        Aborting~input~\msg_line_context:.
      }
    \msg_warning:nn { scontents } { already-loaded }
    \ExplSyntaxOff
    \file_input_stop:
  }
⟨/plain⟩
⟨/loader⟩
```

12.2.4 Checking proper loader

Checking that the package was loaded with the proper loader code. This code was copied from expl3-code.tex.

```
⟨*core⟩
\def\ScontentsCoreFileDate{2026-11-01}%
\begingroup
  \catcode32=10
  \endlinechar=32
  \def\next{\endgroup}%
  \expandafter\ifx\csname PackageError\endcsname\relax
    \begingroup
      \def\next{\endgroup\endgroup}%
      \def\PackageError#1#2#3%
        {%
          \endgroup
          \errhelp{#3}%
          \errmessage{#1 Error: #2!}%
        }%
  \fi
  \expandafter\ifx\csname ScontentsFileDate\endcsname\relax
    \def\next
      {%
        \PackageError{scontents}{No scontents loader detected}
          {%
            You have attempted to use the scontents code directly rather
            than using the correct loader. Loading of scontents will abort.
          }%
        \endgroup
        \endinput
      }%
  \else
    \ifx\ScontentsFileDate\ScontentsCoreFileDate
    \else
      \def\next
```

```
        {%
          \PackageError{scontents}{Mismatched scontents files detected}
            {%
              You have attempted to load scontents with mismatched files:
              probably you have one or more files 'locally installed' which
              are in conflict. Loading of scontents will abort.
            }%
          \endgroup
          \endinput
        }%
    \fi
  \fi
\next
⟨/core⟩
```

12.3 Variables and functions by format

We define and set the variables and the function that must be handled separately in order to work properly with plain TEX, LATEX and ConTEXt.

__scontents_format_case:nnn

The function __scontents_format_case:nnn leaves the argument for the current format: #1 for LATEX, #2 for plain TEX and #3 for ConTEXt.

```
⟨*loader⟩
\cs_new:Npn \__scontents_format_case:nnn #1 #2 #3
⟨latex⟩   {#1} % LaTeX
⟨plain⟩   {#2} % Plain/Generic
⟨context⟩   {#3} % ConTeXt
```

(*End of definition for* __scontents_format_case:nnn.)

__scontents_file_if_exist:nTF

The conditional __scontents_file_if_exist:nTF tests if the ⟨*output file*⟩ passed in #1 exists in the working directory. In LATEX and plain TEX it is \file_if_exist:nTF. In ConTEXt the file search keeps the result found at the start of the run, so a file written by the package in the same run is not found; there the test is done with `lfs.isfile` of LuaTEX, the ⟨*output file*⟩ is always in the working directory.

```
⟨!context⟩\cs_new_eq:NN \__scontents_file_if_exist:nTF \file_if_exist:nTF
⟨*context⟩
\prg_new_conditional:Npnn \__scontents_file_if_exist:n #1 { TF }
  {
    \str_if_eq:eeTF
      {
        \lua_now:e
          { tex.write(lfs.isfile("\lua_escape:e {#1}")~and~"1"~or~"0") }
      }
      { 1 }
      { \prg_return_true: }
      { \prg_return_false: }
  }
⟨/context⟩
```

(*End of definition for* __scontents_file_if_exist:nTF.)

\c__scontents_newline_tl

The constant token list \c__scontents_newline_tl stores the character '^^J' used as end of line in the ⟨*stored content*⟩.

```
⟨*!context⟩
\tl_const:Ne \c__scontents_newline_tl { \iow_char:N \^^J }
⟨/!context⟩
```

In ConTEXt the package does not work under LMTX, so loading is aborted there. The test with \contextlmtxmode is adapted from the file t-lua-widow-control.mkxl of Max Chernoff's lua-widow-control package, \contextlmtxmode is described in https://source.contextgarden.net/tex/context/base/mkxl/context.mkxl.

```
⟨*context⟩
\bool_if:NTF \contextlmtxmode
  {
    \msg_new:nnn { scontents } { luametatex }
      {
        The~'scontents'~package~doesn't~work~under~LMTX.
      }
    \msg_fatal:nn { scontents } { luametatex }
```

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```
  }
  {
    \tl_const:Ne \c__scontents_newline_tl { \iow_char:N \^^J }
  }
⟨/context⟩
```

(*End of definition for* `\c__scontents_newline_tl`.)

`\g__scontents_end_verbatimsc_tl` `\c__scontents_end_env_tl` `\l__scontents_env_name_tl`

The global token list `\g__scontents_end_verbatimsc_tl` matches the end of the `verbatimsc` environment (§12.15) in each format.

```
\tl_new:N \g__scontents_end_verbatimsc_tl
\tl_gset_rescan:Nnn \g__scontents_end_verbatimsc_tl
  {
    \char_set_catcode_other:N \\
⟨*latex⟩
    \char_set_catcode_other:N \{
    \char_set_catcode_other:N \}
⟨/latex⟩
  }
⟨latex⟩   { \end{verbatimsc} }
⟨plain⟩   { \endverbatimsc }
⟨context⟩   { \stopverbatimsc }
```

The constant token list `\c__scontents_end_env_tl` matches the end of the environments defined by `\newenvsc` and the token list `\l__scontents_env_name_tl` stores the *name* of the environment being processed (§12.13).

```
\tl_new:N    \l__scontents_env_name_tl
\tl_const:Ne \c__scontents_end_env_tl
  {
    \c_backslash_str
⟨latex|plain⟩    end
⟨context⟩    stop
⟨latex⟩    \c_left_brace_str
      \exp_not:N \l__scontents_env_name_tl
⟨latex⟩    \c_right_brace_str
  }
```

(*End of definition for* `\g__scontents_end_verbatimsc_tl`, `\c__scontents_end_env_tl`, *and* `\l__scontents_env_name_tl`.)

12.4 Loading the package core

Now we load the core SCONTENTS code:

```
\file_input:n { scontents-code.tex }
⟨/loader⟩
```

12.5 Some kernel variants

`\tl_replace_all:NeV` `\tl_retokenize:e`

Some nonstandard kernel variants.

```
⟨*core⟩
\cs_generate_variant:Nn \tl_replace_all:Nnn { NeV }
\cs_generate_variant:Nn \tl_retokenize:n { e }
```

(*End of definition for* `\tl_replace_all:NeV` *and* `\tl_retokenize:e`.)

12.6 General variables

`\q__scontents_stop` `\q__scontents_mark` `\s__scontents_stop` `\s__scontents_mark`

Quarks and scan marks used along the code as delimiters.

```
\quark_new:N \q__scontents_stop
\quark_new:N \q__scontents_mark
\scan_new:N  \s__scontents_stop
\scan_new:N  \s__scontents_mark
```

(*End of definition for* `\q__scontents_stop` *and others.*)

`\l__scontents_cmd_name_tl`

The token list `\l__scontents_cmd_name_tl` stores the *name* of the commands `\Scontents`, `\foreachsc`, `\typestored`, `\meaningsc` and `\mergesc`, it is used in the messages for unknown ⟨*keys*⟩ (§12.9).

```
\tl_new:N \l__scontents_cmd_name_tl
```

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(*End of definition for* `\l__scontents_cmd_name_tl`.)

`\l__scontents_file_name_tl` `\l__scontents_file_write_iow` `\l__scontents_writing_bool` `\l__scontents_storing_bool` `\l__scontents_writable_bool`

The token list `\l__scontents_file_name_tl` stores the name of the ⟨*output file*⟩ set by the `write-env`, `write-out` and `write-cmd` keys and `\l__scontents_file_write_iow` is the *output stream* used to write it.

The boolean variables `\l__scontents_writing_bool` and `\l__scontents_storing_bool`, set by the same keys, determine whether the content is written to the ⟨*output file*⟩ and whether it is stored, `\l__scontents_storing_bool` is true by default. The boolean variable `\l__scontents_writable_bool` keeps track of whether the ⟨*output file*⟩ was opened by `\__scontents_file_if_writable:nTF` (§12.10).

```
\tl_new:N   \l__scontents_file_name_tl
\iow_new:N  \l__scontents_file_write_iow
\bool_new:N \l__scontents_writing_bool
\bool_new:N \l__scontents_storing_bool
\bool_new:N \l__scontents_writable_bool
\bool_set_true:N \l__scontents_storing_bool
```

(*End of definition for* `\l__scontents_file_name_tl` *and others.*)

`\l__scontents_seq_item_seq` `\g__scontents_name_sc!internal_seq`

The sequence `\l__scontents_seq_item_seq` stores the ⟨*indexes*⟩ of the ⟨*stored content*⟩ requested to `\typestored`, `\meaningsc` or `\mergesc` and the sequence `\g__scontents_name_sc!internal_seq` is the one assembled by `\mergesc` (§12.22).

```
\seq_new:N \l__scontents_seq_item_seq
\seq_new:c { g__scontents_name_sc!internal_seq }
```

(*End of definition for* `\l__scontents_seq_item_seq` *and* `\g__scontents_name_sc!internal_seq`.)

`\g__scontents_last_stored_tl`

The token list `\g__scontents_last_stored_tl` is used by the function `\__scontents_lastfrom_seq:n` (§12.11) to execute the last ⟨*stored content*⟩ outside the group.

```
\tl_new:N \g__scontents_last_stored_tl
```

(*End of definition for* `\g__scontents_last_stored_tl`.)

`\c__scontents_hidden_space_str`

The constant string `\c__scontents_hidden_space_str` hides the ⟨*forced space*⟩ added by TEX when recording content in a macro. It contains the *reserved phrase* '`%^^Ascheol%`', which is added to the end of the ⟨*stored content*⟩ when the key `force-eol` is false.

```
\str_const:Ne \c__scontents_hidden_space_str
  { \c_percent_str \c_circumflex_str \c_circumflex_str A scheol \c_percent_str }
```

(*End of definition for* `\c__scontents_hidden_space_str`.)

`\l__scontents_save_sf_int` `\l__scontents_save_skip`

Variables used by the functions `\__scontents_bsphack:` and `\__scontents_esphack:` (§12.10.2).

```
\int_new:N  \l__scontents_save_sf_int
\skip_new:N \l__scontents_save_skip
```

(*End of definition for* `\l__scontents_save_sf_int` *and* `\l__scontents_save_skip`.)

12.7 General messages

12.7.1 Definition of error messages

`undefined-storage` `index-out-of-range`

The "error" `undefined-storage` is raised when the *sequence* passed to `\getstored`, `\foreachsc`, `\typestored`, `\meaningsc`, `\mergesc` or `\countsc` does not exist (§12.11, §12.18, §12.22, §12.24) and the "error" `index-out-of-range` is raised when the ⟨*index*⟩ is zero or its absolute value is greater than the number of ⟨*stored contents*⟩ (§12.11, §12.22).

```
\msg_new:nnnn { scontents } { undefined-storage }
  { Storage~named~'#1'~is~not~defined. }
  { No~content~has~been~stored~in~a~sequence~named~'#1'. }
\msg_new:nnnn { scontents } { index-out-of-range }
  {
    \int_compare:nNnTF {#1} = { 0 }
      { Index~of~sequence~'#2'~cannot~be~zero. }
      {
        Index~'#1'~out~of~range~for~'#2',~the~
        \int_compare:nNnTF {#1} > { 0 }
          { maximum~is~#3. }
          { minimum~is~-#3. }
```

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```
      }
  }
  {
    The~sequence~'#2'~has~#3~stored~contents,~use~an~index~from~
    1~to~#3~or~from~-#3~to~-1.
  }
```

(*End of definition for* `undefined-storage` *and* `index-out-of-range`.)

positive-integer
empty-store-name

The "error" `positive-integer` is raised when the keys `width-tab`, `start`, `step` or `stop` receive a value that is not a ⟨*positive integer*⟩ (§12.8, §12.18.2) and the "error" `empty-store-name` is raised when the name of the *sequence* is blank (§12.11).

```
\msg_new:nnnn { scontents } { positive-integer }
  { The~key~'#1'~needs~a~positive~integer,~'#2'~given~\msg_line_context:. }
  { The~value~'#2'~is~ignored~and~the~key~'#1'~keeps~its~previous~value. }
\msg_new:nnnn { scontents } { empty-store-name }
  { Empty~sequence~name~for~storing~content~\msg_line_context:. }
  {
    The~keys~'store-env',~'store-cmd'~and~'store-all'~need~a~non-empty~
    sequence~name.~The~content~will~not~be~stored.
  }
```

(*End of definition for* `positive-integer` *and* `empty-store-name`.)

12.7.2 Definition of messages for writing files

writing-file
overwrite-file
not-writing

The "info" messages `writing-file` and `overwrite-file` are written to the `.log` file when an ⟨*output file*⟩ is written or overwritten and the "warning" `not-writing` is issued when the ⟨*output file*⟩ already exists and the key `overwrite` is not active (§12.10).

```
\msg_new:nnn { scontents } { writing-file }
  { Writing~file~'#1'. }
\msg_new:nnn { scontents } { overwrite-file }
  { Overwriting~file~'#1'. }
\msg_new:nnn { scontents } { not-writing }
  { File~'#1'~already~exists.~Not~writing~\msg_line_context:. }
⟨/core⟩
```

(*End of definition for* `writing-file`, `overwrite-file`, *and* `not-writing`.)

12.8 Keys for the package

\l__scontents_tab_width_int
__scontents_set_positive_int:Nnn

The integer variable `\l__scontents_tab_width_int` stores the value of the `width-tab` key. The function `\__scontents_set_positive_int:Nnn` sets the integer variable passed in `#1` to the value passed in `#3` only if it is a ⟨*positive integer*⟩, otherwise it raises the "error" `positive-integer` naming the ⟨*key*⟩ passed in `#2` and the value is ignored. It is used by the `width-tab` key and by the `start`, `step` and `stop` keys of `\foreachsc` (§12.18.2). The core code does not use '`!`' in boolean expressions, so it does not depend on the category code of '`!`' in ConTEXt (§12.2.1).

```
⟨*core⟩
\int_new:N \l__scontents_tab_width_int
\cs_new_protected:Npn \__scontents_set_positive_int:Nnn #1 #2 #3
  {
    \bool_lazy_or:nnTF
      { \tl_if_blank_p:n {#3} }
      { \int_compare_p:nNn {#3} < { 1 } }
      { \msg_error:nnnn { scontents } { positive-integer } {#2} {#3} }
      { \int_set:Nn #1 {#3} }
  }
```

(*End of definition for* `\l__scontents_tab_width_int` *and* `\__scontents_set_positive_int:Nnn`.)

store-env
store-cmd
verb-font
print-env
print-cmd
force-eol
overwrite
width-tab
print-all
store-all
debug-var

We define the common ⟨*keys*⟩ used by the options passed to the package, by `\setupsc` and by the environments and commands. The unknown ⟨*keys*⟩ are passed to `\__scontents_unknown_keys_cmd:n` (§12.9), which names the command from `\l__scontents_cmd_name_tl`: `\usepackage` for the options of the package, `\setupsc` or `\newenvsc` for the ⟨*initial keys*⟩ of an environment.

```
\keys_define:nn { scontents }
  {
    store-env .tl_set:N         = \l__scontents_name_seq_env_tl,
    store-env .initial:n        = contents,
    store-env .value_required:n = true,
```

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```
    store-cmd .tl_set:N         = \l__scontents_name_seq_cmd_tl,
    store-cmd .initial:n        = contents,
    store-cmd .value_required:n = true,
    verb-font .tl_set:N         = \l__scontents_verb_font_tl,
    verb-font .value_required:n = true,
    print-env .bool_set:N       = \l__scontents_print_env_bool,
    print-env .initial:n        = false,
    print-env .default:n        = true,
    print-cmd .bool_set:N       = \l__scontents_print_cmd_bool,
    print-cmd .initial:n        = false,
    print-cmd .default:n        = true,
    force-eol .bool_set:N       = \l__scontents_forced_eol_bool,
    force-eol .initial:n        = false,
    force-eol .default:n        = true,
    overwrite .bool_set:N       = \l__scontents_overwrite_bool,
    overwrite .initial:n        = false,
    overwrite .default:n        = true,
    width-tab .code:n           =
      { \__scontents_set_positive_int:Nnn \l__scontents_tab_width_int { width-tab } {#1} },
    width-tab .initial:n        = 1,
    width-tab .value_required:n = true,
    print-all .meta:n           = { print-env = #1, print-cmd = #1 },
    print-all .default:n        = true,
    store-all .meta:n           = { store-env = #1, store-cmd = #1 },
    store-all .value_required:n = true,
    debug-var .bool_set:N       = \l__scontents_debug_var_bool,
    debug-var .initial:n        = true,
    debug-var .default:n        = true,
    remember  .code:n           = { \__scontents_remember:n {#1} },
    remember  .value_required:n = true,
    remember  .usage:n          = preamble,
    unknown   .code:n           = { \__scontents_unknown_keys_cmd:n {#1} },
  }
⟨/core⟩
```

The default value of the `verb-font` key depends on the format.

```
⟨loader⟩\keys_define:nn { scontents }
⟨latex⟩  { verb-font .initial:n = \ttfamily }
⟨plain|context⟩  { verb-font .initial:n = \tt }
```

In LaTeX the ⟨*keys*⟩ passed as options to the package are processed and an unknown ⟨*key*⟩ returns an "error".

```
⟨*latex⟩
\tl_set:Nn \l__scontents_cmd_name_tl { usepackage }
\ProcessKeyOptions [ scontents ]
⟨/latex⟩
```

(*End of definition for* `store-env` *and others.*)

12.9 Handling unknown keys

The ⟨*keys*⟩ that are not known by the environments and the commands are passed to the functions defined here. The name of the ⟨*key*⟩ is taken from `\l_keys_key_str` and the value, if any, is passed as argument.

12.9.1 Definition of error messages for unknown keys

env-key-unknown
env-key-value-unknown

The "errors" `env-key-unknown` and `env-key-value-unknown` are raised when an unknown ⟨*key*⟩, without or with value, is passed to an environment defined by `\newenvsc` (§12.13.4).

```
⟨*core⟩
\msg_new:nnnn { scontents } { env-key-unknown }
  {
    The~key~'#1'~is~unknown~by~environment~
    '\l__scontents_env_name_tl'~and~is~being~ignored.
  }
  {
    The~environment~'\l__scontents_env_name_tl'~does~not~have~a~key~called~'#1'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
\msg_new:nnnn { scontents } { env-key-value-unknown }
  {
    The~key~'#1=#2'~is~unknown~by~environment~
```

```
    '\l__scontents_env_name_tl'~and~is~being~ignored.
  }
  {
    The~environment~'\l__scontents_env_name_tl'~does~not~have~a~key~called~'#1'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
```

(*End of definition for* `env-key-unknown` *and* `env-key-value-unknown`.)

`cmd-key-unknown`
`cmd-key-value-unknown`

The "errors" `cmd-key-unknown` and `cmd-key-value-unknown` are raised when an unknown ⟨*key*⟩, without or with value, is passed to the commands `\Scontents`, `\foreachsc`, `\typestored`, `\meaningsc`, `\mergesc` or `\setupsc`, to the ⟨*initial keys*⟩ of `\newenvsc` or as an option of the package (§12.8, §12.16.3, §12.18.2, §12.19.2).

```
\msg_new:nnnn { scontents } { cmd-key-unknown }
  {
    The~key~'#1'~is~unknown~by~command~
    \c_backslash_str \l__scontents_cmd_name_tl \c_space_tl
    and~is~being~ignored~\msg_line_context:.
  }
  {
    The~command~\c_backslash_str \l__scontents_cmd_name_tl \c_space_tl
    does~not~have~a~key~called~'#1'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
\msg_new:nnnn { scontents } { cmd-key-value-unknown }
  {
    The~key~'#1=#2'~is~unknown~by~command~
    \c_backslash_str \l__scontents_cmd_name_tl \c_space_tl
    and~is~being~ignored~\msg_line_context:.
  }
  {
    The~command~\c_backslash_str \l__scontents_cmd_name_tl \c_space_tl
    does~not~have~a~key~called~'#1'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
```

(*End of definition for* `cmd-key-unknown` *and* `cmd-key-value-unknown`.)

12.9.2 Internal functions for unknown keys

`\__scontents_unknown_keys_env:n`
`\__scontents_unknown_keys_env:nn`

The function `\__scontents_unknown_keys_env:n` raises the "error" for an unknown ⟨*key*⟩ passed to an environment defined by `\newenvsc`.

```
\cs_new_protected:Npn \__scontents_unknown_keys_env:n #1
  { \exp_args:NV \__scontents_unknown_keys_env:nn \l_keys_key_str {#1} }
\cs_new_protected:Npn \__scontents_unknown_keys_env:nn #1 #2
  {
    \tl_if_blank:nTF {#2}
      { \msg_error:nnn { scontents } { env-key-unknown } {#1} }
      { \msg_error:nnnn { scontents } { env-key-value-unknown } {#1} {#2} }
  }
```

(*End of definition for* `\__scontents_unknown_keys_env:n` *and* `\__scontents_unknown_keys_env:nn`.)

`\__scontents_unknown_keys_cmd:n`
`\__scontents_unknown_keys_cmd:nn`

The function `\__scontents_unknown_keys_cmd:n` raises the "error" for an unknown ⟨*key*⟩ passed to the commands `\Scontents`, `\foreachsc` and `\setupsc`, to the ⟨*initial keys*⟩ of `\newenvsc` and to the package.

```
\cs_new_protected:Npn \__scontents_unknown_keys_cmd:n #1
  { \exp_args:NV \__scontents_unknown_keys_cmd:nn \l_keys_key_str {#1} }
\cs_new_protected:Npn \__scontents_unknown_keys_cmd:nn #1 #2
  {
    \tl_if_blank:nTF {#2}
      { \msg_error:nnn { scontents } { cmd-key-unknown } {#1} }
      { \msg_error:nnnn { scontents } { cmd-key-value-unknown } {#1} {#2} }
  }
```

(*End of definition for* `\__scontents_unknown_keys_cmd:n` *and* `\__scontents_unknown_keys_cmd:nn`.)

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12.10 Utility functions

__scontents_use_delimit_by_s_stop:nw
__scontents_use_i_delimit_by_s_stop:nw
__scontents_use_none_delimit_by_s_stop:w
__scontents_use_none_delimit_by_q_stop:w

Functions used in the implementation of \newenvsc (§12.13.5) and \mergesc (§12.22).

```
\cs_new:Npn \__scontents_use_delimit_by_s_stop:nw #1 \s__scontents_stop {#1}
\cs_new:Npn \__scontents_use_i_delimit_by_s_stop:nw #1 #2 \s__scontents_stop {#1}
\cs_new:Npn \__scontents_use_none_delimit_by_s_stop:w #1 \s__scontents_stop { }
\cs_new:Npn \__scontents_use_none_delimit_by_q_stop:w #1 \q__scontents_stop { }
```

(*End of definition for* __scontents_use_delimit_by_s_stop:nw *and others.*)

__scontents_tl_if_head_is_q_mark:nTF

The conditional __scontents_tl_if_head_is_q_mark:nTF tests if the head of the token list is \q__scontents_mark.

```
\prg_new_protected_conditional:Npnn \__scontents_tl_if_head_is_q_mark:n #1
  { F, TF }
  {
    \exp_after:w \if_meaning:w
      \exp_after:w
        \q__scontents_mark \__scontents_use_i_delimit_by_s_stop:nw #1 ? \s__scontents_stop
      \prg_return_true:
    \else:
      \prg_return_false:
    \fi:
  }
```

(*End of definition for* __scontents_tl_if_head_is_q_mark:nTF.)

__scontents_remove_trailing_eol:N
__scontents_remove_trailing_eol:w

The function __scontents_remove_trailing_eol:N removes the string \c__scontents_hidden_space_str from the end of the token list passed in #1. It is used by \typestored (§12.20) and \mergesc (§12.22).

```
\cs_new_protected:Npe \__scontents_remove_trailing_eol:N #1
  {
    \exp_not:N \exp_after:w \exp_not:N \__scontents_remove_trailing_eol:w
      #1 \s__scontents_stop \c__scontents_hidden_space_str \s__scontents_stop \s__scontents_mark
  }
\use:e
  {
    \cs_new_protected:Npn \exp_not:N \__scontents_remove_trailing_eol:w #1
        \c__scontents_hidden_space_str \s__scontents_stop #2 \s__scontents_mark #3
  }   {
        \tl_set:Ne #3
          {
            \tl_if_empty:nTF {#2}
              { \exp_not:o { \__scontents_use_delimit_by_s_stop:nw #1 } }
              { \exp_not:n {#1} }
          }
      }
```

(*End of definition for* __scontents_remove_trailing_eol:N *and* __scontents_remove_trailing_eol:w.)

__scontents_file_if_writable:nTF

The conditional __scontents_file_if_writable:nTF tests if the ⟨*output file*⟩ passed in #1 can be written, according to the write-env, write-cmd, write-out and overwrite keys, and issues the messages for writing files (§12.7). The ⟨*output file*⟩ is searched with __scontents_file_if_exist:nTF.

```
\prg_new_protected_conditional:Npnn \__scontents_file_if_writable:n #1 { T, TF }
  {
    \bool_if:NTF \l__scontents_writing_bool
      {
        \__scontents_file_if_exist:nTF {#1}
          {
            \bool_if:NTF \l__scontents_overwrite_bool
              {
                \msg_info:nne { scontents } { overwrite-file } {#1}
                \prg_return_true:
              }
              {
                \msg_warning:nne { scontents } { not-writing } {#1}
                \prg_return_false:
              }
          }
          {
            \msg_info:nne { scontents } { writing-file } {#1}
```

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```
            \prg_return_true:
          }
      }
      { \prg_return_false: }
  }
```

(*End of definition for* `\__scontents_file_if_writable:nTF`.)

`\__scontents_file_write_cmd:nn` `\__scontents_file_write_cmd:VV`

The function `\__scontents_file_write_cmd:nn` writes the content passed in `#2` to the ⟨*output file*⟩ passed in `#1`, it is used by the commands.

```
\cs_new_protected:Npn \__scontents_file_write_cmd:nn #1 #2
  {
    \__scontents_file_if_writable:nT {#1}
      {
        \iow_open:Nn \l__scontents_file_write_iow {#1}
        \iow_now:Nn  \l__scontents_file_write_iow {#2}
        \iow_close:N \l__scontents_file_write_iow
      }
  }
\cs_generate_variant:Nn \__scontents_file_write_cmd:nn { VV }
```

(*End of definition for* `\__scontents_file_write_cmd:nn`.)

`\__scontents_set_write_out:n` `\__scontents_set_write_store:n`

The function `\__scontents_set_write_out:n` sets the name of the ⟨*output file*⟩ passed in `#1` and only writes the content, it is used by the `write-out` key. The function `\__scontents_set_write_store:n` also stores the content, it is used by the `write-env` and `write-cmd` keys (§12.13.4, §12.16.3, §12.19.2).

```
\cs_new_protected:Npn \__scontents_set_write_out:n #1
  {
    \bool_set_false:N \l__scontents_storing_bool
    \bool_set_true:N \l__scontents_writing_bool
    \tl_set:Nn \l__scontents_file_name_tl {#1}
  }
\cs_new_protected:Npn \__scontents_set_write_store:n #1
  {
    \__scontents_set_write_out:n {#1}
    \bool_set_true:N \l__scontents_storing_bool
  }
```

(*End of definition for* `\__scontents_set_write_out:n` *and* `\__scontents_set_write_store:n`.)

12.10.1 Functions for TAB and verbatimsc

`\__scontents_tab:` `\__scontents_par:`

Control sequences to replace the tab '`^^I`' and form feed '`^^L`' characters.

```
\cs_new:Ne \__scontents_tab: { \c_space_tl }
\cs_new:Nn \__scontents_par: { ^^J ^^J }
```

(*End of definition for* `\__scontents_tab:` *and* `\__scontents_par:`.)

`\__scontents_tabs_to_spaces:`

In a *verbatim* context the TAB character is made active and set equal to `\__scontents_tabs_to_spaces:`, to produce as many spaces as the `width-tab` key was set to. The spaces are *active* characters, the same as the spaces of the *verbatim* context: a space token at the beginning of a line would be dropped in vertical mode and the TAB that indent a line would be lost. The *active* space is made with `\tex_lowercase:D` from the *active* character '`^^@`', since `\char_generate:nn` would expand it while defining the function.

```
\group_begin:
  \char_set_catcode_active:N \^^@
  \char_set_lccode:nn { 0 } { 32 }
  \tex_lowercase:D
    {
      \group_end:
      \cs_new:Npn \__scontents_tabs_to_spaces:
        { \prg_replicate:nn { \l__scontents_tab_width_int } { ^^@ } }
    }
```

(*End of definition for* `\__scontents_tabs_to_spaces:`.)

`\__scontents_do_noligs:N`

The function `\__scontents_do_noligs:N` is an alternative definition for LaTeX 2_ε's `\do@noligs` which makes sure to not consume the following space tokens. The LaTeX 2_ε version ends with `\char`#1`, which leaves TeX still looking for an ⟨*optional space*⟩.

```
\cs_new_protected:Npn \__scontents_do_noligs:N #1
```

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```
413   {
414     \char_set_catcode_active:N #1
415     \cs_set:cpe { __scontents_active_char_ \token_to_str:N #1 : }
416       {
417         \mode_leave_vertical:
418         \tex_kern:D \c_zero_dim
419         \tex_char:D `\exp_not:N #1
420       }
421     \char_set_active_eq:Nc #1 { __scontents_active_char_ \token_to_str:N #1 : }
422   }
```

(*End of definition for* `\__scontents_do_noligs:N`.)

`\__scontents_set_active_eq:NN`
`\__scontents_plain_disable_outer_par:`
`\__scontents_make_control_chars_active:`

Shortcut definitions for common catcode changes. The '`^^L`' needs a special treatment in plain TEX because it is an `\outer` token there.

```
423 \cs_new_protected:Npn \__scontents_set_active_eq:NN #1
424   {
425     \char_set_catcode_active:N #1
426     \char_set_active_eq:NN #1
427   }
428 ⟨/core⟩
429 ⟨*loader⟩
430 \group_begin:
431 ⟨plain⟩   \char_set_catcode_active:n { `\* }
432   \cs_new_protected:Nn \__scontents_plain_disable_outer_par:
433 ⟨*plain⟩
434     {
435       \group_begin:
436         \char_set_lccode:nn { `\* } { `\^^L }
437         \tex_lowercase:D { \group_end:
438         \tex_let:D * \scan_stop:
439       }
440     }
441 ⟨/plain⟩
442 ⟨latex|context⟩     { }
443 \group_end:
444 ⟨/loader⟩
445 ⟨*core⟩
446 \cs_new_protected:Nn \__scontents_make_control_chars_active:
447   {
448     \__scontents_plain_disable_outer_par:
449     \__scontents_set_active_eq:NN \^^I \__scontents_tab:
450     \__scontents_set_active_eq:NN \^^L \__scontents_par:
451     \__scontents_set_active_eq:NN \^^M \__scontents_ret:w
452   }
```

(*End of definition for* `\__scontents_set_active_eq:NN`, `\__scontents_plain_disable_outer_par:`, *and* `\__scontents_make_control_chars_active:`.)

12.10.2 Functions `\@bsphack` and `\@esphack`

`\__scontents_bsphack:`
`\__scontents_esphack:`

We emulate `\@bsphack` and `\@esphack` for plain TEX and ConTEXt, this is necessary to prevent *unwanted spaces* when the `print-cmd` key is false. In LATEX the kernel definitions are used.

```
453 \cs_new_protected:Nn \__scontents_bsphack:
454   {
455     \scan_stop:
456     \mode_if_horizontal:T
457       {
458         \skip_set_eq:NN \l__scontents_save_skip \tex_lastskip:D
459         \int_set_eq:NN \l__scontents_save_sf_int \tex_spacefactor:D
460       }
461   }
462 \cs_new_protected:Nn \__scontents_esphack:
463   {
464     \scan_stop:
465     \mode_if_horizontal:T
466       {
467         \int_set_eq:NN \tex_spacefactor:D \l__scontents_save_sf_int
468         \dim_compare:nNnT { \l__scontents_save_skip } > { \c_zero_skip }
469           {
470             \skip_if_eq:nnT { \tex_lastskip:D } { \c_zero_skip }
```

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```
                {
                  \nobreak
                  \skip_horizontal:n { \c_zero_skip }
                }
              \tex_ignorespaces:D
            }
        }
    }
⟨/core⟩
⟨*latex⟩
\cs_gset_eq:NN \__scontents_bsphack: \@bsphack
\cs_gset_eq:NN \__scontents_esphack: \@esphack
⟨/latex⟩
```

(*End of definition for* `\__scontents_bsphack:` *and* `\__scontents_esphack:`.)

12.11 Functions for sequences

The *storage system* of the package SCONTENTS is done using `seq` variables. Here we set up the functions that manage them.

`\__scontents_append_contents:nn` `\__scontents_append_contents:Ve`

The function `\__scontents_append_contents:nn` creates the *sequence* passed in `#1` if it does not exist and appends the {⟨*body env*⟩} of an environment or the {⟨*argument*⟩} of a command, passed in `#2`, to the right of the *sequence*.

```
⟨*core⟩
\cs_new_protected:Npn \__scontents_append_contents:nn #1 #2
  {
    \seq_if_exist:cF { g__scontents_name_#1_seq }
      { \seq_new:c { g__scontents_name_#1_seq } }
    \seq_gput_right:cn { g__scontents_name_#1_seq } {#2}
  }
\cs_generate_variant:Nn \__scontents_append_contents:nn { Ve }
```

(*End of definition for* `\__scontents_append_contents:nn`.)

`\__scontents_store_to_seq:NN`

The function `\__scontents_store_to_seq:NN` writes the recorded content in `#1` to the `.log` file if `debug-var` is true and stores it in the *sequence* named by `#2`.

```
\cs_new_protected:Npn \__scontents_store_to_seq:NN #1 #2
  {
    \bool_if:NT \l__scontents_debug_var_bool
      {
        \tl_log:N #1
      }
    \__scontents_append_contents:Ve #2 { \exp_not:V #1 }
  }
```

(*End of definition for* `\__scontents_store_to_seq:NN`.)

`\__scontents_finish_storing:NNN`

The function `\__scontents_finish_storing:NNN` works only in the *storing mode*, when the `write-out` key is not active. If the key `force-eol` is false, its default value, `\c__scontents_hidden_space_str` is added to the end of the token list passed in `#1`, which contains the {⟨*body env*⟩} of an environment or the {⟨*argument*⟩} of `\Scontents`.

Then `\__scontents_store_to_seq:NN` stores the content in the *sequence* named by `#2` and, if the boolean passed in `#3`, set by the `print-env` or `print-cmd` key, is true, the content is printed from the *sequence* by `\__scontents_lastfrom_seq:V`. If the *name* of the *sequence* is blank, the "error" `empty-store-name` is raised and nothing is stored or printed.

```
\cs_new_protected:Npn \__scontents_finish_storing:NNN #1 #2 #3
  {
    \bool_if:NT \l__scontents_storing_bool
      {
        \tl_if_blank:VTF #2
          { \msg_error:nn { scontents } { empty-store-name } }
          {
            \bool_if:NF \l__scontents_forced_eol_bool
              { \tl_put_right:Ne #1 { \c__scontents_hidden_space_str } }
            \__scontents_store_to_seq:NN #1 #2
            \bool_if:NT #3 { \__scontents_lastfrom_seq:V #2 }
          }
      }
  }
```

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(*End of definition for* `\__scontents_finish_storing:NNN`.)

`\__scontents_seq_count:n`
`\__scontents_seq_item:nn`
`\__scontents_seq_if_exist:nTF`

A *sequence* set by the `remember` key has a second copy with the ⟨*stored contents*⟩ of the previous compilation, whose name uses `prev` instead of `name` (§12.12). The conditional `\__scontents_seq_if_exist:nTF` tests if any of the two *sequences* named by `#1` exists, the function `\__scontents_seq_count:n` returns the larger number of ⟨*stored contents*⟩ of the two, or `0` if none exists, and the function `\__scontents_seq_item:nn` returns the ⟨*stored content*⟩ at the ⟨*index*⟩ passed in `#2`, from the previous compilation if it has it and from the current one otherwise. Without `remember`, and always in plain TEX and ConTEXt, only the current *sequence* exists.

```
\prg_new_conditional:Npnn \__scontents_seq_if_exist:n #1 { TF }
  {
    \bool_lazy_or:nnTF
      { \seq_if_exist_p:c { g__scontents_name_#1_seq } }
      { \seq_if_exist_p:c { g__scontents_prev_#1_seq } }
      { \prg_return_true: }
      { \prg_return_false: }
  }
\cs_new:Npn \__scontents_seq_count:n #1
  {
    \int_max:nn
      {
        \seq_if_exist:cTF { g__scontents_name_#1_seq }
          { \seq_count:c { g__scontents_name_#1_seq } }
          { 0 }
      }
      {
        \seq_if_exist:cTF { g__scontents_prev_#1_seq }
          { \seq_count:c { g__scontents_prev_#1_seq } }
          { 0 }
      }
  }
\cs_new:Npn \__scontents_seq_item:nn #1 #2
  {
    \bool_lazy_and:nnTF
      { \seq_if_exist_p:c { g__scontents_prev_#1_seq } }
      {
        \int_compare_p:nNn { \int_abs:n {#2} }
          < { \seq_count:c { g__scontents_prev_#1_seq } + 1 }
      }
      { \seq_item:cn { g__scontents_prev_#1_seq } {#2} }
      { \seq_item:cn { g__scontents_name_#1_seq } {#2} }
  }
```

(*End of definition for* `\__scontents_seq_count:n`, `\__scontents_seq_item:nn`, *and* `\__scontents_seq_if_exist:nTF`.)

`\__scontents_getfrom_seq:Nn`
`\__scontents_getfrom_seq:nNn`
`\__scontents_getfrom_seq_aux:nnn`
`\__scontents_getfrom_seq:nn`
`\__scontents_getfrom_seq:nnn`

The function `\__scontents_getfrom_seq:Nn` retrieves the ⟨*stored contents*⟩ whose ⟨*indexes*⟩ are in the sequence passed in `#1` from the *sequence* named by `#2` and the function `\__scontents_getfrom_seq:nn` retrieves the one whose ⟨*index*⟩ is passed in `#1`, both with `\__scontents_seq_item:nn`. Both raise the "errors" `undefined-storage` and `index-out-of-range` (§12.7).

```
\cs_new:Npn \__scontents_getfrom_seq:Nn #1 #2
  {
    \__scontents_seq_if_exist:nTF {#2}
      {
        \exp_args:Nf \__scontents_getfrom_seq:nNn
          { \__scontents_seq_count:n {#2} } #1 {#2}
      }
      { \msg_expandable_error:nnn { scontents } { undefined-storage } {#2} }
  }
\cs_new:Npn \__scontents_getfrom_seq:nNn #1 #2 #3
  {
    \seq_map_tokens:Nn #2 { \__scontents_getfrom_seq_aux:nnn {#1} {#3} }
  }
\cs_new:Npn \__scontents_getfrom_seq_aux:nnn #1 #2 #3
  {
    \exp_args:Nnf \use:n
      { \__scontents_getfrom_seq:nnn {#1} } { \int_eval:n {#3} } {#2}
  }
\cs_new:Npn \__scontents_getfrom_seq:nn #1 #2
  {
```

```
    \__scontents_seq_if_exist:nTF {#2}
      {
        \exp_args:Nf \__scontents_getfrom_seq:nnn
          { \__scontents_seq_count:n {#2} }
          {#1} {#2}
      }
      { \msg_expandable_error:nnn { scontents } { undefined-storage } {#2} }
  }
\cs_new:Npn \__scontents_getfrom_seq:nnn #1 #2 #3
  {
    \bool_lazy_or:nnTF
      { \int_compare_p:nNn {#2} = { 0 } }
      { \int_compare_p:nNn { \int_abs:n {#2} } > {#1} }
      {
        \msg_expandable_error:nnnnn { scontents } { index-out-of-range }
          {#2} {#3} {#1}
      }
      { \__scontents_seq_item:nn {#3} {#2} }
  }
```

(*End of definition for* `\__scontents_getfrom_seq:Nn` *and others.*)

`\__scontents_lastfrom_seq:n`
`\__scontents_lastfrom_seq:V`

The function `\__scontents_lastfrom_seq:n` saves the last ⟨*stored content*⟩ of the *sequence* named by `#1` in `\g__scontents_last_stored_tl` and rescans it with `\tl_retokenize:V` after the group, when the keys `print-env` or `print-cmd` are active.

```
\cs_new_protected:Npn \__scontents_lastfrom_seq:n #1
  {
    \tl_gset:Ne \g__scontents_last_stored_tl
      {
        \seq_item:cn { g__scontents_name_#1_seq } { -1 }
      }
    \group_insert_after:N \tl_retokenize:V
    \group_insert_after:N \g__scontents_last_stored_tl
    \group_insert_after:N \tl_gclear:N
    \group_insert_after:N \g__scontents_last_stored_tl
  }
\cs_generate_variant:Nn \__scontents_lastfrom_seq:n { V }
⟨/core⟩
```

(*End of definition for* `\__scontents_lastfrom_seq:n`.)

12.12 Remembering contents between compilations

The `remember` key (§12.8) makes the ⟨*stored contents*⟩ of some *sequences* available before they are stored, as `\label` and `\ref` do with the references. At the end of the document the ⟨*stored contents*⟩ of these *sequences* are written to the file `\jobname.sct` and at the beginning of the next compilation they are read in a second *sequence*, that is used by `\__scontents_seq_item:nn` (§12.11). Each line of the file has the name of the *sequence* and one ⟨*stored content*⟩, both converted to hexadecimal digits and separated by '=', so that any character can be written and read again. This is only available in LaTeX.

12.12.1 Definition of variables for remembering contents

`\g__scontents_remember_seq`
`\g__scontents_remember_pending_bool`

The global *sequence* `\g__scontents_remember_seq` stores, as strings, the names of the *sequences* set by the `remember` key and the boolean `\g__scontents_remember_pending_bool` is true when a ⟨*stored content*⟩ of one of them is requested and it is not available yet.

```
⟨*core⟩
\seq_new:N \g__scontents_remember_seq
\bool_new:N \g__scontents_remember_pending_bool
⟨/core⟩
```

(*End of definition for* `\g__scontents_remember_seq` *and* `\g__scontents_remember_pending_bool`.)

`\g__scontents_remember_ior`
`\g__scontents_remember_iow`
`\g__scontents_remember_old_str`
`\l__scontents_remember_new_str`
`\l__scontents_remember_name_str`
`\l__scontents_remember_item_str`

The streams `\g__scontents_remember_ior` and `\g__scontents_remember_iow` read and write the file `\jobname.sct`. The string `\g__scontents_remember_old_str` saves the lines read from the file and `\l__scontents_remember_new_str` the lines written, to compare them at the end of the document, and the strings `\l__scontents_remember_name_str` and `\l__scontents_remember_item_str` save the name of the *sequence* and the ⟨*stored content*⟩ of a line.

```
⟨*latex⟩
\ior_new:N \g__scontents_remember_ior
```

```
\iow_new:N \g__scontents_remember_iow
\str_new:N \g__scontents_remember_old_str
\str_new:N \l__scontents_remember_new_str
\str_new:N \l__scontents_remember_name_str
\str_new:N \l__scontents_remember_item_str
⟨/latex⟩
```

(*End of definition for* `\g__scontents_remember_ior` *and others.*)

12.12.2 Definition of error messages for remembering contents

remember-latex-only

The "error" `remember-latex-only` is raised when the `remember` key is used in plain TEX or ConTEXt.

```
⟨*core⟩
\msg_new:nnnn { scontents } { remember-latex-only }
  { The~key~'remember'~is~only~available~in~LaTeX~\msg_line_context:. }
  {
    The~stored~contents~can~only~be~used~after~they~are~stored~
    in~this~format.~The~key~is~ignored.
  }
⟨/core⟩
```

(*End of definition for* `remember-latex-only`.)

12.12.3 Definition of messages for remembering contents

remember-rerun
remember-undefined

The "warning" `remember-rerun` is issued at the end of the document when the lines written to `\jobname.sct` are not the ones read from it, and otherwise the "warning" `remember-undefined` when a ⟨*stored content*⟩ was requested and it was not available.

```
⟨*latex⟩
\msg_new:nnn { scontents } { remember-rerun }
  { Stored~contents~may~have~changed.~Rerun~to~get~them~right. }
\msg_new:nnn { scontents } { remember-undefined }
  { There~were~undefined~stored~contents. }
⟨/latex⟩
```

(*End of definition for* `remember-rerun` *and* `remember-undefined`.)

12.12.4 Internal functions for remembering contents

__scontents_remember:n

The function `\__scontents_remember:n`, used by the `remember` key, adds each name of the comma separated list passed in `#1` to `\g__scontents_remember_seq` as a string, only once. In plain TEX and ConTEXt it raises the "error" `remember-latex-only`.

```
⟨*core⟩
\cs_new_protected:Npn \__scontents_remember:n #1
  {
    \__scontents_format_case:nnn
      {
        \clist_map_inline:nn {#1}
          {
            \seq_if_in:NeF \g__scontents_remember_seq { \tl_to_str:n {##1} }
              { \seq_gput_right:Ne \g__scontents_remember_seq { \tl_to_str:n {##1} } }
          }
      }
      { \msg_error:nn { scontents } { remember-latex-only } }
      { \msg_error:nn { scontents } { remember-latex-only } }
  }
```

(*End of definition for* `\__scontents_remember:n`.)

__scontents_if_pending:nTF
__scontents_if_pending:neF

The conditional `\__scontents_if_pending:nnTF` tests if the ⟨*stored content*⟩ at the ⟨*index*⟩ passed in `#2` of the remembered *sequence* named by `#1` is not available yet, that is, if its absolute value is greater than the number of ⟨*stored contents*⟩ of both copies. In that case `\g__scontents_remember_pending_bool` is set to true and the commands print nothing. For the *sequences* that are not remembered it is always false and the "errors" of the commands are raised as usual.

```
\prg_new_protected_conditional:Npnn \__scontents_if_pending:nn #1 #2 { T, F }
  {
    \bool_lazy_and:nnTF
      { \seq_if_exist_p:c { g__scontents_prev_#1_seq } }
      { \int_compare_p:nNn { \int_abs:n {#2} } > { \__scontents_seq_count:n {#1} } }
      {
```

```
      \bool_gset_true:N \g__scontents_remember_pending_bool
      \prg_return_true:
    }
    { \prg_return_false: }
  }
\prg_generate_conditional_variant:Nnn \__scontents_if_pending:nn { ne } { F }
⟨/core⟩
```

(*End of definition for* `\__scontents_if_pending:nTF` *and* `\__scontents_if_pending:neF`.)

`\__scontents_remember_read:`
`\__scontents_remember_read_line:n`
`\__scontents_remember_read_line:w`

The function `\__scontents_remember_read:` reads the file `\jobname.sct`, if it exists, line by line with `\__scontents_remember_read_line:n`, and creates an empty copy of the remembered *sequences* that are not in the file. The function `\__scontents_remember_read_line:w` converts the name of the *sequence* from hexadecimal digits and, if it is remembered, saves the line in `\g__scontents_remember_old_str` and adds the ⟨*stored content*⟩, converted from hexadecimal digits, to the copy of the previous compilation. The lines of the *sequences* that are no longer remembered are ignored.

```
⟨*latex⟩
\cs_new_protected:Npn \__scontents_remember_read:
  {
    \ior_open:NnT \g__scontents_remember_ior { \c_sys_jobname_str .sct }
      {
        \ior_str_map_inline:Nn \g__scontents_remember_ior
          { \__scontents_remember_read_line:n {##1} }
        \ior_close:N \g__scontents_remember_ior
      }
    \seq_map_inline:Nn \g__scontents_remember_seq
      {
        \seq_if_exist:cF { g__scontents_prev_##1_seq }
          { \seq_new:c { g__scontents_prev_##1_seq } }
      }
  }
\cs_new_protected:Npn \__scontents_remember_read_line:n #1
  {
    \str_if_empty:nF {#1}
      { \__scontents_remember_read_line:w #1 = = \s__scontents_stop }
  }
\cs_new_protected:Npn \__scontents_remember_read_line:w #1 = #2 = #3 \s__scontents_stop
  {
    \str_set_convert:Nnnn \l__scontents_remember_name_str {#1} { utf8/hex } { }
    \seq_if_in:NVT \g__scontents_remember_seq \l__scontents_remember_name_str
      {
        \str_gput_right:Nn \g__scontents_remember_old_str { #1 = #2 , }
        \str_set_convert:Nnnn \l__scontents_remember_item_str {#2} { utf8/hex } { }
        \seq_if_exist:cF { g__scontents_prev_ \l__scontents_remember_name_str _seq }
          { \seq_new:c { g__scontents_prev_ \l__scontents_remember_name_str _seq } }
        \seq_gput_right:cV { g__scontents_prev_ \l__scontents_remember_name_str _seq }
          \l__scontents_remember_item_str
      }
  }
```

(*End of definition for* `\__scontents_remember_read:`, `\__scontents_remember_read_line:n`, *and* `\__scontents_remember_read_line:w`.)

`\__scontents_remember_write:`
`\__scontents_remember_write_seq:n`
`\__scontents_remember_write_item:n`

The function `\__scontents_remember_write:` writes the file `\jobname.sct` if some *sequence* is remembered: for each one that exists, `\__scontents_remember_write_seq:n` converts its name to hexadecimal digits and `\__scontents_remember_write_item:n` writes a line for each ⟨*stored content*⟩, also saved in `\l__scontents_remember_new_str`. If the lines written are not the ones read, the "warning" `remember-rerun` is issued, otherwise `remember-undefined` if a ⟨*stored content*⟩ was not available.

```
\cs_new_protected:Npn \__scontents_remember_write:
  {
    \seq_if_empty:NF \g__scontents_remember_seq
      {
        \str_clear:N \l__scontents_remember_new_str
        \iow_open:Nn \g__scontents_remember_iow { \c_sys_jobname_str .sct }
        \seq_map_function:NN \g__scontents_remember_seq \__scontents_remember_write_seq:n
        \iow_close:N \g__scontents_remember_iow
        \str_if_eq:NNTF \g__scontents_remember_old_str \l__scontents_remember_new_str
```

```
694           {
695             \bool_if:NT \g__scontents_remember_pending_bool
696               { \msg_warning:nn { scontents } { remember-undefined } }
697           }
698           { \msg_warning:nn { scontents } { remember-rerun } }
699       }
700   }
701 \cs_new_protected:Npn \__scontents_remember_write_seq:n #1
702   {
703     \seq_if_exist:cT { g__scontents_name_#1_seq }
704       {
705         \str_set_convert:Nnnn \l__scontents_remember_name_str {#1} { } { utf8/hex }
706         \seq_map_function:cN { g__scontents_name_#1_seq }
707           \__scontents_remember_write_item:n
708       }
709   }
710 \cs_new_protected:Npn \__scontents_remember_write_item:n #1
711   {
712     \str_set_convert:Nnnn \l__scontents_remember_item_str {#1} { } { utf8/hex }
713     \iow_now:Ne \g__scontents_remember_iow
714       { \l__scontents_remember_name_str = \l__scontents_remember_item_str }
715     \str_put_right:Ne \l__scontents_remember_new_str
716       { \l__scontents_remember_name_str = \l__scontents_remember_item_str , }
717   }
```

(*End of definition for* `\__scontents_remember_write:`, `\__scontents_remember_write_seq:n`, *and* `\__scontents_remember_write_item:n`.)

The file is read at the beginning of the document and written at the end of it.

```
718 \hook_gput_code:nnn { begindocument } { scontents }
719   { \__scontents_remember_read: }
720 \hook_gput_code:nnn { enddocument } { scontents }
721   { \__scontents_remember_write: }
722 ⟨/latex⟩
```

12.13 The command \newenvsc

The user-level command `\newenvsc` creates a new environment with the same behavior as the base environment `scontents`, which is itself defined with `\newenvsc` (§12.14). It takes a mandatory argument, the {⟨*env name*⟩}, and an optional argument with the ⟨*initial keys*⟩. The references to `\begin{scontents}` or `\end{scontents}` in this section apply to any environment defined with `\newenvsc`.

12.13.1 Definition of variables for \newenvsc

`\l__scontents_save_every_body_lines_tl`
`\l__scontents_processed_body_lines_tl`
`\l__scontents_environment_keys_tl`
`\l__scontents_nesting_env_int`
`\l__scontents_nesting_aux_int`

The token list `\l__scontents_save_every_body_lines_tl` holds the {⟨*body env*⟩} of the environment while it is being read and the token list `\l__scontents_processed_body_lines_tl` saves it once the environment is closed. The token list `\l__scontents_environment_keys_tl` saves the ⟨*keys*⟩ passed to the *optional argument* and the integer variables `\l__scontents_nesting_env_int` and `\l__scontents_nesting_aux_int` are used to analyze the nesting of the environment.

```
723 ⟨*core⟩
724 \tl_new:N  \l__scontents_save_every_body_lines_tl
725 \tl_new:N  \l__scontents_processed_body_lines_tl
726 \tl_new:N  \l__scontents_environment_keys_tl
727 \int_new:N \l__scontents_nesting_env_int
728 \int_new:N \l__scontents_nesting_aux_int
```

(*End of definition for* `\l__scontents_save_every_body_lines_tl` *and others.*)

12.13.2 Definition of error messages for \newenvsc

`\__scontents_env_begin_str:n`
`\__scontents_env_end_str:n`

The functions `\__scontents_env_begin_str:n` and `\__scontents_env_end_str:n` return the start and the end of the environment named by `#1` as they are written in each format: `\begin{#1}` and `\end{#1}` in LaTeX, `\#1` and `\end#1` in plain TeX and `\start#1` and `\stop#1` in ConTeXt. They are used in the messages of the environments.

```
729 \cs_new:Npn \__scontents_env_begin_str:n #1
730   {
731     \__scontents_format_case:nnn
732       { \c_backslash_str begin \c_left_brace_str #1 \c_right_brace_str }
733       { \c_backslash_str #1 }
734       { \c_backslash_str start #1 }
735   }
```

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```
\cs_new:Npn \__scontents_env_end_str:n #1
  {
    \__scontents_format_case:nnn
      { \c_backslash_str end \c_left_brace_str #1 \c_right_brace_str }
      { \c_backslash_str end #1 }
      { \c_backslash_str stop #1 }
  }
```

(*End of definition for* `\__scontents_env_begin_str:n` *and* `\__scontents_env_end_str:n`.)

junk-after-begin
env-already-defined

The "error" `junk-after-begin` is raised when there are characters after the start of the environment or its *optional argument* on the same line and the "error" `env-already-defined` is raised when `\newenvsc` is used with the name of an environment that already exists.

```
\msg_new:nnnn { scontents } { junk-after-begin }
  {
    Junk~characters~#1~\msg_line_context: :
    \\ \\
    #2
  }
  {
    Nothing~is~allowed~after~the~start~of~the~environment~on~the~same~
    line.~These~characters~are~ignored.
  }
\msg_new:nnnn { scontents } { env-already-defined }
  { Environment~'#1'~already~defined! }
  {
    You~have~used~\token_to_str:N \newenvsc \c_space_tl
    with~an~environment~that~already~has~a~definition. \\ \\
    The~existing~definition~of~'#1'~will~not~be~altered.
  }
```

(*End of definition for* `junk-after-begin` *and* `env-already-defined`.)

12.13.3 Definition of messages for `\newenvsc`

empty-environment
rescanning-text
multiple-begin

The "warning" `empty-environment` is issued when the environment is empty, the "warning" `rescanning-text` when there are characters after the end of the environment on the same line and the "warning" `multiple-begin` when the environment starts more than once on the same line. The start and the end of the environment are written with `\__scontents_env_begin_str:n` and `\__scontents_env_end_str:n`.

```
\msg_new:nnn { scontents } { empty-environment }
  { Environment~'#1'~is~empty~\msg_line_context:. }
\msg_new:nnn { scontents } { rescanning-text }
  {
    Rescanning~text~'#1'~after~\__scontents_env_end_str:n {#2}~
    \msg_line_context:.
  }
\msg_new:nnn { scontents } { multiple-begin }
  {
    Multiple~\__scontents_env_begin_str:n { \l__scontents_env_name_tl }~
    \msg_line_context:.
  }
```

(*End of definition for* `empty-environment`, `rescanning-text`, *and* `multiple-begin`.)

12.13.4 Definition of keys for `\newenvsc`

write-env
write-out
print-env
store-env
force-eol
overwrite
unknown

We define the ⟨*keys*⟩ for the environments defined by `\newenvsc`, the unknown ⟨*keys*⟩ are passed to `\__scontents_unknown_keys_env:n` (§12.9).

```
\keys_define:nn { scontents / scontents }
  {
    write-env .code:n           = { \__scontents_set_write_store:n {#1} },
    write-out .code:n           = { \__scontents_set_write_out:n {#1} },
    write-env .value_required:n = true,
    write-out .value_required:n = true,
    print-env .meta:nn          = { scontents } { print-env = #1 },
    print-env .default:n        = true,
    store-env .meta:nn          = { scontents } { store-env = #1 },
    force-eol .meta:nn          = { scontents } { force-eol = #1 },
    force-eol .default:n        = true,
```

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```
    overwrite .meta:nn          = { scontents } { overwrite = #1 },
    overwrite .default:n        = true,
    unknown   .code:n           = { \__scontents_unknown_keys_env:n {#1} },
  }
```

(*End of definition for* `write-env` *and others.*)

12.13.5 Internal functions for `\newenvsc`

`\__scontents_grab_opt_arg:w`
`\__scontents_grab_opt_arg:n`

The function `\__scontents_grab_opt_arg:w` is called with the tokens following `\begin{scontents}` when the next character is a '`[`'. It is defined with `ltcmd`[11] to use its delimited argument processor.

```
\NewDocumentCommand \__scontents_grab_opt_arg:w { r[] }
  {
    \__scontents_grab_opt_arg:n {#1}
  }
```

The function `\__scontents_grab_opt_arg:n` is called where '`^^M`' is active, so `\__scontents_normalise_line_e` replaces the active '`^^M`' characters by spaces before setting the ⟨*keys*⟩.

```
\cs_new_protected:Npn \__scontents_grab_opt_arg:n #1
  {
    \tl_set:Nn \l__scontents_environment_keys_tl {#1}
    \__scontents_normalise_line_ends:N \l__scontents_environment_keys_tl
    \keys_set:nV { scontents / scontents } \l__scontents_environment_keys_tl
    \__scontents_start_after_option:w
  }
```

(*End of definition for* `\__scontents_grab_opt_arg:w` *and* `\__scontents_grab_opt_arg:n`.)

`\__scontents_normalise_line_ends:N`
`\__scontents_start_environment:w`
`\__scontents_start_after_option:w`
`\__scontents_check_line_process:nn`
`\__scontents_stop_environment:`

Here '`^^I`', '`^^L`' and '`^^M`' are active characters, so that the end of line can be used as a delimiter and TEX does not remove the space-like characters.

The function `\__scontents_start_environment:w` checks if the next token after `\begin{scontents}` is a '`[`', in that case `\__scontents_grab_opt_arg:w` processes the *optional argument* and calls `\__scontents_start_after_option:w`. In all cases the function `\__scontents_check_line_process:nn` raises the "error" `junk-after-begin` if there is something else on the line and then calls `\__scontents_file_tl_write_start:V`, which reads the content of the environment and stores it in a token list or writes it to an ⟨*external file*⟩.

At the end of the environment, the function `\__scontents_stop_environment:` calls `\__scontents_file_write_stop:N` and issues the "warning" `empty-environment` if nothing was stored. This part of the code is adapted from the code of the package `xsim`[13] by Clemens Niederberger.

```
\group_begin:
  \char_set_catcode_active:N \^^I
  \char_set_catcode_active:N \^^L
  \char_set_catcode_active:N \^^M
  \cs_new_protected:Npn \__scontents_normalise_line_ends:N #1
    { \tl_replace_all:Nnn #1 { ^^M } { ~ } }
  \cs_new_protected:Npn \__scontents_start_environment:w #1 ^^M
    {
      \tl_if_head_is_N_type:nTF {#1}
        {
          \str_if_eq:eeTF { \tl_head:n {#1} } { [ }
            { \__scontents_grab_opt_arg:w #1 ^^M }
            { \__scontents_check_line_process:nn { } {#1} }
        }
        { \__scontents_check_line_process:nn { } {#1} }
    }
  \cs_new_protected:Npn \__scontents_start_after_option:w #1 ^^M
    { \__scontents_check_line_process:nn { [...] } {#1} }
  \cs_new_protected:Npn \__scontents_check_line_process:nn #1 #2
    {
      \tl_if_blank:nF {#2}
        {
          \msg_error:nnee { scontents } { junk-after-begin }
            { after~ \__scontents_env_begin_str:n { \l__scontents_env_name_tl } #1 } {#2}
        }
      \__scontents_make_control_chars_active:
      \__scontents_file_tl_write_start:V \l__scontents_file_name_tl
    }
  \cs_new_protected:Nn \__scontents_stop_environment:
```

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```
    {
      \__scontents_file_write_stop:N \l__scontents_processed_body_lines_tl
      \bool_lazy_and:nnT
        { \l__scontents_storing_bool }
        { \tl_if_empty_p:N \l__scontents_processed_body_lines_tl }
        {
          \msg_warning:nne { scontents } { empty-environment }
            { \l__scontents_env_name_tl }
        }
    }
```

(*End of definition for* `\__scontents_normalise_line_ends:N` *and others.*)

`\__scontents_file_tl_write_start:n`
`\__scontents_file_tl_write_start:V`
`\__scontents_setup_verb_processor:`
`\__scontents_verb_processor_iterate:w`
`\__scontents_verb_processor_iterate:nnn`
`\__scontents_env_end_function:`
`\__scontents_file_write_stop:N`
`\__scontents_remove_leading_nl:n`
`\__scontents_remove_leading_nl:nn`
`\__scontents_remove_leading_nl:w`

This is the main function to collect the {⟨*body env*⟩} of the environment. The function `\__scontents_file_tl_write_start:n` starts a *group*, opens the ⟨*output file*⟩ if necessary, sets the *verbatim catcodes* and then issues '`^^M`', set equal to `\__scontents_ret:w`, to read the {⟨*body env*⟩} line by line until its end. Each line read is appended to the output token list after an active '`^^J`' and written to the ⟨*output file*⟩, if any.

The function `\__scontents_setup_verb_processor:` defines `\__scontents_verb_processor_iterate:w`, delimited by the end of the current environment, and `\__scontents_verb_processor_iterate:nnn` decides if the line is a nested `\end{scontents}` or the end of the environment, in which case `\__scontents_env_end_function:` leaves the closing command for the current format and the characters after it are rescanned with the "warning" `rescanning-text`.

At the end of the environment, the function `\__scontents_file_write_stop:N` closes the ⟨*output file*⟩, if it was opened, and smuggles the output token list out of the group. The leading '`^^J`' is removed with `\__scontents_remove_leading_nl:n`, which expects an active '`^^J`' at the head of the token list, otherwise a low level TEX "error" is raised.

```
\cs_new_protected:Npn \__scontents_file_tl_write_start:n #1
  {
    \group_begin:
      \__scontents_file_if_writable:nTF {#1}
        {
          \bool_set_true:N \l__scontents_writable_bool
          \iow_open:Nn \l__scontents_file_write_iow {#1}
        }
        { \bool_set_false:N \l__scontents_writable_bool }
      \tl_clear:N \l__scontents_save_every_body_lines_tl
      \seq_map_function:NN \l_char_special_seq \char_set_catcode_other:N
      \int_step_function:nnnN { 128 } { 1 } { 255 } \char_set_catcode_letter:n
      \cs_set_protected:Npe \__scontents_ret:w ##1 ^^M
        {
          \exp_not:N \__scontents_verb_processor_iterate:w
          ##1 \c__scontents_end_env_tl
              \c__scontents_end_env_tl
              \exp_not:N \q__scontents_stop
        }
      \__scontents_make_control_chars_active:
      \__scontents_ret:w
  }
\cs_new_protected:Nn \__scontents_setup_verb_processor:
  {
    \use:e
      {
        \cs_set_protected:Npn \exp_not:N \__scontents_verb_processor_iterate:w
          ##1 \c__scontents_end_env_tl
          ##2 \c__scontents_end_env_tl
          ##3 \exp_not:N \q__scontents_stop
      }   { \__scontents_verb_processor_iterate:nnn {##1} {##2} {##3} }
  }
\cs_new_protected:Npn \__scontents_verb_processor_iterate:nnn #1 #2 #3
  {
    \tl_if_blank:nTF {#3}
      {
        \__scontents_analyse_nesting:n {#1}
        \__scontents_verb_processor_output:n {#1}
      }
      {
        \__scontents_if_nested:TF
          {
```

```
            \__scontents_nesting_decr:
            \__scontents_verb_processor_output:e
              { \exp_not:n {#1} \c__scontents_end_env_tl \exp_not:n {#2} }
          }
          {
            \tl_if_blank:nF {#1}
              { \__scontents_verb_processor_output:n {#1} }
            \cs_set_protected:Npe \__scontents_ret:w
              {
                \__scontents_env_end_function:
                \bool_lazy_or:nnF
                  { \tl_if_blank_p:n {#2} }
                  { \str_if_eq_p:ee {#2} { \c_percent_str } }
                  {
                    \str_if_eq:VnF \c__scontents_hidden_space_str {#2}
                      {
                        \msg_warning:nnnn { scontents } { rescanning-text }
                          {#2} { \tl_use:N \l__scontents_env_name_tl }
                      }
                    \tl_retokenize:n {#2}
                  }
              }
            \char_set_active_eq:NN ^^M \__scontents_ret:w
          }
      }
    ^^M
  }
\cs_new:Nn \__scontents_env_end_function:
  {
    \__scontents_format_case:nnn
      { \exp_not:N \end { \if_false: } \fi: }
      { \exp_after:w \exp_not:N \cs:w end }
      { \exp_after:w \exp_not:N \cs:w stop }
    \tl_use:N \l__scontents_env_name_tl
    \__scontents_format_case:nnn
      { \if_false: { \fi: } }
      { \cs_end: }
      { \cs_end: }
  }
\cs_new_protected:Npn \__scontents_file_write_stop:N #1
  {
    \bool_if:NT \l__scontents_writable_bool
      { \iow_close:N \l__scontents_file_write_iow }
    \use:e
      {
        \group_end:
        \bool_if:NT \l__scontents_storing_bool
          {
            \tl_set:Nn \exp_not:N #1
              {
                \exp_args:NV
                  \__scontents_remove_leading_nl:n \l__scontents_save_every_body_lines_tl
              }
          }
      }
  }
\cs_new:Npn \__scontents_remove_leading_nl:n #1
  {
    \tl_if_head_is_N_type:nTF {#1}
      {
        \exp_args:Nf
          \__scontents_remove_leading_nl:nn
            { \tl_head:n {#1} } {#1}
      }
      { \exp_not:n {#1} }
  }
\cs_new:Npn \__scontents_remove_leading_nl:nn #1 #2
  {
    \token_if_eq_meaning:NNTF ^^J #1
      { \exp_not:o { \__scontents_remove_leading_nl:w #2 } }
      { \exp_not:n {#2} }
```

```
    }
  \cs_new:Npn \__scontents_remove_leading_nl:w ^^J { }
```

(*End of definition for* `\__scontents_file_tl_write_start:n` *and others.*)

`\__scontents_verb_processor_output:n`
`\__scontents_verb_processor_output:e`

The function `\__scontents_verb_processor_output:n` does the output of each line read, to the token list and to the ⟨*output file*⟩, according to the booleans `\l__scontents_writable_bool` and `\l__scontents_storing_bool`.

```
  \cs_new_protected:Npn \__scontents_verb_processor_output:n #1
    {
      \bool_if:NT \l__scontents_writable_bool
        { \iow_now:Nn \l__scontents_file_write_iow {#1} }
      \bool_if:NT \l__scontents_storing_bool
        { \tl_put_right:Nn \l__scontents_save_every_body_lines_tl { ^^J #1 } }
    }
\group_end:
\cs_generate_variant:Nn \__scontents_verb_processor_output:n { e }
\cs_generate_variant:Nn \__scontents_file_tl_write_start:n { V }
```

(*End of definition for* `\__scontents_verb_processor_output:n`.)

`\__scontents_analyse_nesting:n`
`\__scontents_nesting_incr:`
`\__scontents_nesting_decr:`
`\__scontents_if_nested:TF`

The function `\__scontents_analyse_nesting:n` scans the nested `\begin{scontents}` in a line and steps the counter `\l__scontents_nesting_env_int`, issuing the "warning" `multiple-begin` if there is more than one. The conditional `\__scontents_if_nested:TF` tests if we are in a nested environment and `\__scontents_nesting_decr:` reduces the nesting level when an `\end{scontents}` is found.

```
\cs_new_protected:Npn \__scontents_analyse_nesting:n #1
  {
    \int_zero:N \l__scontents_nesting_aux_int
    \__scontents_analyse_nesting_format:n {#1}
    \int_compare:nNnT { \l__scontents_nesting_aux_int } > { 1 }
      { \msg_warning:nn { scontents } { multiple-begin } }
  }
\cs_new_protected:Nn \__scontents_nesting_incr:
  {
    \int_incr:N \l__scontents_nesting_env_int
    \int_incr:N \l__scontents_nesting_aux_int
  }
\cs_new_protected:Nn \__scontents_nesting_decr:
  {
    \int_decr:N \l__scontents_nesting_env_int
  }
\prg_new_protected_conditional:Npnn \__scontents_if_nested: { TF }
  {
    \int_compare:nNnTF { \l__scontents_nesting_env_int } > { \c_zero_int }
      { \prg_return_true: }
      { \prg_return_false: }
  }
```

(*End of definition for* `\__scontents_analyse_nesting:n` *and others.*)

Multiple `\end{scontents}` on the same line are NOT supported.

`\__scontents_analyse_nesting_latex:n`
`\__scontents_analyse_nesting_latex:w`
`\__scontents_analyse_nesting_generic:nn`
`\__scontents_analyse_nesting_generic:w`
`ontents_analyse_nesting_generic_process:nn`
`contents_define_generic_nesting_function:n`
`\__scontents_analyse_nesting_format:n`

In LaTeX the environments start with `\begin{«env»}` and no '}' can appear inside «env», so a macro delimited by '}' is enough to find the nested `\begin{scontents}`.

```
\use:e
  {
    \cs_new_protected:Npn \exp_not:N \__scontents_analyse_nesting_latex:w #1
      \c_backslash_str begin \c_left_brace_str #2 \c_right_brace_str
  }  {
       \__scontents_tl_if_head_is_q_mark:nTF {#2}
         { \__scontents_use_none_delimit_by_q_stop:w }
         {
           \str_if_eq:VnT \l__scontents_env_name_tl {#2}
             { \__scontents_nesting_incr: }
           \__scontents_analyse_nesting_latex:w
         }
     }
\cs_new_protected:Npe \__scontents_analyse_nesting_latex:n #1
  {
```

```
    \__scontents_analyse_nesting_latex:w #1
      \c_backslash_str begin
        \c_left_brace_str \exp_not:N \q__scontents_mark \c_right_brace_str
    \exp_not:N \q__scontents_stop
  }
```

In the other formats there is no "end anchor" to delimit the name of the environment, so we search for the whole command that starts the environment, `\scontents` or `\startscontents`.

```
\cs_new_protected:Npn \__scontents_analyse_nesting_generic_process:nn #1 #2
  {
    \tl_if_head_is_N_type:nTF {#2}
      {
        \__scontents_tl_if_head_is_q_mark:nF {#2}
          {
            \__scontents_nesting_incr:
            \__scontents_analyse_nesting_generic:w #2 \q__scontents_stop
          }
      }
      { \__scontents_analyse_nesting_generic:w #2 \q__scontents_stop }
  }
\cs_new_protected:Npn \__scontents_analyse_nesting_generic:nn #1 #2
  {
    \__scontents_define_generic_nesting_function:n {#1}
    \use:e
      {
        \exp_not:N \__scontents_analyse_nesting_generic:w #2
          \c_backslash_str #1 \tl_use:N \l__scontents_env_name_tl
            \exp_not:N \q__scontents_mark \exp_not:N \q__scontents_stop
      }
  }
\cs_new_protected:Npn \__scontents_define_generic_nesting_function:n #1
  {
    \use:e
      {
        \cs_set_protected:Npn \exp_not:N \__scontents_analyse_nesting_generic:w ##1
          \c_backslash_str #1 \tl_use:N \l__scontents_env_name_tl
            ##2 \exp_not:N \q__scontents_stop
      }   { \__scontents_analyse_nesting_generic_process:nn {##1} {##2} }
  }
⟨/core⟩
```

The function `\__scontents_analyse_nesting_format:n` uses the version for the current format.

```
⟨*loader⟩
⟨latex⟩\cs_new_eq:NN \__scontents_analyse_nesting_format:n
⟨latex⟩  \__scontents_analyse_nesting_latex:n
⟨!latex⟩\cs_new_protected:Npn \__scontents_analyse_nesting_format:n
⟨plain⟩  { \__scontents_analyse_nesting_generic:nn { } }
⟨context⟩  { \__scontents_analyse_nesting_generic:nn { start } }
⟨/loader⟩
```

(*End of definition for* `\__scontents_analyse_nesting_latex:n` *and others.*)

`\__scontents_generic_begin:`
`\__scontents_generic_end:`

The function `\__scontents_generic_begin:` leaves the '`^^M`' character active and calls `\__scontents_start_environment:w`. The function `\__scontents_generic_end:` calls `\__scontents_stop_environment:` and then `\__scontents_finish_storing:NNN`, which stores the {⟨*body env*⟩} in the *sequence* and prints it if the `print-env` key is active (§12.11).

```
⟨*core⟩
\cs_new_protected:Npn \__scontents_generic_begin:
  {
    \char_set_catcode_active:N \^^M
    \__scontents_start_environment:w
  }
\cs_new_protected:Npn \__scontents_generic_end:
  {
    \__scontents_stop_environment:
    \__scontents_finish_storing:NNN
      \l__scontents_processed_body_lines_tl
      \l__scontents_name_seq_env_tl
      \l__scontents_print_env_bool
  }
```

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(*End of definition for* `\__scontents_generic_begin:` *and* `\__scontents_generic_end:`.)

`\__scontents_setting_env:nn`
`\__scontents_define_env:nnn`

The function `\__scontents_setting_env:nn` receives the *name* of the environment in `#1` and the ⟨*initial keys*⟩ in `#2`. It creates the functions `\__scontents_#1_begin:` and `\__scontents_#1_end:`, which set `\l__scontents_env_name_tl` and the ⟨*initial keys*⟩ and call `\__scontents_generic_begin:` and `\__scontents_generic_end:`, and passes them to `\__scontents_define_env:nnn`.

```
\cs_new_protected:Npn \__scontents_setting_env:nn #1 #2
  {
    \cs_new_protected:cpn { __scontents_#1_begin: }
      {
        \tl_set:Nn \l__scontents_env_name_tl {#1}
        \tl_set:Nn \l__scontents_cmd_name_tl { newenvsc }
        \keys_set:nn { scontents } {#2}
        \__scontents_setup_verb_processor:
        \__scontents_generic_begin:
      }
    \cs_new_protected:cpn { __scontents_#1_end: }
      { \__scontents_generic_end: }
    \exp_args:Nooo \__scontents_define_env:nnn
      { \tl_to_str:n {#1} }
      { \cs:w __scontents_#1_begin: \cs_end: }
      { \cs:w __scontents_#1_end: \cs_end: }
  }
⟨/core⟩
```

The function `\__scontents_define_env:nnn` creates the environment for LaTeX, plain TeX and ConTeXt.

```
⟨*loader⟩
\cs_new_protected:Npn \__scontents_define_env:nnn #1 #2 #3
  {
⟨latex|plain⟩    \NewDocumentEnvironment {#1} { }
⟨context⟩    \cs_new_protected:cpn { start #1 }
      {
⟨!latex⟩        \group_begin:
          #2
      }
⟨context⟩    \cs_new_protected:cpn { stop #1 }
      {
          #3
⟨!latex⟩        \group_end:
      }
  }
```

(*End of definition for* `\__scontents_setting_env:nn` *and* `\__scontents_define_env:nnn`.)

`\newenvsc`

Defines the user command `\newenvsc` for LaTeX, plain TeX and ConTeXt, raising the "error" `env-already-defined` if the environment exists.

```
\NewDocumentCommand \newenvsc { m O{} }
  {
⟨latex|plain⟩    \cs_if_exist:cTF {#1}
⟨context⟩    \cs_if_exist:cTF { start #1 }
      { \msg_error:nnn { scontents } { env-already-defined } {#1} }
      { \__scontents_setting_env:nn {#1} {#2} }
  }
⟨/loader⟩
```

(*End of definition for* `\newenvsc`. *This function is documented on page 5.*)

12.14 The environment `scontents`

`scontents`
`\scontents`
`\endscontents`
`\startscontents`
`\stopscontents`

The base environment `scontents` is defined with `\newenvsc`, in plain TeX it is `\scontents` ... `\endscontents` and in ConTeXt `\startscontents` ... `\stopscontents`.

```
⟨*loader⟩
\newenvsc{scontents}
⟨/loader⟩
```

(*End of definition for* `scontents` *and others. These functions are documented on page 4.*)

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12.15 The environment verbatimsc

The verbatimsc environment is a customized version of the standard verbatim environment of LaTeX used by \typestored (§12.20). For plain TeX and ConTeXt we must add some additional functions.

12.15.1 Definition of variables for verbatimsc

\l__scontents_plain_bool

The boolean variable \l__scontents_plain_bool is used in the plain TeX version of the verbatimsc environment.

```
⟨*plain⟩
\bool_new:N \l__scontents_plain_bool
⟨/plain⟩
```

(*End of definition for* \l__scontents_plain_bool.)

12.15.2 Internal functions for verbatimsc

\dospecials

The verbatim environment requires \dospecials. In case it does not exist when SCONTENTS is loaded, we define it with \l_char_special_seq.

```
⟨*!latex⟩
\cs_if_exist:NF \dospecials
  {
    \cs_new:Npn \dospecials
      { \seq_map_function:NN \l_char_special_seq \do }
  }
⟨/!latex⟩
```

(*End of definition for* \dospecials.)

__scontents_xverb:w
__scontents_xverb:

The environment verbatimsc needs to find literally its end, this is done by __scontents_xverb:w, delimited by \g__scontents_end_verbatimsc_tl, for plain TeX and LaTeX. The function __scontents_xverb: makes the TAB character active, equal to __scontents_tabs_to_spaces:, and calls __scontents_xverb:w.

```
⟨*loader⟩
⟨*!context⟩
\use:e
  {
    \cs_new_protected:Npn \exp_not:N \__scontents_xverb:w
      #1 \g__scontents_end_verbatimsc_tl
⟨latex⟩       { #1 \exp_not:N \end{verbatimsc} }
⟨plain⟩       { #1 \exp_not:N \endverbatimsc }
  }
⟨/!context⟩
⟨/loader⟩
⟨*core⟩
\cs_new_protected:Nn \__scontents_xverb:
  {
    \char_set_catcode_active:n { 9 }
    \char_set_active_eq:nN { 9 } \__scontents_tabs_to_spaces:
    \__scontents_xverb:w
  }
⟨/core⟩
```

(*End of definition for* __scontents_xverb:w *and* __scontents_xverb:.)

12.15.3 plain TeX version of verbatimsc

__scontents_verbatimsc_aux:
__scontents_nolig_list:
__scontents_vobeyspaces:
__scontents_xobeysp:

In plain TeX we emulate the verbatim environment of LaTeX.

```
⟨*plain⟩
\cs_new_protected:Nn \__scontents_verbatimsc_aux:
  {
    \skip_vertical:N \parskip
    \dim_zero:N \parindent
    \skip_set:Nn \parfillskip { 0pt plus 1fil }
    \skip_set:Nn \parskip { 0pt plus0pt minus0pt }
    \tex_par:D
    \bool_set_false:N \l__scontents_plain_bool
    \cs_set:Npn \par
      {
        \bool_if:NTF \l__scontents_plain_bool
          {
```

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```
            \mode_leave_vertical:
            \null
            \tex_par:D
            \penalty \interlinepenalty
          }
          {
            \bool_set_true:N \l__scontents_plain_bool
            \mode_if_horizontal:T
              { \tex_par:D \penalty \interlinepenalty }
          }
      }
    \cs_set_eq:NN \do \char_set_catcode_other:N
    \dospecials \obeylines
    \tl_use:N \l__scontents_verb_font_tl
    \cs_set_eq:NN \do \__scontents_do_noligs:N
    \__scontents_nolig_list:
    \tex_everypar:D \exp_after:w
      { \tex_the:D \tex_everypar:D \tex_unpenalty:D }
  }
\cs_new_protected:Nn \__scontents_nolig_list:
  { \do\`\do\<\do\>\do\,\do\'\do\- }
\cs_new_protected:Nn \__scontents_vobeyspaces:
  { \__scontents_set_active_eq:NN \  \__scontents_xobeysp: }
\cs_new_protected:Nn \__scontents_xobeysp:
  { \mode_leave_vertical: \nobreak \ }
```

(*End of definition for* `\__scontents_verbatimsc_aux:` *and others.*)

`\__scontents_verbatimsc_line:w`

The function `\__scontents_verbatimsc_line:w` discards the rest of the line of `\verbatimsc` up to the active end of line. When `\typestored` rescans the ⟨*stored content*⟩ a space follows `\verbatimsc` and, once the spaces are active, it would start an empty line at the top of the environment.

```
\group_begin:
  \char_set_catcode_active:N \^^M
  \cs_new_protected:Npn \__scontents_verbatimsc_line:w #1 ^^M
    {
      \__scontents_verbatimsc_aux: \frenchspacing \__scontents_vobeyspaces:
      \__scontents_xverb:
    }
\group_end:
```

(*End of definition for* `\__scontents_verbatimsc_line:w`.)

`\verbatimsc` `\endverbatimsc`

Defines the environment `verbatimsc` for plain TEX. As in the environments of ConTEXt, the rest of the line of `\verbatimsc` is ignored.

```
\cs_new_protected:Npn \verbatimsc
  {
    \group_begin:
      \char_set_catcode_active:N \^^M
      \__scontents_verbatimsc_line:w
  }
\cs_new_protected:Npn \endverbatimsc
  { \group_end: }
⟨/plain⟩
```

(*End of definition for* `\verbatimsc` *and* `\endverbatimsc`. *These functions are documented on page* 8.)

12.15.4 ConTEXt version of `verbatimsc`

`\startverbatimsc` `\stopverbatimsc`

In ConTEXt we use its own tool `\definetyping`.

```
⟨*loader⟩
⟨context⟩\definetyping[verbatimsc]
⟨/loader⟩
```

(*End of definition for* `\startverbatimsc` *and* `\stopverbatimsc`. *These functions are documented on page* ??.)

12.15.5 LATEX version of `verbatimsc`

`\__scontents_verbatimsc_instance:`

To be compatible with *tagged* PDF the environment `verbatimsc` is defined with the `xtemplate`[11] module of the LATEX kernel and the `blocks-code` of `latex-lab`[14]. The function `\__scontents_verbatimsc_instance:` declares the instance of `blockenv` used by the environment,

only once at the beginning of the document and only if *tagged* PDF is active. This code is adapted from the answer of Ulrike Fischer to New verbatim environment with block code (tagged-pdf).

```
⟨*loader⟩
⟨*latex⟩
\cs_new_protected:Nn \__scontents_verbatimsc_instance:
  {
    \DeclareInstance{blockenv}{verbatimsc}{std}
      {
        name                   = verbatimsc,
        tag-name               = \UseStructureName{block/verbatim},
        tagging-recipe         = standard,
        tagging-suppress-paras = true,
        block-instance         = verbatim,
        para-instance          = justify,
        final-code             = \legacyverbatimsetup{invisible},
      }
  }
\hook_gput_code:nnn { begindocument } { scontents }
  {
    \IfDocumentMetadataTF { \__scontents_verbatimsc_instance: } { }
  }
```

(*End of definition for* `\__scontents_verbatimsc_instance:`.)

`verbatimsc` Defines the environment `verbatimsc` for LaTeX. The `\endverbatim` in the second argument is only needed for compatibility with the `verbatim`[15] package.

```
\NewDocumentEnvironment { verbatimsc } { }
  {
    \IfDocumentMetadataTF
      {
        \UseInstance{blockenv}{verbatimsc}{}
        \@setupverbinvisiblespace\frenchspacing\@vobeyspaces
        \__scontents_xverb:
      }
      {
        \cs_set_eq:cN { @xverbatim } \__scontents_xverb:
        \verbatim
      }
  }
  {
    \IfDocumentMetadataTF
      {
        \BlockEnvEnd
      }
      { \endverbatim }
  }
⟨/latex⟩
⟨/loader⟩
```

(*End of definition for* `verbatimsc`. *This function is documented on page 8.*)

12.16 The command \Scontents

The user-level command `\Scontents` stores its {⟨*argument*⟩} in a *sequence*. It takes an optional *starred argument*, an optional argument with the ⟨*keys*⟩ and the {⟨*argument*⟩}, which is read in standard mode or, with the *starred argument*, under the *verbatim catcode* regimen. It is adapted from code by Ulrich Diez in Stringify input - \string on token list and code by user `@siracusa` in Convert a macro from Latex2e to expl3.

12.16.1 Definition of variables for \Scontents

`\l__scontents_Scontents_arg_tl` The token list `\l__scontents_Scontents_arg_tl` saves the {⟨*argument*⟩} passed to `\Scontents`.

```
⟨*core⟩
\tl_new:N \l__scontents_Scontents_arg_tl
```

(*End of definition for* `\l__scontents_Scontents_arg_tl`.)

12.16.2 Definition of messages for \Scontents

`empty-argument-command` The "warning" `empty-argument-command` is issued when the {⟨*argument*⟩} of `\Scontents` is empty.

```
\msg_new:nnn { scontents } { empty-argument-command }
  { Empty~argument~for~\c_backslash_str Scontents~\msg_line_context:. }
```

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(*End of definition for* `empty-argument-command`.)

12.16.3 Definition of keys for `\Scontents`

write-cmd
write-out
print-cmd
store-cmd
force-eol
overwrite
unknown

We define the ⟨*keys*⟩ for both versions of `\Scontents`, the unknown ⟨*keys*⟩ are passed to `\__scontents_unknown_keys_cmd:n` (§12.9).

```
\keys_define:nn { scontents / Scontents }
  {
    write-cmd .code:n           = { \__scontents_set_write_store:n {#1} },
    write-out .code:n           = { \__scontents_set_write_out:n {#1} },
    write-cmd .value_required:n = true,
    write-out .value_required:n = true,
    print-cmd .meta:nn          = { scontents } { print-cmd = #1 },
    print-cmd .default:n        = true,
    store-cmd .meta:nn          = { scontents } { store-cmd = #1 },
    force-eol .meta:nn          = { scontents } { force-eol = #1 },
    force-eol .default:n        = true,
    overwrite .meta:nn          = { scontents } { overwrite = #1 },
    overwrite .default:n        = true,
    unknown   .code:n           = { \__scontents_unknown_keys_cmd:n {#1} },
  }
```

(*End of definition for* `write-cmd` *and others.*)

12.16.4 Internal functions for `\Scontents`

`\__scontents_Scontents_code:nn`

The function `\__scontents_Scontents_code:nn` executes `\__scontents_bsphack:`, opens a group, sets the ⟨*keys*⟩ passed in `#2` and leaves the `TAB` character active. With the *starred argument* passed in `#1` it calls `\__scontents_Scontents_verb_arg:w`, otherwise `\__scontents_Scontents_norm_arg:n`.

```
\cs_new_protected:Npn \__scontents_Scontents_code:nn #1 #2
  {
    \__scontents_bsphack:
    \group_begin:
      \keys_set:nn { scontents / Scontents } {#2}
      \char_set_catcode_active:n { 9 }
      \bool_if:NTF #1
        { \__scontents_Scontents_verb_arg:w }
        { \__scontents_Scontents_norm_arg:n }
  }
```

(*End of definition for* `\__scontents_Scontents_code:nn`.)

`\__scontents_Scontents_verb_arg:w`

The function `\__scontents_Scontents_verb_arg:w` saves the {⟨*argument*⟩} under the *verbatim catcode* regimen, using a v-type argument of `ltcmd`, in `\l__scontents_Scontents_arg_tl` and replaces every '`^^M`' and `\obeyedline`, added in LaTeX release 2024-06-01, by '`^^J`'. The command `\obeyedline` is redefined with `\RenewDocumentCommand` because it can be modified by the user, which would return a low-level error.

```
\NewDocumentCommand \__scontents_Scontents_verb_arg:w { +v }
  {
    \tl_set:Nn \l__scontents_Scontents_arg_tl {#1}
    \cs_if_exist:NT \obeyedline
      {
        \RenewDocumentCommand \obeyedline { } { \iow_char:N \^^M }
        \tl_replace_all:Nee \l__scontents_Scontents_arg_tl
          { \obeyedline } { \iow_char:N \^^M }
      }
    \tl_replace_all:NeV \l__scontents_Scontents_arg_tl
      { \iow_char:N \^^M } \c__scontents_newline_tl
    \__scontents_Scontents_finish:
  }
```

(*End of definition for* `\__scontents_Scontents_verb_arg:w`.)

`\__scontents_Scontents_norm_arg:n`

The function `\__scontents_Scontents_norm_arg:n` saves the {⟨*argument*⟩} read in the *standard catcode* regimen.

```
\cs_new_protected:Npn \__scontents_Scontents_norm_arg:n #1
  {
    \tl_set:Nn \l__scontents_Scontents_arg_tl {#1}
    \__scontents_Scontents_finish:
  }
```

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(*End of definition for* `\__scontents_Scontents_norm_arg:n`.)

`\__scontents_Scontents_finish:` The function `\__scontents_Scontents_finish:` issues the "warning" `empty-argument-command` if the {⟨*argument*⟩} is empty, writes it with `\__scontents_file_write_cmd:VV` for the `write-out` and `write-cmd` keys and stores and prints it with `\__scontents_finish_storing:NNN`. Finally it closes the group and runs `\__scontents_esphack:` if the `print-cmd` key is not active.

```
\cs_new_protected:Nn \__scontents_Scontents_finish:
  {
    \tl_if_empty:NT \l__scontents_Scontents_arg_tl
      { \msg_warning:nn { scontents } { empty-argument-command } }
    \__scontents_file_write_cmd:VV \l__scontents_file_name_tl \l__scontents_Scontents_arg_tl
    \__scontents_finish_storing:NNN
      \l__scontents_Scontents_arg_tl \l__scontents_name_seq_cmd_tl \l__scontents_print_cmd_bool
    \use:e
      {
        \group_end:
        \bool_if:NF \l__scontents_print_cmd_bool { \__scontents_esphack: }
      }
  }
```

(*End of definition for* `\__scontents_Scontents_finish:`.)

`\Scontents` Defines the user command `\Scontents`.

```
\NewDocumentCommand \Scontents { !s !O{} }
  {
    \tl_set:Nn \l__scontents_cmd_name_tl { Scontents }
    \__scontents_Scontents_code:nn {#1} {#2}
  }
```

(*End of definition for* `\Scontents`. *This function is documented on page* 5.)

12.17 The command `\getstored`

The user-level command `\getstored` takes an optional argument with the ⟨*index*⟩, the last ⟨*stored content*⟩ by default, and a mandatory argument with the {⟨*seq name*⟩}. It is robust.

12.17.1 Internal functions for `\getstored`

`\__scontents_getstored:nn` The function `\__scontents_getstored:nn` rescans with `\tl_retokenize:e` the ⟨*stored content*⟩ at the ⟨*index*⟩ passed in `#1` in the *sequence* named by `#2`, given by `\__scontents_getfrom_seq:nn` (§12.11), unless it is not available yet in a remembered *sequence* (§12.12).

```
\cs_new_protected:Npn \__scontents_getstored:nn #1 #2
  {
    \__scontents_if_pending:neF {#2} { \int_eval:n {#1} }
      {
        \tl_retokenize:e
          {
            \__scontents_getfrom_seq:nn {#1} {#2}
          }
      }
  }
```

(*End of definition for* `\__scontents_getstored:nn`.)

`\getstored` Defines the user command `\getstored`.

```
\NewDocumentCommand \getstored { O{-1} m }
  {
    \__scontents_getstored:nn {#1} {#2}
  }
```

(*End of definition for* `\getstored`. *This function is documented on page* 6.)

12.18 The command `\foreachsc`

The user-level command `\foreachsc` runs `\getstored` on the ⟨*stored contents*⟩ of a *sequence*. It takes an optional argument with the ⟨*keys*⟩ and a mandatory argument with the {⟨*seq name*⟩}, raising the "error" `undefined-storage` if the *sequence* does not exist.

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12.18.1 Definition of variables for \foreachsc

\l__scontents_foreachsc_arg_tl
\l__scontents_foreach_print_seq
\g__scontents_foreach_exec_tl
\l__scontents_foreach_before_tl
\l__scontents_foreach_before_bool
\l__scontents_foreach_after_tl
\l__scontents_foreach_after_bool
\l__scontents_foreach_start_int
\l__scontents_foreach_step_int
\l__scontents_foreach_stop_int
\l__scontents_foreach_stop_bool
\l__scontents_foreach_wrapper_bool
\l__scontents_foreach_body_tl
\l__scontents_foreach_cmd_tl

The token list \l__scontents_foreachsc_arg_tl saves the {⟨*seq name*⟩}, the sequence \l__scontents_foreach_print_seq collects the code for each ⟨*stored content*⟩, built in the token list \l__scontents_foreach_body_tl, and the token list \g__scontents_foreach_exec_tl executes it outside the group. The token list \l__scontents_foreach_cmd_tl saves the command executed on each ⟨*stored content*⟩, \getstored by default. The other variables are set by the ⟨*keys*⟩ of \foreachsc.

```
\tl_new:N    \l__scontents_foreachsc_arg_tl
\seq_new:N   \l__scontents_foreach_print_seq
\tl_new:N    \g__scontents_foreach_exec_tl
\tl_new:N    \l__scontents_foreach_before_tl
\bool_new:N  \l__scontents_foreach_before_bool
\tl_new:N    \l__scontents_foreach_after_tl
\bool_new:N  \l__scontents_foreach_after_bool
\int_new:N   \l__scontents_foreach_start_int
\int_new:N   \l__scontents_foreach_step_int
\int_new:N   \l__scontents_foreach_stop_int
\bool_new:N  \l__scontents_foreach_stop_bool
\bool_new:N  \l__scontents_foreach_wrapper_bool
\tl_new:N    \l__scontents_foreach_body_tl
\tl_new:N    \l__scontents_foreach_cmd_tl
\tl_set:Nn   \l__scontents_foreach_cmd_tl { \getstored }
```

(*End of definition for* \l__scontents_foreachsc_arg_tl *and others.*)

12.18.2 Definition of keys for \foreachsc

before
after
start
stop
step
wrapper
sep
unknown

We define the ⟨*keys*⟩ for \foreachsc, the keys start, step and stop use __scontents_set_positive_int:Nnn (§12.8) and the unknown ⟨*keys*⟩ are passed to __scontents_unknown_keys_cmd:n (§12.9).

```
\keys_define:nn { scontents / foreachsc }
  {
    before  .code:n           = {
                                  \bool_set_true:N \l__scontents_foreach_before_bool
                                  \tl_set:Nn \l__scontents_foreach_before_tl {#1}
                                },
    before  .value_required:n = true,
    after   .code:n           = {
                                  \bool_set_true:N \l__scontents_foreach_after_bool
                                  \tl_set:Nn \l__scontents_foreach_after_tl {#1}
                                },
    after   .value_required:n = true,
    start   .code:n           = {
                                  \__scontents_set_positive_int:Nnn
                                    \l__scontents_foreach_start_int { start } {#1}
                                },
    start   .value_required:n = true,
    start   .initial:n        = 1,
    stop    .code:n           = {
                                  \__scontents_set_positive_int:Nnn
                                    \l__scontents_foreach_stop_int { stop } {#1}
                                  \bool_set:Nn \l__scontents_foreach_stop_bool
                                    {
                                      \int_compare_p:nNn
                                        { \l__scontents_foreach_stop_int } > { 0 }
                                    }
                                },
    stop    .value_required:n = true,
    step    .code:n           = {
                                  \__scontents_set_positive_int:Nnn
                                    \l__scontents_foreach_step_int { step } {#1}
                                },
    step    .value_required:n = true,
    step    .initial:n        = 1,
    wrapper .code:n           = {
                                  \bool_set_true:N \l__scontents_foreach_wrapper_bool
                                  \cs_set_protected:Npn
                                    \__scontents_foreach_wrapper:n ##1 {#1}
                                },
    wrapper .value_required:n = true,
    sep     .tl_set:N         = \l__scontents_foreach_sep_tl,
```

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```
    sep      .initial:n        = { },
    sep      .value_required:n = true,
    unknown  .code:n           = { \__scontents_unknown_keys_cmd:n {#1} },
  }
```

(*End of definition for* `before` *and others.*)

getstored
typestored
meaningsc

The ⟨*keys*⟩ `getstored`, `typestored` and `meaningsc` set the command saved in `\l__scontents_foreach_cmd_tl`.

```
\keys_define:nn { scontents / foreachsc }
  {
    getstored  .code:n            =
      { \tl_set:Nn \l__scontents_foreach_cmd_tl { \getstored } },
    getstored  .value_forbidden:n = true,
    typestored .code:n            =
      { \tl_set:Nn \l__scontents_foreach_cmd_tl { \typestored } },
    typestored .value_forbidden:n = true,
    meaningsc  .code:n            =
      { \tl_set:Nn \l__scontents_foreach_cmd_tl { \meaningsc } },
    meaningsc  .value_forbidden:n = true,
  }
```

(*End of definition for* `getstored`, `typestored`, *and* `meaningsc`.)

12.18.3 Internal functions for `\foreachsc`

__scontents_foreachsc:nn
__scontents_foreach_add_body:n
__scontents_foreach_wrap:n
__scontents_foreach_wrap:V

The function `\__scontents_foreachsc:nn` sets the ⟨*keys*⟩ passed in `#1` and steps from `start` to `stop`, the number of ⟨*stored contents*⟩ by default, with `\__scontents_foreach_add_body:n`, which adds the command saved in `\l__scontents_foreach_cmd_tl`, with the code of the `before`, `after` and `wrapper` keys, to `\l__scontents_foreach_print_seq`. The result, separated by `sep`, is executed after the group. The function `\__scontents_foreach_wrap:n` expands `\__scontents_foreach_wrapper:n` once inside the group, so the code of the `wrapper` key is already in place when the result is executed; the function `\__scontents_foreach_wrapper:n` is local and no longer exists after the group.

```
\cs_new_protected:Npn \__scontents_foreachsc:nn #1 #2
  {
    \group_begin:
      \tl_if_novalue:nF {#1} { \keys_set:nn { scontents / foreachsc } {#1} }
      \tl_set:Nn \l__scontents_foreachsc_arg_tl {#2}
      \seq_clear:N \l__scontents_foreach_print_seq
      \bool_if:NF \l__scontents_foreach_stop_bool
        {
          \int_set:Nn \l__scontents_foreach_stop_int
            { \__scontents_seq_count:n {#2} }
        }
      \int_step_function:nnnN
        { \l__scontents_foreach_start_int }
        { \l__scontents_foreach_step_int }
        { \l__scontents_foreach_stop_int }
        \__scontents_foreach_add_body:n
      \tl_gset:Ne \g__scontents_foreach_exec_tl
        {
          \exp_args:NNV \seq_use:Nn
            \l__scontents_foreach_print_seq \l__scontents_foreach_sep_tl
        }
    \group_end:
    \exp_after:w \tl_gclear:N
    \exp_after:w \g__scontents_foreach_exec_tl
    \g__scontents_foreach_exec_tl
  }
\cs_new_protected:Npn \__scontents_foreach_add_body:n #1
  {
    \tl_set:Ne \l__scontents_foreach_body_tl
      {
        \exp_not:V \l__scontents_foreach_cmd_tl [#1]
          { \exp_not:V \l__scontents_foreachsc_arg_tl }
      }
    \bool_if:NT \l__scontents_foreach_wrapper_bool
      { \__scontents_foreach_wrap:V \l__scontents_foreach_body_tl }
    \seq_put_right:Ne \l__scontents_foreach_print_seq
      {
```

```
        \bool_if:NT \l__scontents_foreach_before_bool
          { \exp_not:V \l__scontents_foreach_before_tl }
        \exp_not:V \l__scontents_foreach_body_tl
        \bool_if:NT \l__scontents_foreach_after_bool
          { \exp_not:V \l__scontents_foreach_after_tl }
      }
  }
\cs_new_protected:Npn \__scontents_foreach_wrap:n #1
  { \tl_set:No \l__scontents_foreach_body_tl { \__scontents_foreach_wrapper:n {#1} } }
\cs_generate_variant:Nn \__scontents_foreach_wrap:n { V }
```

(*End of definition for* `\__scontents_foreachsc:nn`, `\__scontents_foreach_add_body:n`, *and* `\__scontents_foreach_wrap:n`.)

`\foreachsc` Defines the user command `\foreachsc`.

```
\NewDocumentCommand \foreachsc { +o m }
  {
    \tl_set:Nn \l__scontents_cmd_name_tl { foreachsc }
    \__scontents_seq_if_exist:nTF {#2}
      { \__scontents_foreachsc:nn {#1} {#2} }
      { \msg_error:nnn { scontents } { undefined-storage } {#2} }
  }
```

(*End of definition for* `\foreachsc`. *This function is documented on page 6.*)

12.19 The optional argument of `\typestored`, `\meaningsc` and `\mergesc`

The commands `\typestored` and `\meaningsc`, and `\mergesc` which uses them internally, accept a single *optional argument* with the ⟨*index*⟩, ⟨*1-end*⟩ or the range ⟨*start-stop*⟩ of the ⟨*stored contents*⟩ in the *sequence* along with the ⟨*keys*⟩, this avoids the awkward `\typestored[...][`⟨*keys*⟩`]{...}` syntax. The ⟨*indexes*⟩ and ranges are processed as unknown ⟨*keys*⟩ of `l3keys`[16, 17].

12.19.1 Definition of variables for the optional argument

`\l__scontents_current_seq_name_str` `\l__scontents_index_str` `\l__scontents_range_start_tl` `\l__scontents_range_stop_tl`

The string variable `\l__scontents_current_seq_name_str` stores the name of the *sequence* passed to `\typestored` or `\meaningsc`, it is used by `\__scontents_parse_type_meaning_key:nn`. The string variable `\l__scontents_index_str` is used to check that an ⟨*index*⟩ only has digits and the token lists `\l__scontents_range_start_tl` and `\l__scontents_range_stop_tl` save the limits of a range ⟨*start-stop*⟩.

```
\str_new:N \l__scontents_current_seq_name_str
\str_new:N \l__scontents_index_str
\tl_new:N  \l__scontents_range_start_tl
\tl_new:N  \l__scontents_range_stop_tl
```

(*End of definition for* `\l__scontents_current_seq_name_str` *and others.*)

12.19.2 Definition of keys for the optional argument

`width-tab` `write-out` `overwrite` `print-cmd` `unknown`

We define the common ⟨*keys*⟩ for `\typestored` and `\meaningsc`. The `write-out`, `overwrite` and `print-cmd` keys are necessary for the implementation of `\mergesc` (§12.22) and the unknown ⟨*keys*⟩ are passed to `\__scontents_parse_type_meaning_key:n`.

```
\keys_define:nn { scontents / typemeaning }
  {
    width-tab .meta:nn          = { scontents } { width-tab = #1 },
    write-out .code:n           = { \__scontents_set_write_out:n {#1} },
    write-out .value_required:n = true,
    overwrite .meta:nn          = { scontents } { overwrite = #1 },
    overwrite .default:n        = true,
    print-cmd .bool_set:N       = \l__scontents_print_verb_style_bool,
    print-cmd .initial:n        = true,
    print-cmd .default:n        = true,
    unknown   .code:n           = { \__scontents_parse_type_meaning_key:n {#1} },
  }
```

(*End of definition for* `width-tab` *and others.*)

12.19.3 Internal functions for the optional argument

`\__scontents_parse_type_meaning_key:n` `\__scontents_parse_type_meaning_key:nn` `\__scontents_range_parser:nnn` `\__scontents_range_parser:nnnw` `\__scontents_range_limit:nn`

The function `\__scontents_parse_type_meaning_key:n` checks that the unknown ⟨*key*⟩ has no value, otherwise something odd as `1=1` was used and the "error" `cmd-key-value-unknown` is raised. The conditional `\__scontents_if_digits:nTF` tests if `#1` is not blank and only has digits, it removes

the digits from \l__scontents_index_str, so a text that is not a number never reaches an *integer expression* and only the "error" of the package is raised.

The function __scontents_range_parser:nnn receives the ⟨*index*⟩ or the range in #1, the number of ⟨*stored contents*⟩ in #2 and the code that raises the "error" in #3, and splits #1 at the hyphens with __scontents_range_parser:nnnw: without hyphen it is an ⟨*index*⟩, with a hyphen at the beginning it is a negative ⟨*index*⟩ and otherwise it is a range ⟨*start-stop*⟩, where __scontents_range_limit:nn replaces **end** by the number of ⟨*stored contents*⟩ and __scontents_range_step:nn adds the ⟨*indexes*⟩ of the range, ascending or descending, to \l__scontents_seq_item_seq. Anything else is passed to the code in #3.

```
\cs_new_protected:Npn \__scontents_parse_type_meaning_key:n #1
  { \exp_args:NV \__scontents_parse_type_meaning_key:nn \l_keys_key_str {#1} }
\cs_new_protected:Npn \__scontents_parse_type_meaning_key:nn #1 #2
  {
    \tl_if_blank:nTF {#2}
      {
        \__scontents_range_parser:nnn {#1}
          { \__scontents_seq_count:n { \l__scontents_current_seq_name_str } }
          { \msg_error:nnn { scontents } { cmd-key-unknown } }
      }
      { \msg_error:nnnn { scontents } { cmd-key-value-unknown } {#1} {#2} }
  }
\prg_new_protected_conditional:Npnn \__scontents_if_digits:n #1 { TF }
  {
    \str_set:Ne \l__scontents_index_str { \tl_trim_spaces:n {#1} }
    \clist_map_inline:nn { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 }
      { \str_remove_all:Nn \l__scontents_index_str {##1} }
    \tl_if_blank:nTF {#1}
      { \prg_return_false: }
      {
        \str_if_empty:NTF \l__scontents_index_str
          { \prg_return_true: }
          { \prg_return_false: }
      }
  }
\prg_generate_conditional_variant:Nnn \__scontents_if_digits:n { V } { TF }
\cs_new_protected:Npn \__scontents_range_parser:nnn #1 #2 #3
  { \__scontents_range_parser:nnnw {#1} {#2} {#3} #1 - \q__scontents_mark - \s__scontents_mark }
\cs_new_protected:Npn \__scontents_range_parser:nnnw #1 #2 #3 #4 - #5 - #6 \s__scontents_mark
  {
    \__scontents_tl_if_head_is_q_mark:nTF {#5}
      {
        \__scontents_if_digits:nTF {#4}
          { \seq_put_right:Ne \l__scontents_seq_item_seq { \int_eval:n {#4} } }
          { #3 {#1} }
      }
      {
        \__scontents_tl_if_head_is_q_mark:nTF {#6}
          {
            \tl_if_blank:nTF {#4}
              {
                \__scontents_if_digits:nTF {#5}
                  {
                    \seq_put_right:Ne \l__scontents_seq_item_seq
                      { \int_eval:n { - #5 } }
                  }
                  { #3 {#1} }
              }
              {
                \tl_set:Ne \l__scontents_range_start_tl
                  { \__scontents_range_limit:nn {#4} {#2} }
                \tl_set:Ne \l__scontents_range_stop_tl
                  { \__scontents_range_limit:nn {#5} {#2} }
                \__scontents_if_digits:VTF \l__scontents_range_start_tl
                  {
                    \__scontents_if_digits:VTF \l__scontents_range_stop_tl
                      {
                        \exp_args:NVV \__scontents_range_step:nn
                          \l__scontents_range_start_tl \l__scontents_range_stop_tl
                      }
                      { #3 {#1} }
```

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```
                    }
                    { #3 {#1} }
                }
            }
            { #3 {#1} }
        }
    }
\cs_new:Npn \__scontents_range_limit:nn #1 #2
  {
    \str_if_eq:eeTF { \tl_trim_spaces:n {#1} } { end }
      {#2}
      { \tl_trim_spaces:n {#1} }
  }
\cs_new_protected:Npn \__scontents_range_step:nn #1 #2
  {
    \int_step_inline:nnnn {#1}
      { \int_compare:nNnTF {#2} < {#1} { -1 } { 1 } }
      {#2}
      { \seq_put_right:Nn \l__scontents_seq_item_seq {##1} }
  }
```

(*End of definition for* `\__scontents_parse_type_meaning_key:n` *and others.*)

`\__scontents_typemeaning_setup:nn` `\l__scontents_pending_bool`

The function `\__scontents_typemeaning_setup:nn` is the common start of `\typestored` and `\meaningsc`, it saves the name of the *sequence* passed in `#2`, sets the ⟨*keys*⟩ and the ⟨*indexes*⟩ passed in `#1` and uses the first ⟨*stored content*⟩ if no ⟨*index*⟩ was given. The boolean `\l__scontents_pending_bool` is true if any of them is not available yet in a remembered *sequence* (§12.12).

```
\bool_new:N \l__scontents_pending_bool
\cs_new_protected:Npn \__scontents_typemeaning_setup:nn #1 #2
  {
    \seq_clear:N \l__scontents_seq_item_seq
    \str_set:Ne \l__scontents_current_seq_name_str {#2}
    \tl_if_novalue:nF {#1}
      { \keys_set:nn { scontents / typemeaning } {#1} }
    \seq_if_empty:NT \l__scontents_seq_item_seq
      { \seq_set_from_clist:Nn \l__scontents_seq_item_seq { 1 } }
    \bool_set_false:N \l__scontents_pending_bool
    \seq_map_inline:Nn \l__scontents_seq_item_seq
      {
        \__scontents_if_pending:nnT {#2} {##1}
          { \bool_set_true:N \l__scontents_pending_bool }
      }
  }
```

(*End of definition for* `\__scontents_typemeaning_setup:nn` *and* `\l__scontents_pending_bool`.)

12.20 The command `\typestored`

The user-level command `\typestored` prints the ⟨*stored contents*⟩ of a *sequence* in the `verbatimsc` environment (§12.15). It takes an optional argument (§12.19) and a mandatory argument with the {⟨*seq name*⟩}.

12.20.1 Definition of variables for `\typestored`

`\l__scontents_typestored_arg_tl`

The token list `\l__scontents_typestored_arg_tl` saves the ⟨*stored contents*⟩ to print.

```
\tl_new:N \l__scontents_typestored_arg_tl
```

(*End of definition for* `\l__scontents_typestored_arg_tl`.)

12.20.2 Definition of error messages for `\typestored`

`empty-variable`

The "error" `empty-variable` is raised when the ⟨*stored contents*⟩ to print are blank.

```
\msg_new:nnnn { scontents } { empty-variable }
  { Blank~stored~content~to~print~\msg_line_context:. }
  {
    The~stored~content~only~has~spaces,~so~there~is~nothing~to~print~
    in~verbatim~style.~Check~the~stored~content~or~the~index.
  }
```

(*End of definition for* `empty-variable`.)

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12.20.3 Internal functions for \typestored

__scontents_typestored:nn

The function __scontents_typestored:nn sets the ⟨*keys*⟩ and the ⟨*indexes*⟩ passed in #1 with __scontents_typemeaning_setup:nn, gets the ⟨*stored contents*⟩ of the *sequence* named by #2, writes them to the .log file if debug-var is true and prints them with __scontents_typestored:N and writes them with __scontents_file_write_cmd:VV.

```
\cs_new_protected:Npn \__scontents_typestored:nn #1 #2
  {
    \__scontents_bsphack:
    \group_begin:
      \__scontents_typemeaning_setup:nn {#1} {#2}
      \bool_if:NF \l__scontents_pending_bool
        {
          \tl_set:Ne \l__scontents_typestored_arg_tl
            { \__scontents_getfrom_seq:Nn \l__scontents_seq_item_seq {#2} }
          \__scontents_remove_trailing_eol:N \l__scontents_typestored_arg_tl
          \tl_replace_all:NeV \l__scontents_typestored_arg_tl
            { \c__scontents_hidden_space_str } \c__scontents_newline_tl
          \bool_if:NT \l__scontents_debug_var_bool
            {
              \tl_log:N \l__scontents_typestored_arg_tl
            }
          \tl_if_empty:NF \l__scontents_typestored_arg_tl
            {
              \bool_if:NT \l__scontents_print_verb_style_bool
                {
                  \__scontents_typestored:N \l__scontents_typestored_arg_tl
                }
            }
          \__scontents_file_write_cmd:VV \l__scontents_file_name_tl \l__scontents_typestored_arg_
        }
      \use:e
        {
          \group_end:
          \bool_if:NF \l__scontents_print_verb_style_bool { \__scontents_esphack: }
        }
  }
```

(*End of definition for* __scontents_typestored:nn.)

__scontents_typestored:N
__scontents_verb_print_EOL:

The function __scontents_typestored:N is defined with active carriage return characters to mimic an actual verbatim environment "on the loose". The content of the token list passed in #1 is placed in a verbatimsc environment and rescanned with \tl_retokenize:e.

```
\group_begin:
  \char_set_catcode_active:N \^^M
  \cs_new_protected:Npn \__scontents_typestored:N #1
    {
      \tl_if_blank:VT #1
        { \msg_error:nn { scontents } { empty-variable } }
      \cs_set_eq:NN \__scontents_verb_print_EOL: ^^M
      \cs_set_eq:NN ^^M \scan_stop:
      \cs_set_eq:cN { do@noligs } \__scontents_do_noligs:N
      \int_set:Nn \tex_newlinechar:D { `\^^J }
      \tl_retokenize:e
        {
          \__scontents_format_case:nnn
            { \exp_not:N \begin{verbatimsc} } % LaTeX
            { \verbatimsc } % Plain/Generic
            { \startverbatimsc } % ConTeXt
            ^^M
          \exp_not:V #1 ^^M
          \g__scontents_end_verbatimsc_tl
        }
      \cs_set_eq:NN ^^M \__scontents_verb_print_EOL:
    }
\group_end:
```

(*End of definition for* __scontents_typestored:N *and* __scontents_verb_print_EOL:.)

\typestored

Defines the user command \typestored.

```
\NewDocumentCommand \typestored { o m }
```

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```
  {
    \tl_set:Nn \l__scontents_cmd_name_tl { typestored }
    \__scontents_typestored:nn {#1} {#2}
  }
```

(*End of definition for* `\typestored`. *This function is documented on page* 7.)

12.21 The command \meaningsc

The user-level command `\meaningsc` prints the meaning of the ⟨*stored contents*⟩ of a *sequence*. It takes the same arguments as `\typestored` (§12.20).

12.21.1 Definition of variables for \meaningsc

`\l__scontents_meaningsc_arg_tl` The token list `\l__scontents_meaningsc_arg_tl` saves the ⟨*stored contents*⟩ to print.

```
\tl_new:N \l__scontents_meaningsc_arg_tl
```

(*End of definition for* `\l__scontents_meaningsc_arg_tl`.)

12.21.2 Internal functions for \meaningsc

`\__scontents_meaningsc:nn` The function `\__scontents_meaningsc:nn` sets the ⟨*keys*⟩ and the ⟨*indexes*⟩ passed in `#1` with `\__scontents_typemeaning_setup:nn` and calls `\__scontents_meaningsc:n` with the *sequence* named by `#2`.

```
\cs_new_protected:Npn \__scontents_meaningsc:nn #1 #2
  {
    \__scontents_bsphack:
    \group_begin:
      \__scontents_typemeaning_setup:nn {#1} {#2}
      \bool_if:NF \l__scontents_pending_bool { \__scontents_meaningsc:n {#2} }
      \use:e
        {
          \group_end:
          \bool_if:NF \l__scontents_print_verb_style_bool { \__scontents_esphack: }
        }
  }
```

(*End of definition for* `\__scontents_meaningsc:nn`.)

`\__scontents_meaningsc:n` The function `\__scontents_meaningsc:n` is defined with the TAB character active. It gets the ⟨*stored contents*⟩, replaces the ends of line and `\c__scontents_hidden_space_str` by spaces and each TAB character, active or read from a remembered *sequence* (§12.12), by as many spaces as the `width-tab` key was set to, prints them with `\cs_replacement_spec:N` and writes them with `\__scontents_file_write_cmd:VV`.

```
\group_begin:
  \char_set_catcode_active:N \^^I
  \cs_new_protected:Npn \__scontents_meaningsc:n #1
    {
      \tl_set:Ne \l__scontents_meaningsc_arg_tl
        { \__scontents_getfrom_seq:Nn \l__scontents_seq_item_seq {#1} }
      \tl_replace_all:Nen \l__scontents_meaningsc_arg_tl
        { \iow_char:N \^^J } { ~ }
      \tl_replace_all:Nen \l__scontents_meaningsc_arg_tl
        { \c__scontents_hidden_space_str } { ~ }
      \bool_if:NT \l__scontents_debug_var_bool
        {
          \tl_log:N \l__scontents_meaningsc_arg_tl
        }
      \tl_use:N \l__scontents_verb_font_tl
      \tl_replace_all:Nne \l__scontents_meaningsc_arg_tl
        { ^^I } { \prg_replicate:nn { \l__scontents_tab_width_int } { ~ } }
      \tl_replace_all:Nee \l__scontents_meaningsc_arg_tl
        { \iow_char:N \^^I }
        { \prg_replicate:nn { \l__scontents_tab_width_int } { ~ } }
      \tl_if_empty:NF \l__scontents_meaningsc_arg_tl
        {
          \bool_if:NT \l__scontents_print_verb_style_bool
            {
              \cs_replacement_spec:N \l__scontents_meaningsc_arg_tl
            }
        }
```

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```
      \__scontents_file_write_cmd:VV \l__scontents_file_name_tl \l__scontents_meaningsc_arg_tl
    }
\group_end:
```

(*End of definition for* `\__scontents_meaningsc:n`.)

`\meaningsc` Defines the user command `\meaningsc`.

```
\NewDocumentCommand \meaningsc { o m }
  {
    \tl_set:Nn \l__scontents_cmd_name_tl { meaningsc }
    \__scontents_meaningsc:nn {#1} {#2}
  }
```

(*End of definition for* `\meaningsc`. *This function is documented on page* 7.)

12.22 The command \mergesc

The user-level command `\mergesc` takes an optional argument with the ⟨*keys*⟩ and a mandatory argument with a *comma separated list* of {⟨*seq name*⟩} followed by an optional ⟨*index*⟩ or range. It assembles the ⟨*stored contents*⟩ in a temporary *internal sequence* and passes it to `\typestored` or `\meaningsc`, or rescans it as `\getstored` does.

12.22.1 Definition of variables for \mergesc

`\l__scontents_mergesc_arg_tl` `\l__scontents_mergesc_keys_tl` The token list `\l__scontents_mergesc_arg_tl` saves the last ⟨*stored content*⟩ assembled and the token list `\l__scontents_mergesc_keys_tl` saves the ⟨*keys*⟩ passed to `\typestored` or `\meaningsc`.

```
\tl_new:N \l__scontents_mergesc_arg_tl
\tl_new:N \l__scontents_mergesc_keys_tl
```

(*End of definition for* `\l__scontents_mergesc_arg_tl` *and* `\l__scontents_mergesc_keys_tl`.)

12.22.2 Definition of error messages for \mergesc

mergesc-missing-key The "error" mergesc-missing-key is raised when none of the typestored, meaningsc and getstored keys is passed to `\mergesc`.

```
\msg_new:nnnn { scontents } { mergesc-missing-key }
  {
    Missing~mandatory~key~'typestored',~'meaningsc'~or~'getstored'~
    for~\c_backslash_str mergesc~\msg_line_context:.
  }
  {
    The~command~\c_backslash_str mergesc~needs~one~of~the~keys~
    'typestored',~'meaningsc'~or~'getstored'.\\
    Check~that~you~have~spelled~the~key~name~correctly.
  }
```

(*End of definition for* mergesc-missing-key.)

12.22.3 Definition of keys for \mergesc

typestored meaningsc getstored `\__scontents_mergesc_cmd:nn` The ⟨*keys*⟩ typestored, meaningsc and getstored are *mandatory*, they set the function `\__scontents_mergesc_cmd:nn`, which by default raises the "error" mergesc-missing-key.

```
\keys_define:nn { scontents / mergesc }
  {
    typestored .code:n            =
      {
        \cs_set_eq:NN \__scontents_mergesc_cmd:nn \__scontents_typestored:nn
      },
    typestored .value_forbidden:n = true,
    meaningsc  .code:n            =
      {
        \cs_set_eq:NN \__scontents_mergesc_cmd:nn \__scontents_meaningsc:nn
      },
    meaningsc  .value_forbidden:n = true,
    getstored  .code:n            =
      {
        \cs_set_eq:NN \__scontents_mergesc_cmd:nn \__scontents_mergesc_getstored:nn
      },
    getstored  .value_forbidden:n = true,
  }
\cs_new_protected:Npn \__scontents_mergesc_cmd:nn #1 #2
```

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```
  {
    \msg_error:nn { scontents } { mergesc-missing-key }
  }
```

(*End of definition for* `typestored` *and others.*)

12.22.4 Internal functions for `\mergesc`

`\__scontents_mergesc_code:nn`

The function `\__scontents_mergesc_code:nn` sets the known ⟨*keys*⟩ passed in `#1`, saves the others in `\l__scontents_mergesc_keys_tl`, assembles the *internal sequence* from the list passed in `#2` and, if it is not empty, passes it to `\__scontents_mergesc_cmd:nn`.

```
\cs_new_protected:Npn \__scontents_mergesc_code:nn #1 #2
  {
    \group_begin:
      \tl_clear:N \l__scontents_mergesc_keys_tl
      \tl_if_novalue:nF {#1}
        {
          \keys_set_known:nnN { scontents / mergesc } {#1}
            \l__scontents_mergesc_keys_tl
        }
      \seq_gclear:c { g__scontents_name_sc!internal_seq }
      \__scontents_mergesc_parse_list:n {#2}
      \seq_if_empty:cF { g__scontents_name_sc!internal_seq }
        {
          \exp_args:Ne \__scontents_mergesc_cmd:nn
            { 1-end, \exp_not:V \l__scontents_mergesc_keys_tl } { sc!internal }
        }
    \group_end:
  }
```

(*End of definition for* `\__scontents_mergesc_code:nn`.)

`\__scontents_mergesc_parse_list:n` `\__scontents_mergesc_parse_item:n` `\__scontents_parse_mergesc:nw` `\__scontents_parse_mergesc_aux:nw` `\__scontents_mergesc_put_item:nn`

The function `\__scontents_mergesc_parse_list:n` parses each item of the list with `\__scontents_mergesc_parse_item:n`, which passes the {⟨*seq name*⟩} between braces to `\__scontents_parse_mergesc:nw` when the item has no ⟨*index*⟩, because the *comma separated list* removes the braces around an item. The function `\__scontents_parse_mergesc:nw` raises the "error" `undefined-storage` if the *sequence* does not exist and uses all the ⟨*stored contents*⟩ of the *sequence* if there is no ⟨*index*⟩. The function `\__scontents_parse_mergesc_aux:nw` uses `\__scontents_range_parser:nnn` (§12.19) and `\__scontents_mergesc_put_item:nn` adds each ⟨*stored content*⟩ to the *internal sequence* or raises the "error" `index-out-of-range`, except for the ⟨*stored contents*⟩ that are not available yet in a remembered *sequence* (§12.12). Finally the end of line of the last ⟨*stored content*⟩ is removed with `\__scontents_remove_trailing_eol:N`.

```
\cs_new_protected:Npn \__scontents_mergesc_parse_list:n #1
  {
    \clist_map_inline:nn {#1} { \__scontents_mergesc_parse_item:n {##1} }
    \seq_if_empty:cF { g__scontents_name_sc!internal_seq }
      {
        \seq_gpop_right:cN { g__scontents_name_sc!internal_seq } \l__scontents_mergesc_arg_tl
        \__scontents_remove_trailing_eol:N \l__scontents_mergesc_arg_tl
        \seq_gput_right:cV { g__scontents_name_sc!internal_seq } \l__scontents_mergesc_arg_tl
      }
  }
\cs_new_protected:Npn \__scontents_mergesc_parse_item:n #1
  {
    \tl_if_in:nnTF {#1} { [ } % ]
      { \__scontents_parse_mergesc:nw #1 \s__scontents_stop }
      { \__scontents_parse_mergesc:nw {#1} \s__scontents_stop }
  }
\cs_new_protected:Npn \__scontents_parse_mergesc:nw #1
  {
    \__scontents_seq_if_exist:nTF {#1}
      {
        \peek_remove_spaces:n
          {
            \peek_charcode:NTF [ % ]
              { \__scontents_parse_mergesc_aux:nw {#1} }
              { \__scontents_parse_mergesc_aux:nw {#1} [ 1-end ] }
          }
      }
      {
```

```
        \msg_error:nnn { scontents } { undefined-storage } {#1}
        \__scontents_use_none_delimit_by_s_stop:w
      }
  }
\cs_new_protected:Npn \__scontents_parse_mergesc_aux:nw #1 [#2]
  {
    \seq_clear:N \l__scontents_seq_item_seq
    \clist_map_inline:nn {#2}
      {
        \__scontents_range_parser:nnn {##1} { \__scontents_seq_count:n {#1} }
          { \msg_error:nnn { scontents } { cmd-key-unknown } }
      }
    \seq_map_inline:Nn \l__scontents_seq_item_seq
      { \__scontents_mergesc_put_item:nn {#1} {##1} }
    \__scontents_use_none_delimit_by_s_stop:w
  }
\cs_new_protected:Npn \__scontents_mergesc_put_item:nn #1 #2
  {
    \__scontents_if_pending:nnF {#1} {#2}
      {
        \bool_lazy_or:nnTF
          { \int_compare_p:nNn {#2} = { 0 } }
          {
            \int_compare_p:nNn { \int_abs:n {#2} }
              > { \__scontents_seq_count:n {#1} }
          }
          {
            \msg_error:nneee { scontents } { index-out-of-range }
              {#2} {#1} { \__scontents_seq_count:n {#1} }
          }
          {
            \seq_gput_right:ce { g__scontents_name_sc!internal_seq }
              { \__scontents_seq_item:nn {#1} {#2} }
          }
      }
  }
```

(*End of definition for* `\__scontents_mergesc_parse_list:n` *and others.*)

`\__scontents_mergesc_getstored:nn`

The function `\__scontents_mergesc_getstored:nn`, set by the `getstored` key, joins the ⟨*stored contents*⟩ of the *internal sequence* passed in `#2` in `\g__scontents_last_stored_tl` separated by an end of line, adds `\c__scontents_hidden_space_str` at the end, because the last one has no end of line, and rescans them with `\tl_retokenize:V` after the group of `\mergesc`, as `\__scontents_lastfrom_seq:n` does (§12.11). The result is the same as using `\getstored` on each ⟨*stored content*⟩. The ⟨*indexes*⟩ passed in `#1` are always `1`-end and the other ⟨*keys*⟩, saved in `\l__scontents_mergesc_keys_tl`, are unknown for it and raise the "errors" of the unknown ⟨*keys*⟩ (§12.9).

```
\keys_define:nn { scontents / getstored }
  {
    unknown .code:n = { \__scontents_unknown_keys_cmd:n {#1} },
  }
\cs_new_protected:Npn \__scontents_mergesc_getstored:nn #1 #2
  {
    \keys_set:nV { scontents / getstored } \l__scontents_mergesc_keys_tl
    \tl_gset:Ne \g__scontents_last_stored_tl
      {
        \exp_args:NnV \seq_use:cn { g__scontents_name_#2_seq } \c__scontents_newline_tl
        \c__scontents_hidden_space_str
      }
    \group_insert_after:N \tl_retokenize:V
    \group_insert_after:N \g__scontents_last_stored_tl
    \group_insert_after:N \tl_gclear:N
    \group_insert_after:N \g__scontents_last_stored_tl
  }
```

(*End of definition for* `\__scontents_mergesc_getstored:nn`.)

`\mergesc`

Defines the user command `\mergesc`.

```
\NewDocumentCommand \mergesc { o m }
  {
    \tl_set:Nn \l__scontents_cmd_name_tl { mergesc }
```

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```
    \__scontents_mergesc_code:nn {#1} {#2}
  }
```

(*End of definition for* `\mergesc`. *This function is documented on page 7.*)

12.23 The command \setupsc

`\setupsc` Defines the user command `\setupsc`, which sets the ⟨*keys*⟩ of the package (§12.8).

```
\NewDocumentCommand \setupsc { +m }
  {
    \tl_set:Nn \l__scontents_cmd_name_tl { setupsc }
    \keys_set:nn { scontents } {#1}
  }
```

(*End of definition for* `\setupsc`. *This function is documented on page 3.*)

12.24 The command \countsc

`\countsc` Defines the user command `\countsc`, which counts the ⟨*stored contents*⟩ in the *sequence*. If the *sequence* does not exist, the "error" `undefined-storage` is raised and `0` is returned.

```
\NewExpandableDocumentCommand \countsc { m }
  {
    \__scontents_seq_if_exist:nTF {#1}
      { \__scontents_seq_count:n {#1} }
      {
        \msg_expandable_error:nnn { scontents } { undefined-storage } {#1}
        0
      }
  }
```

(*End of definition for* `\countsc`. *This function is documented on page 8.*)

12.25 The command \cleanseqsc

`\cleanseqsc` Defines the user command `\cleanseqsc`, which removes all the ⟨*stored contents*⟩ in the *sequence*.

```
\NewDocumentCommand \cleanseqsc { m }
  {
    \seq_gclear_new:c { g__scontents_name_#1_seq }
  }
⟨/core⟩
```

(*End of definition for* `\cleanseqsc`. *This function is documented on page 8.*)

12.26 Programming interface

The public functions of SCONTENTS for `expl3` (§6.3), they use the functions for sequences (§12.11).

12.26.1 Definition of variables for the programming interface

`\g__scontents_map_int` `\l__scontents_map_item_tl` The integer variable `\g__scontents_map_int` is the nesting level of `\scontents_map_inline:nn`, used in the name of the function that saves each {⟨*inline function*⟩}, and the token list `\l__scontents_map_item_tl` saves each ⟨*stored content*⟩.

```
⟨*core⟩
\int_new:N \g__scontents_map_int
\tl_new:N \l__scontents_map_item_tl
```

(*End of definition for* `\g__scontents_map_int` *and* `\l__scontents_map_item_tl`.)

12.26.2 Public functions of the programming interface

`\scontents_if_exist_p:n` `\scontents_if_exist:nTF` `\scontents_count:n` The conditional `\scontents_if_exist:nTF` and the function `\scontents_count:n` use `\__scontents_seq_if_exist:nTF` and `\__scontents_seq_count:n`.

```
\prg_new_conditional:Npnn \scontents_if_exist:n #1 { p, T, F, TF }
  {
    \__scontents_seq_if_exist:nTF {#1}
      { \prg_return_true: }
      { \prg_return_false: }
  }
\cs_new:Npn \scontents_count:n #1 { \__scontents_seq_count:n {#1} }
```

(*End of definition for* `\scontents_if_exist:nTF` *and* `\scontents_count:n`. *These functions are documented on page 9.*)

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\scontents_get:nnN
__scontents_get:nnN

The function \scontents_get:nnN evaluates the ⟨*index*⟩ and passes it to __scontents_get:nnN, which sets the token list passed in #3 to the ⟨*stored content*⟩ given by __scontents_seq_item:nn and removes \c__scontents_hidden_space_str from its end with __scontents_remove_trailing_eol:N. Otherwise the token list is empty and the "errors" undefined-storage or index-out-of-range are raised, except for a ⟨*stored content*⟩ that is not available yet in a remembered *sequence* (§12.12).

```
\cs_new_protected:Npn \scontents_get:nnN #1 #2 #3
  { \exp_args:Nne \__scontents_get:nnN {#1} { \int_eval:n {#2} } #3 }
\cs_new_protected:Npn \__scontents_get:nnN #1 #2 #3
  {
    \tl_clear:N #3
    \__scontents_seq_if_exist:nTF {#1}
      {
        \__scontents_if_pending:nnF {#1} {#2}
          {
            \bool_lazy_or:nnTF
              { \int_compare_p:nNn {#2} = { 0 } }
              {
                \int_compare_p:nNn { \int_abs:n {#2} }
                  > { \__scontents_seq_count:n {#1} }
              }
              {
                \msg_error:nneee { scontents } { index-out-of-range }
                  {#2} {#1} { \__scontents_seq_count:n {#1} }
              }
              {
                \tl_set:Ne #3 { \__scontents_seq_item:nn {#1} {#2} }
                \__scontents_remove_trailing_eol:N #3
              }
          }
      }
      { \msg_error:nnn { scontents } { undefined-storage } {#1} }
  }
```

(*End of definition for* \scontents_get:nnN *and* __scontents_get:nnN*. This function is documented on page 9.*)

\scontents_map_inline:nn
__scontents_map_inline:nn

The function \scontents_map_inline:nn increases \g__scontents_map_int, defines with the {⟨*inline function*⟩} passed in #2 a function whose name has the nesting level and calls __scontents_map_inline:nn, which gets each ⟨*stored content*⟩ with __scontents_get:nnN and passes it to that function. Each nesting level has its own function, so the {⟨*inline function*⟩} can use \scontents_map_inline:nn again.

```
\cs_new_protected:Npn \scontents_map_inline:nn #1 #2
  {
    \__scontents_seq_if_exist:nTF {#1}
      {
        \int_gincr:N \g__scontents_map_int
        \cs_gset_protected:cpn { __scontents_map_ \int_use:N \g__scontents_map_int :n }
          ##1 {#2}
        \exp_args:Nne \__scontents_map_inline:nn {#1} { \int_use:N \g__scontents_map_int }
        \int_gdecr:N \g__scontents_map_int
      }
      { \msg_error:nnn { scontents } { undefined-storage } {#1} }
  }
\cs_new_protected:Npn \__scontents_map_inline:nn #1 #2
  {
    \int_step_inline:nn { \__scontents_seq_count:n {#1} }
      {
        \__scontents_get:nnN {#1} {##1} \l__scontents_map_item_tl
        \exp_args:Nc \exp_args:NV { __scontents_map_ #2 :n } \l__scontents_map_item_tl
      }
  }
⟨/core⟩
```

(*End of definition for* \scontents_map_inline:nn *and* __scontents_map_inline:nn*. This function is documented on page 9.*)

12.27 Finish package

Finish package implementation, in ConTEXt '!' is a letter again (§12.2.1).

```
⟨context⟩\char_set_catcode_letter:N \!
⟨plain|context⟩\ExplSyntaxOff
```

```
⟨plain|context⟩\endinput
⟨latex|core⟩\file_input_stop:
```

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13 Index of Implementation

The italic numbers denote the pages where the corresponding entry is described, the numbers underlined and all others indicate the line on which they are implemented in the package code.

A

after 1323

B

before 1323

C

\cleanseqsc *8*, *60*, 1813
cmd-key-unknown 287
cmd-key-value-unknown 287
Commands provided by SCONTENTS:
\Scontents *49*
\cleanseqsc *60*
\countsc *60*
\foreachsc *52*
\getstored *49*
\meaningsc *57*
\mergesc *59*
\newenvsc *44*
\setupsc *60*
\typestored *55*
\countsc *8*, *60*, 1804

D

debug-var 222
\dospecials 1104

E

empty-argument-command 1231
empty-environment 760
empty-store-name 196
empty-variable 1548
\endscontents *4*, 1098
\endverbatimsc 1176
env-already-defined 743
env-key-unknown 268
env-key-value-unknown 268
Environments provided by SCONTENTS:
verbatimsc *46*, *47*

F

force-eol 222, 772, 1233
\foreachsc *6*, *49*, 1427

G

\getstored *6*, *49*, 1304
getstored 1368, 1673

I

index-out-of-range 178

J

junk-after-begin 743

M

\meaningsc *7*, *52*, *56*, 1656
meaningsc 1368, 1673
\mergesc *7*, *15*, *52*, *57*, 1794
mergesc-missing-key 1663
multiple-begin 760

N

\newenvsc *5*, *37*, 1090
not-writing 205

O

overwrite 222, 772, 1233, 1438
overwrite-file 205

P

Packages :
expl3 . . . . . *21*, *22*, *60*
l3keys *52*
latex-lab *46*
ltcmd *21*, *39*, *48*
lua-widow-control *23*
verbatim *47*
xparse *21*
xsim *39*
xtemplate *46*
positive-integer 196
print-all 222
print-cmd 222, 1233, 1438
print-env 222, 772

Q

quark internal commands:
\q__scontents_mark 160
\q__scontents_stop 160

R

remember 222
remember-latex-only 611
remember-rerun 619
remember-undefined 619
rescanning-text 760

S

scan internal commands:
\s__scontents_mark 160
\s__scontents_stop 160
\Scontents *5*, *47*, 1289
\scontents *4*, 1098
scontents *4*, 1098
scontents commands:
\scontents_count:n *9*, 1821
\scontents_get:nnN *9*, 1828
\scontents_if_exist:nTF *9*, 1821
\scontents_if_exist_p:n *9*, 1821
\scontents_map_inline:nn *9*, 1855
scontents internal commands:
__scontents_analyse_nesting:n 962
__scontents_analyse_nesting_format:n . 984
__scontents_analyse_nesting_generic:nn 984
__scontents_analyse_nesting_generic:w 984
__scontents_analyse_nesting_generic_-
process:nn 984
__scontents_analyse_nesting_latex:n . . 984
__scontents_analyse_nesting_latex:w . . 984
__scontents_append_contents:nn 484

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__scontents_bsphack: 453
\l__scontents_char_value_int 12
__scontents_check_line_process:nn . . . 798
\l__scontents_cmd_name_tl 164
\l__scontents_current_seq_name_str . . . 1434
__scontents_define_env:nnn 1057
__scontents_define_generic_nesting_-
function:n 984
__scontents_do_noligs:N 412
\c__scontents_end_env_tl 133
\g__scontents_end_verbatimsc_tl 133
__scontents_env_begin_str:n 729
__scontents_env_end_function: 837
__scontents_env_end_str:n 729
\l__scontents_env_name_tl 133
\l__scontents_environment_keys_tl 723
__scontents_esphack: 453
__scontents_file_if_exist:nTF 103
__scontents_file_if_writable:nTF 357
\l__scontents_file_name_tl 165
__scontents_file_tl_write_start:n . . . 837
__scontents_file_write_cmd:nn 380
\l__scontents_file_write_iow 165
__scontents_file_write_stop:N 837
__scontents_finish_storing:NNN 500
__scontents_foreach_add_body:n 1380
\l__scontents_foreach_after_bool 1308
\l__scontents_foreach_after_tl 1308
\l__scontents_foreach_before_bool 1308
\l__scontents_foreach_before_tl 1308
\l__scontents_foreach_body_tl 1308
\l__scontents_foreach_cmd_tl 1308
\g__scontents_foreach_exec_tl 1308
\l__scontents_foreach_print_seq 1308
\l__scontents_foreach_start_int 1308
\l__scontents_foreach_step_int 1308
\l__scontents_foreach_stop_bool 1308
\l__scontents_foreach_stop_int 1308
__scontents_foreach_wrap:n 1380
\l__scontents_foreach_wrapper_bool . . . 1308
__scontents_foreachsc:nn 1380
\l__scontents_foreachsc_arg_tl 1308
__scontents_format_case:nnn 98
__scontents_generic_begin: 1043
__scontents_generic_end: 1043
__scontents_get:nnN 1828
__scontents_getfrom_seq:Nn 547
__scontents_getfrom_seq:nn 547
__scontents_getfrom_seq:nNn 547
__scontents_getfrom_seq:nnn 547
__scontents_getfrom_seq_aux:nnn 547
__scontents_getstored:nn 1294
__scontents_grab_opt_arg:n 787
__scontents_grab_opt_arg:w 787
\c__scontents_hidden_space_str 174
__scontents_if_digits:nTF 1450
__scontents_if_nested:TF 962
__scontents_if_pending:nnTF 639
__scontents_if_pending:nTF 639
\l__scontents_index_str 1434
\g__scontents_last_stored_tl 173
__scontents_lastfrom_seq:n 586
__scontents_make_control_chars_active: 423
__scontents_map_inline:nn 1855
\g__scontents_map_int 1818
\l__scontents_map_item_tl 1818
__scontents_meaningsc:n 1626
__scontents_meaningsc:nn 1614
\l__scontents_meaningsc_arg_tl 1613
\l__scontents_mergesc_arg_tl 1661
__scontents_mergesc_cmd:nn 1673
__scontents_mergesc_code:nn 1695
__scontents_mergesc_getstored:nn 1777
\l__scontents_mergesc_keys_tl 1661
__scontents_mergesc_parse_item:n 1713
__scontents_mergesc_parse_list:n 1713
__scontents_mergesc_put_item:nn 1713
\g__scontents_name_sc!internal_seq . . . 171
\l__scontents_nesting_aux_int 723
__scontents_nesting_decr: 962
\l__scontents_nesting_env_int 723
__scontents_nesting_incr: 962
\c__scontents_newline_tl 117
__scontents_nolig_list: 1130
__scontents_normalise_line_ends:N . . . 798
__scontents_par: 401
__scontents_parse_mergesc:nw 1713
__scontents_parse_mergesc_aux:nw 1713
__scontents_parse_type_meaning_key:n . 1450
__scontents_parse_type_meaning_key:nn 1450
\l__scontents_pending_bool 1531
\l__scontents_plain_bool 1101
__scontents_plain_disable_outer_par: . 423
\l__scontents_processed_body_lines_tl . 723
__scontents_range_limit:nn 1450
__scontents_range_parser:nnn 1450
__scontents_range_parser:nnnw 1450
\l__scontents_range_start_tl 1434
__scontents_range_step:nn 1450
\l__scontents_range_stop_tl 1434
__scontents_remember:n 625
\g__scontents_remember_ior 603
\g__scontents_remember_iow 603
\l__scontents_remember_item_str 603
\l__scontents_remember_name_str 603
\l__scontents_remember_new_str 603
\g__scontents_remember_old_str 603
\g__scontents_remember_pending_bool . . . 599
__scontents_remember_read: 652
__scontents_remember_read_line:n 652
__scontents_remember_read_line:w 652
\g__scontents_remember_seq 599
__scontents_remember_write: 685
__scontents_remember_write_item:n . . . 685
__scontents_remember_write_seq:n 685
__scontents_remove_leading_nl:n 837
__scontents_remove_leading_nl:nn 837
__scontents_remove_leading_nl:w 837
__scontents_remove_trailing_eol:N . . . 340
__scontents_remove_trailing_eol:w . . . 340
\l__scontents_save_every_body_lines_tl 723
\l__scontents_save_sf_int 176

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\l__scontents_save_skip 176
\l__scontents_Scontents_arg_tl 1229
__scontents_Scontents_code:nn 1248
__scontents_Scontents_finish: 1276
__scontents_Scontents_norm_arg:n 1271
__scontents_Scontents_verb_arg:w 1258
__scontents_seq_count:n 514
__scontents_seq_if_exist:nTF 514
__scontents_seq_item:nn 514
\l__scontents_seq_item_seq 171
__scontents_set_active_eq:NN 423
__scontents_set_positive_int:Nnn 212
__scontents_set_write_out:n 390
__scontents_set_write_store:n 390
__scontents_setting_env:nn 1057
__scontents_setup_verb_processor: . . . 837
__scontents_start_after_option:w 798
__scontents_start_environment:w 798
__scontents_stop_environment: 798
__scontents_store_to_seq:NN 492
\l__scontents_storing_bool 165
__scontents_tab: 401
\l__scontents_tab_width_int 212
__scontents_tabs_to_spaces: 403
__scontents_tl_if_head_is_q_mark:nTF . 329
__scontents_typemeaning_setup:nn 1531
__scontents_typestored:N 1585
__scontents_typestored:nn 1554
\l__scontents_typestored_arg_tl 1547
__scontents_unknown_keys_cmd:n 317
__scontents_unknown_keys_cmd:nn 317
__scontents_unknown_keys_env:n 309
__scontents_unknown_keys_env:nn 309
__scontents_use_delimit_by_s_stop:nw . 325
__scontents_use_i_delimit_by_s_stop:nw 325
__scontents_use_none_delimit_by_q_-
stop:w 325
__scontents_use_none_delimit_by_s_-
stop:w 325
__scontents_verb_print_EOL: 1585
__scontents_verb_processor_iterate:nnn 837
__scontents_verb_processor_iterate:w . 837
__scontents_verb_processor_output:n . . 952
__scontents_verbatimsc_aux: 1130
__scontents_verbatimsc_instance: 1188
__scontents_verbatimsc_line:w 1168
\c__scontents_version_str 12
__scontents_vobeyspaces: 1130
\l__scontents_writable_bool 165
\l__scontents_writing_bool 165
__scontents_xobeysp: 1130
__scontents_xverb: 1111
__scontents_xverb:w 1111
sep 1323
\setupsc *3*, *60*, 1799
start 1323
\startscontents *5*, 1098
\startverbatimsc 1185
step 1323
stop 1323
\stopscontents *5*, 1098
\stopverbatimsc 1185
store-all 222
store-cmd 222, 1233
store-env 222, 772

T

tl commands:
\tl_replace_all:Nnn 157
\tl_retokenize:n 157
\typestored *7*, *52*, *54*, 1608
typestored 1368, 1673

U

undefined-storage 178
unknown 222, 772, 1233, 1323, 1438

V

verb-font 222
\verbatimsc 1176
verbatimsc *8*, 1207

W

width-tab 222, 1438
wrapper 1323
write-cmd 1233
write-env 772
write-out 772, 1233, 1438
writing-file 205

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